

Vision-Language Models for Embodied Reasoning: A Comprehensive Survey from Perception to Generalist Action

1 Foundations and Evolution of Embodied Intelligence

1.1 Defining the Perception-Cognition-Action Loop

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The fundamental distinction between traditional artificial intelligence and embodied intelligence lies in the transition from static data processing to a dynamic, closed-loop interaction with the physical world. While conventional vision-language models operate primarily as passive observers, analyzing pre-collected datasets to generate responses or classifications, embodied agents must navigate a continuous cycle of perception, cognition, and action. This sensorimotor loop is not merely a sequential pipeline but a tightly coupled system where actions actively alter sensory input, thereby influencing subsequent cognitive states and future decisions. To establish the theoretical boundaries of embodied reasoning, it is essential to contrast this dynamic requirement with the limitations of static tasks, emphasizing that true embodiment demands an agent be an active participant in its environment rather than a detached analyzer. This conceptual foundation is critical for understanding the architectural shifts discussed in the following section, where the move from modular pipelines to unified models is driven by the need to better support this intrinsic loop.

Historically, cognitive systems were often modeled as input/output devices where perceptual and motor functions were encapsulated in modules with limited interaction, adhering to a feedforward, open-loop fashion [1]. However, this modular separation fails to capture the complexity of real-world interactions where external forces and environmental perturbations constantly influence the agent. As argued in recent literature, even architectures attempting closed-loop designs often rely too heavily on the separation principle of control theory, which can lead to fragility when facing unknown external inputs or interference from other agents [1]. A more robust approach involves nonmodular architectures based on active inference, which demonstrate superior resilience by treating perception and action as inseparable components of a unified generative model [1]. This framework posits that agents must continuously minimize prediction errors to maintain life-compatible states, effectively merging the processes of sensing and acting into a single inferential mechanism [2; 3].

The interdependence of strategy, physiology, and sensing forms the core of this embodied loop [4]. Biological organisms have evolved to optimize these three factors in concert, adapting their physiological capabilities and sensing modalities to suit specific environmental strategies. For instance, the “introspective agent” concept highlights that different environments yield vastly different optimal designs; in some contexts, increasing physical ability or refining sensing modalities offers greater long-term gains than enhancing complex planning strategies alone [4]. This challenges the prevailing AI paradigm that equates intelligence solely with increasingly complex cognitive models. Instead, embodied reasoning requires a holistic consideration where the agent’s morphology and sensory apparatus are viewed as integral tools for cognitive offloading and environmental adaptation. Theoretical frameworks such as morphological computation quantify this interaction, revealing an antagonistic yet symbiotic relationship between controller complexity and the body’s ability to handle task dynamics [5]. While a well-adapted morphology can reduce the necessary complexity of the controller, learning to interact effectively with complex surroundings often necessitates an increase in controller complexity, suggesting that embodied intelligence emerges from the dynamic interplay between the brain and the body [5].

Active inference provides a unifying theoretical lens through which to view this perception-cognition-action loop. Under this paradigm, agents are driven by the imperative to minimize free energy, effectively acting to fulfill optimistic predictions about their preferred outcomes [6; 7]. Unlike reinforcement learning, which relies on extrinsic, human-designed reward functions, active inference offers a natural, self-supervised objective where agents act to resolve uncertainty and confirm their internal models of the world [6; 8]. This approach has been successfully scaled to high-dimensional tasks, demonstrating efficient exploration and sample efficiency that rivals strong model-free baselines [8]. By framing action selection as a mechanism to gather evidence for a generative model, active inference bridges the gap between cognitive decision-making and continuous motor control. This is particularly evident in dynamic planning scenarios, where agents must infer and impose motor trajectories based on cognitive decisions while simultaneously understanding and exploiting affordances for object manipulation [3; 9].

Furthermore, the loop extends beyond individual agency to encompass social cognition and collaborative interaction. In multi-agent settings, the dynamics of interaction can constitute social cognition itself, where the boundaries of the mind extend into the environment and the actions of others [10]. Embodied social interaction experiments suggest that co-regulated interactions are critical for agency detection and experiencing the presence of others, challenging folk psychological notions that locate cognition strictly within a single brain [10]. This perspective aligns with the vision of “shared intelligence,” where ecosystems of natural and synthetic agents engage in belief sharing and collective self-evidencing [11]. In such ecosystems, communication protocols and shared narratives become the medium through which individual generative models align, facilitating coordinated action and complex problem-solving in open worlds [11].

The shift from reactive to deliberative agents further underscores the importance of this loop. Reactive agents, which simply imitate behaviors encountered in training data, often fail in long-horizon complex tasks that require proactive reasoning and planning [12]. In contrast, deliberative agents utilize neuro-symbolic frameworks to apply reasoning based on neural and symbolic representations acquired from past experience, significantly outperforming reactive baselines in challenging benchmarks [12; 13]. These agents do not merely respond to stimuli; they engage in active inquiry, clarifying ambiguities and gathering missing environmental information before committing to physical actions [2]. This capacity for active vision and sensing allows agents to coordinate motor and sensory policies, selecting observations that are optimal for inferring necessary motor actions, much like human eye movements guide hand manipulation [14].

Ultimately, defining the perception-cognition-action loop requires recognizing that perception is not a passive compression mechanism but an active process shaped by action. The commutative properties of action sequences with respect to sensory information reveal how agents learn about objects and improve sample efficiency in reinforcement learning [15]. By grounding language in this dynamic loop, where words are linked to percepts and actions through continuous interaction, embodied AI moves closer to the goal of Artificial General Intelligence [16; 17]. This comprehensive framework, integrating active inference, morphological computation, and social dynamics, sets the stage for the architectural evolution detailed in the next section, tracing the field’s progression from rigid, separated components toward scalable, data-driven foundation models capable of seamless adaptation, robust reasoning, and meaningful collaboration in the physical world.

1.2 Architectural Evolution: From Modular Pipelines to Unified Foundations

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Building upon the theoretical imperative of the perception-cognition-action loop established in the previous section, the architectural landscape of embodied artificial intelligence has undergone a paradigmatic shift over the last decade, moving decisively from rigid, modular pipelines toward flexible, unified foundation models. In the early stages of embodied AI research, systems were predominantly constructed as a cascade of distinct, independently trained components, adhering to a traditional “sense-plan-act” cycle. This approach relied on a strict separation of concerns: perception modules processed raw sensor data into semantic representations, symbolic planners generated high-level action sequences based on these representations, and low-level controllers executed physical movements. While this modular design offered interpretability and allowed researchers to optimize individual components in isolation, it fundamentally conflicted with the tightly coupled nature of true embodiment. The decoupling of perception from action often led to severe error propagation, where minor perceptual inaccuracies cascaded into catastrophic planning failures. Furthermore, these systems struggled with the “grounding problem,” as the symbolic representations used by planners frequently lacked the nuanced, continuous information required for robust interaction in dynamic, unstructured environments. As noted in comprehensive analyses of the field, previous survey pursuits have often characterized only one or two dimensions of this trinity—vision, language, or robotics—without providing a holistic view of how these components interact within a unified architecture [18].

The limitations of modular designs became increasingly apparent as tasks grew in complexity, requiring agents to handle long-horizon dependencies and open-world generalization. Early attempts to bridge these gaps often involved complex engineering to interface disparate modules, such as connecting classical task planners with vision-language models to verify action affordances and detect failures [19]. Similarly, neuro-symbolic frameworks like JARVIS emerged to combine the scalability of large language models with the precision of symbolic reasoning, yet they still relied on a fundamentally modular structure where distinct components handled language understanding, map construction, and action generation [13]. Even advanced hierarchical control architectures, such as those utilizing multi-abstractive neural controllers for driving, maintained a degree of separation between high-level behavior planning and low-level motion primitives [20]. While these hybrid approaches improved robustness, they often retained the brittleness inherent in systems where information flow between modules is constrained by predefined interfaces rather than learned jointly, failing to fully realize the nonmodular resilience advocated by active inference theories.

The advent of large-scale foundation models has catalyzed a transition toward end-to-end unified architectures that dissolve the boundaries between perception, cognition, and action, directly addressing the need for a seamless sensorimotor loop. This new paradigm treats embodied reasoning as a single sequence modeling problem, where multimodal inputs—including images, proprioceptive states, and natural language instructions—are tokenized and processed by a shared backbone to directly predict action tokens or executable code. A pioneering example of this shift is PaLM-E, an embodied multimodal language model that directly incorporates continuous sensor modalities into a large language model [16]. By training encodings for visual and state estimation end-to-end alongside a pre-trained language model, PaLM-E establishes a direct link between words and percepts, enabling positive transfer across diverse embodied tasks without the need for separate planning or control modules. This approach demonstrates that a single large model can address a variety of reasoning tasks across multiple embodiments, effectively functioning as a generalist agent

that internalizes the feedback loops essential for adaptation.

Building on this unified philosophy, subsequent works have further refined the integration of planning and control within a single framework to support more complex deliberative processes. EmbodiedGPT, for instance, introduces an end-to-end multimodal foundation model specifically designed for embodied planning and execution [21]. By leveraging a large-scale dataset constructed with “Chain of Thoughts” reasoning, EmbodiedGPT adapts a large language model to generate high-level plans and extract task-related features that form a closed loop with low-level control. This eliminates the disjointed nature of traditional pipelines, allowing the model to learn the interdependencies between visual observations, linguistic goals, and motor actions simultaneously. Similarly, the Octopus framework exemplifies the move toward unified programming agents, where a vision-language model is trained to decipher task objectives and generate executable code sequences directly from environmental feedback [22]. Through reinforcement learning with environmental feedback, Octopus refines its decision-making capabilities within a single architecture, demonstrating proficiency in tasks ranging from daily chores to complex video game interactions.

The evolution also extends to specialized domains such as autonomous driving and 3D manipulation, where unified models are replacing stacked pipelines to better capture physical dynamics. ADriver-I proposes a general world model for autonomous driving that unifies visual features and control signals into interleaved vision-action pairs, using a multimodal large language model and diffusion techniques to autoregressively predict control signals and future frames [23]. In the realm of 3D understanding, 3D-VLA introduces a generative world model that links 3D perception, reasoning, and action, moving beyond the 2D limitations of earlier vision-language-action models [24]. These systems illustrate a broader trend where the distinction between “perception” and “planning” becomes blurred; the model learns a joint representation space where physical dynamics and semantic meaning are co-optimized, creating a robust substrate for the spatial reasoning required in real-world deployment.

Despite the promise of unified models, the transition is not without challenges, and the debate between modularity and unification continues. Some researchers argue for the continued relevance of modular designs in specific contexts, particularly when safety and interpretability are paramount. For example, systems controlling dual-arm robots have successfully employed modular architectures that deploy state-of-the-art large language models for planning alongside specialized vision-language models for perception and point cloud transformers for grasping [25]. This suggests that while the trajectory of the field is toward unification, hybrid approaches that leverage the strengths of both modular robustness and unified learning may persist during the interim. Nevertheless, the overarching architectural evolution is clear: the field is moving away from fragile, hand-engineered pipelines toward scalable, data-driven foundation models that embody intelligence through the seamless integration of sensing, thinking, and acting. This shift provides the necessary computational framework to support the dynamic interactions described in the perception-cognition-action loop, while simultaneously setting the stage for the critical challenge discussed in the following section: ensuring these unified models can effectively process and reason from the unique, immersive egocentric perspective inherent to embodied agents.

1.3 The Critical Role of Egocentric Perspective

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While the architectural evolution described in the previous section establishes a unified framework for integrating perception, cognition, and action, the efficacy of these foundation models hinges on the specific nature of the perceptual input they process. For embodied agents, this necessitates a fundamental shift from the detached, third-person exocentric view typical of traditional computer vision to the immersive, first-person egocentric perspective. This transition is not merely a change in camera angle but represents a profound transformation in how agents must ground language in physical reality. In standard vision tasks, models rely on curated datasets where objects are presented in canonical views, detached from the agent’s immediate sensorimotor loop. In contrast, embodied agents operate within a dynamic stream of sensory data where the visual field is inextricably linked to their own movement and intent. Consequently, the unique challenges of first-person perception arise because the agent must continuously transform abstract, often exocentrically derived knowledge—frequently embedded in pre-trained foundation models—into actionable, egocentric affordances to execute instructions effectively. This process demands sophisticated spatial perspective-taking capabilities that go beyond simple object recognition, requiring a real-time understanding of how the self relates to the environment.

A primary challenge in bridging the gap between unified architectures and physical deployment is the “interaction bias” and the disparity between how humans observe actions from a third-person perspective versus how an agent experiences them from a first-person viewpoint. Humans possess an innate cognitive ability to transform diverse exocentric interactions into invariant egocentric affordances, allowing them to understand how to interact with an object regardless of the viewing angle. Empowering artificial agents with this same capability is critical for robust embodied reasoning. Recent research has explicitly addressed this by proposing tasks focused on affordance grounding from exocentric views, where agents must learn to transfer interaction knowledge gained from observing others (exocentric) to their own operational context (egocentric) [26]. This cross-view knowledge transfer is essential because training data often consists of internet-scale videos or demonstrations filmed from a third-person perspective, while the deployed robot operates from a first-person stance. Without effective mechanisms to bridge this gap, even advanced unified models struggle to localize the specific regions of an object that afford action when the perspective shifts dramatically.

To mitigate these challenges, advanced frameworks have been developed to extract affordance-specific features from exocentric interactions and align them with egocentric observations, thereby enhancing the representational power of embodied models. For instance, methods have been devised to preserve affordance co-relations across views, ensuring that the semantic understanding of “graspable” or “pushable” remains consistent despite geometric distortions caused by perspective changes [27]. Furthermore, the concept of weakly supervised learning has been leveraged to identify matching object parts across images, transferring knowledge from active usage scenarios to inactive egocentric views [28]. These approaches highlight that successful embodiment relies on the agent’s ability to decouple the invariant functional properties of objects from the variable visual appearances caused by the ego-motion of the camera, a prerequisite for the generalization capabilities discussed in subsequent benchmarking analyses.

Beyond object-level affordances, the egocentric perspective is pivotal for constructing comprehensive environment representations that support the long-horizon planning enabled by unified ar-

chitectures. First-person video naturally foregrounds the use of physical space, revealing spatial zones of interaction that are often invisible in global maps or third-person reconstructions. Models like EGO-TOPO demonstrate how egocentric video can be decomposed into topological maps that capture the primary spatial zones of a room and the activities they support, effectively linking transient observations to persistent environmental functionality [29]. This human-centric modeling of space allows agents to anticipate future actions and understand the layout of an environment based on their own trajectory and intent, rather than relying on a static, allocentric blueprint. Similarly, approaches that learn environment-aware representations from egocentric video enable agents to predict unseen local surroundings, bridging the gap between immediate visual input and the broader physical context [30].

The difficulty of reasoning from this perspective is further compounded by the need for active perception and the management of limited fields of view, factors that are often overlooked in static evaluations. Unlike static tasks where the entire scene is available, egocentric agents must actively explore to resolve ambiguities and gather missing information. The evaluation of vision-language models has traditionally focused on third-person benchmarks, leaving a significant gap in assessing an agent’s ability to “think” from a first-person perspective. New benchmarks such as EgoThink have been introduced to specifically evaluate these capabilities, revealing that current models still possess considerable potential for improvement in handling first-person spatial reasoning and context retention [31]. The ability to localize active objects undergoing state changes based on textual instructions is another critical skill, requiring the integration of symbolic world knowledge with visual tracking to robustly identify targets in a cluttered, moving field of view [32]. These limitations in current evaluation protocols foreshadow the systemic issues regarding sim-to-real transfer and metric reliability that will be examined in detail in the following section.

Furthermore, the geometry of the world model itself plays a crucial role in how agents process egocentric information within unified frameworks. Theoretical work suggests that imbuing world models with specific group structures, such as projective geometry, can better capture the nonlinear magnification of information inherent in first-person views, thereby driving more effective curiosity-based exploration and action planning [33]. This geometric alignment is essential for agents to correctly interpret spatial relationships and navigate complex environments. Additionally, the generation of synthetic egocentric data has become a vital tool for overcoming the scarcity of real-world first-person training data, a key bottleneck in developing robust foundation models. Frameworks like EgoGen utilize novel human motion synthesis models to produce realistic egocentric trajectories, ensuring that the training data faithfully represents the visual changes and motion dynamics unique to embodied agents [34].

Ultimately, the egocentric perspective serves as the cornerstone for grounding language in physical action, forcing the convergence of perception, cognition, and control into a unified loop where every visual input is relative to the agent’s current state. Whether through learning navigational visual representations that contrast egocentric views with semantic maps [35] or by developing agents that can answer situated questions based on their specific position and orientation in a 3D scene [36], the field is increasingly recognizing that true embodied intelligence cannot be achieved without mastering the complexities of the first-person view. The ability to seamlessly transform exocentric knowledge into egocentric affordances and to manage spatial perspective-taking remains one of the most critical frontiers in the development of generalist embodied agents. As we move toward rigorous validation of these capabilities, it becomes evident that the limitations of current egocentric benchmarks and the challenges of simulating first-person dynamics are central to the broader systemic limitations facing the field.

1.4 Benchmarking Progress and Systemic Limitations

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The transition from mastering the egocentric perspective, as detailed in the previous section, to deploying robust embodied agents in the physical world is contingent upon rigorous validation. The trajectory of embodied intelligence has been inextricably linked to the development of high-fidelity simulators and benchmarks, which serve as the proving grounds for validating perception-cognition-action loops. Early progress was largely driven by datasets and environments such as ALFRED, TEACH, and EgoPlan-Bench, which standardized the evaluation of agents capable of interpreting natural language instructions and executing complex manipulation or navigation tasks. These platforms provided the necessary infrastructure to move beyond static image captioning toward dynamic, interactive decision-making. However, as the field matures, a critical reassessment of these benchmarks reveals systemic limitations that threaten to stall the transition from simulated proficiency—often achieved under the idealized conditions discussed earlier—to real-world deployment. The primary challenges revolve around the fragility of generalization to unseen environments, the persistent and often misunderstood sim-to-real transfer gap, and the reliability of current evaluation metrics.

A significant portion of current research relies on the assumption that performance gains in simulation will translate directly to physical reality. Yet, empirical evidence suggests a disconnect between simulated success and real-world robustness. For instance, studies investigating PointGoal navigation have demonstrated that high performance in simulators like Habitat often stems from agents learning to exploit specific physics engine artifacts, such as “sliding” along walls, rather than developing genuine navigational competence [37]. This phenomenon highlights a critical flaw in benchmark design: when agents can achieve high scores by overfitting to simulator idiosyncrasies, the benchmark fails to measure the intended cognitive capabilities. Furthermore, the reliance on validation sets within these benchmarks may provide a false sense of security. Research on the ALFRED dataset has shown that models achieving state-of-the-art results on unseen validation splits can still underperform significantly on true test splits, indicating that current validation protocols may not sufficiently account for variance or truly measure generalization [38].

The sim-to-real transfer gap remains one of the most formidable barriers to progress, particularly when considering the complex visual dynamics of egocentric vision. Traditionally, the community has operated under the hypothesis that increasing simulator fidelity—through higher resolution textures, more accurate physics, and detailed sensor models—is the optimal path to closing this gap. However, recent large-scale evaluations challenge this intuition, suggesting that lower-fidelity simulations can sometimes yield better transfer performance. This counter-intuitive finding arises because high-fidelity simulators can lead to overfitting on inaccurate physical dynamics and suffer from slow simulation speeds that limit the scale of learning [39]. Consequently, the pursuit of photorealism in benchmarks like those used for autonomous driving or robotic manipulation may inadvertently hinder the development of robust policies that can adapt to the noisy, unstructured nature of the real world, where the perfect geometric alignments assumed in simulation rarely hold.

To address these discrepancies, researchers have proposed various strategies, including domain randomization and bi-directional domain adaptation. While domain randomization has proven effective in exposing agents to a wide range of physical parameters, theoretical frameworks indicate that its success often depends on the use of history-dependent policies and specific conditions that are not always met in standard benchmark setups [40]. Moreover, the lack of standardized metrics

for quantifying the sim-to-real gap makes it difficult to compare methodologies objectively. Efforts to create unified frameworks for calibrating robotics simulators aim to fill this void by providing consistent tests and metrics, yet the ad-hoc nature of many current evaluations persists [41]. In the context of autonomous driving, the gap is further complicated by the difficulty of replicating rare, long-tail scenarios in simulation, which are critical for safety but often absent from standard benchmarks [42].

Another systemic limitation lies in the evaluation of generalization itself, extending beyond the geometric and physical constraints previously noted. Many benchmarks focus on in-distribution performance or minor variations of training environments, failing to test an agent’s ability to handle out-of-distribution shifts. Theoretical work on transfer risk suggests that optimizing for standard empirical loss does not guarantee robustness to environment shifts, necessitating new evaluation criteria that explicitly measure transferability [43]. Additionally, the generation of diverse training data within simulators is often constrained by human effort. Recent approaches leveraging Large Language Models to automatically generate rich simulation environments and expert demonstrations, such as GenSim, indicate a promising direction for expanding task-level diversity beyond the limited scope of manually curated benchmarks [44]. Such methods aim to overcome the bottleneck of static datasets by creating dynamic curricula that better prepare agents for the unpredictability of open-world scenarios.

Finally, the reliability of evaluation metrics extends beyond success rates to include the assessment of physical commonsense and causal reasoning, which are essential for true embodied intelligence. Current benchmarks often lack the depth to evaluate whether an agent understands the underlying physical mechanisms of a task or is merely memorizing visual patterns. The introduction of more rigorous testing protocols, such as those querying labeled data with scenario programs to validate failure modes, is essential for ensuring that simulated failures correspond to real-world risks [45]. As the field moves toward generalist embodied agents, future benchmarks must evolve to prioritize causal understanding, robustness to domain shifts, and the ability to generalize from limited data, rather than solely maximizing performance on fixed, potentially exploitable, simulated environments. Only by addressing these systemic limitations can the community ensure that progress in embodied AI translates to tangible capabilities in the physical world, bridging the gap between the theoretical potential of egocentric reasoning and practical deployment.

2 Core Architectures for Multimodal Alignment and Efficiency

2.1 Evolution from Dual-Encoders to Unified Multimodal Foundations

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The architectural trajectory of Vision-Language Models (VLMs) has undergone a paradigmatic shift, moving from strictly separated dual-encoder frameworks toward deeply integrated, unified multimodal foundations. This evolution is critical for embodied reasoning, as the structural capacity of the model dictates its potential for the fine-grained visual-linguistic-action alignment discussed in the subsequent section. In the early stages, the dominant design pattern involved dual-encoder architectures, exemplified by models like CLIP, which utilized separate encoders for visual and linguistic modalities mapped into a shared embedding space via contrastive learning. While these approaches demonstrated remarkable zero-shot transfer capabilities, they inherently suffered from a “modality gap,” where representations of images and text remained geometrically distinct within the latent space [46]. This separation fundamentally limited the model’s ability to perform the complex cross-modal synthesis required for physical interaction, as the interaction between modalities was

restricted to the final projection layer rather than occurring throughout the deep processing stages.

To address the limitations of rigid separation and prepare the ground for more sophisticated alignment strategies, researchers began exploring the extent to which transformer parameters could be shared across modalities. Early investigations into Modality-Shared Contrastive Language-Image Pre-training (MS-CLIP) revealed that moving away from completely separate encoders toward a mostly unified encoder architecture significantly improved performance. Studies indicated that sharing the majority of transformer parameters allowed for better transfer of common semantic structures, such as attention patterns, from language to vision, outperforming vanilla dual-encoder baselines by substantial margins [47]. This finding challenged the prevailing assumption that modality-specific encoders were strictly necessary, suggesting instead that a unified backbone could learn more transferable and aligned features while simultaneously reducing the total parameter count—a prerequisite for the efficient token-level operations described later.

Building on the concept of parameter sharing, the field witnessed the emergence of true Large Multimodal Models (LMMs) that treat all inputs as sequences of tokens processed by a single, massive transformer backbone. A pioneering example of this unified approach is Kosmos-1, a model trained from scratch on web-scale multimodal corpora containing arbitrarily interleaved text and images. Unlike previous models that relied on distinct encoders followed by a fusion module, Kosmos-1 employs a single transformer to perceive general modalities, enabling it to learn in-context and follow instructions without gradient updates. This architecture facilitates seamless cross-modal knowledge transfer, allowing the model to leverage linguistic reasoning capabilities to enhance visual perception and vice versa [48]. The success of Kosmos-1 demonstrated that a unified architecture could handle diverse tasks ranging from optical character recognition (OCR)-free NLP to complex visual question answering, establishing a new standard for multimodal intelligence capable of supporting embodied agents.

Further validating the unified architecture paradigm, the MM1 family of models conducted comprehensive ablations to identify crucial design lessons for scaling multimodal pre-training. The research behind MM1 confirmed that for large-scale applications, the specific design of the vision-language connector is of comparatively negligible importance when contrasted with the impact of the image encoder, resolution, and the diversity of pre-training data. By utilizing a careful mix of image-caption, interleaved image-text, and text-only data, MM1 models achieved state-of-the-art few-shot results, reinforcing the notion that a shared transformer backbone, when fed with high-quality and diverse data, enables robust cross-modal understanding [49]. These models also exhibited enhanced in-context learning and multi-image reasoning capabilities, attributes that are difficult to achieve with disconnected dual-encoder systems but essential for the dynamic environments faced by embodied agents.

The transition to unified foundations also necessitated innovations in how different modalities are tokenized and aligned before entering the shared backbone, directly informing the token clustering and fusion techniques discussed in Section 2.2. Approaches like Meta-Transformer proposed a framework where raw input data from various modalities, including 3D point clouds and audio, are mapped into a shared token space, allowing a frozen encoder to extract high-level semantic features without paired multimodal training data. This methodology underscores the flexibility of unified transformers to process heterogeneous data streams within a single architectural structure [50]. Similarly, the 4M (Massively Multimodal Masked Modeling) scheme unified the representation space of text, images, and geometric modalities by mapping them into discrete tokens and performing masked modeling on randomized subsets, demonstrating that a single encoder-decoder could excel at diverse vision tasks and function as a generative model [51].

Despite the advantages of unified backbones, the challenge of aligning modalities without catastrophic forgetting or losing modality-specific nuances remains a central concern that motivates the advanced alignment strategies detailed in the following subsection. Techniques such as Co-Articulated Multi-Modal Learning (COMMA) have been developed to ensure that prompts from both vision and language branches are fully correlated, preventing the degradation of generic knowledge learned during pre-training [52]. Furthermore, the integration of fine-grained visual knowledge into these unified models has been addressed by frameworks like LION, which injects spatial-aware visual knowledge and high-level semantic evidence into the MLLM through progressive incorporation and soft prompting, thereby overcoming the information loss often associated with coarse image-text pre-training [53].

In summary, the evolution from dual-encoders to unified multimodal foundations represents a fundamental restructuring of how AI systems perceive and reason about the world, setting the architectural stage for the specific alignment mechanisms required in embodied AI. By collapsing the barriers between vision and language into a shared transformer backbone, models like Kosmos-1, MM1, and Meta-Transformer have achieved a level of cross-modal synergy that dual-encoder systems could not attain. This architectural unification not only improves performance on downstream tasks but also provides the necessary substrate for the heterogeneous representational harmonization—bridging sensory inputs, language, and action spaces—that defines the next critical challenge in developing competent embodied agents.

2.2 Strategies for Visual-Linguistic-Action Space Alignment

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Building upon the unified transformer backbones established in the previous section, the critical challenge for embodied Vision-Language Models (VLMs) shifts to bridging the heterogeneous representational gaps between high-dimensional sensory inputs and the discrete, symbolic spaces of language and action. While Section 2.1 demonstrated the architectural benefits of collapsing vision and language into a shared backbone, effective embodied reasoning requires more than mere co-location; it demands a deep structural harmonization where visual features, linguistic semantics, and motor commands inhabit a truly unified latent space. The evolution of alignment strategies has therefore progressed from simplistic linear mappings to sophisticated, dynamic mechanisms that enable fine-grained interaction and efficient information flow across these distinct domains, directly addressing the modality-specific nuances that unified backbones alone cannot fully resolve.

Early approaches to multimodal alignment predominantly relied on linear projections or simple Multi-Layer Perceptron (MLP) connectors to map visual embeddings into the textual feature space of a pre-trained Large Language Model (LLM). While computationally efficient, these rigid methods often struggle to capture the complex, non-linear relationships necessary for grounded physical reasoning. To address these limitations, recent research has explored adaptive alignment paradigms that operate directly at the token level. For instance, the concept of “Align before Fuse” suggests that aligning image and text representations via contrastive losses prior to fusion enables more grounded representation learning, avoiding the pitfalls of unaligned token interactions [54]. This principle is particularly critical for embodied agents, where the precise correspondence between a visual object and its affordance-related textual description dictates successful action execution, serving as a necessary precursor to the discrete tokenization strategies discussed in Section 2.3.

A significant advancement in alignment efficiency involves the use of discrete tokenization and clustering mechanisms to reduce redundancy while preserving semantic integrity, foreshadowing

the full discretization paradigm detailed later. In video-based embodied tasks, consecutive frames often contain homogeneous information that inflates computational costs without adding semantic value. Methods like CenterCLIP employ multi-segment token clustering to identify and retain only the most representative visual tokens, thereby enhancing segment-level semantic alignment between video and text representations while drastically reducing memory footprints [55]. Similarly, the introduction of Finite Discrete Tokens (FDT) proposes a set of learnable tokens representing specific visual-semantic concepts, enforcing a sparse activation constraint that ensures matched visual and semantic concepts are represented by the same discrete tokens, effectively reducing the granularity gap between modalities [56]. These techniques are essential for real-time embodied agents that must process continuous video streams and generate actions with low latency, acting as a bridge between continuous feature spaces and the fully discrete token streams of future architectures.

Beyond static alignment, dynamic and hierarchical strategies have emerged to handle the temporal and spatial complexities of embodied environments. Hierarchical Alignment Networks (HANet) demonstrate the efficacy of decomposing video and text into multiple semantic levels—event, action, and entity—to achieve fine-to-coarse correlations, ensuring that local motion details and global contextual narratives are simultaneously aligned [57]. This multi-level approach is mirrored in training schemes that explicitly shape the encoding process through global and segment-level alignment pairs, ensuring that the network captures both high-level context and fine-grained semantics crucial for navigation and manipulation tasks [58]. Furthermore, implicit learnable alignment mechanisms have been proposed to minimize temporal modeling efforts by predicting interactive points within frame pairs, implicitly aligning features around these rich mutual information regions without the need for costly temporal attention modules [59].

For embodied agents, the alignment must extend beyond perception to explicitly include action spaces. The integration of action tokens into the language feature space often requires specialized adapters that can interpret physical dynamics. The “Align before Adapt” paradigm emphasizes exploiting entity-to-region alignments prior to adapting to video representation learning, which helps in understanding complex activities by bridging the gap with underlying entity semantics [60]. Moreover, advanced frameworks like ModaVerse introduce Input/Output (I/O) alignment mechanisms that operate directly at the natural language level, aligning the LLM’s output with the input of generative models to simplify the synchronization of latent spaces across diverse modalities including audio and video [61]. This shifts the burden from complex latent feature alignment to a more interpretable linguistic interface, facilitating the translation of high-level instructions into low-level motor commands and setting the stage for the action-token integration seen in Section 2.3.

The pursuit of a unified representation space has also led to the development of modality-agnostic alignment centers capable of handling the diverse sensor suites of embodied agents. UniBind, for example, constructs a unified and balanced representation space by leveraging LLM-augmented class-wise embedding centers, allowing for the alignment of seven diverse modalities including point clouds and thermal data without biasing toward a single central modality like images [62]. This is particularly relevant for embodied AI, where agents must synthesize data from depth sensors, tactile feedback, and egocentric vision. Additionally, methods such as TokenFusion dynamically detect uninformative tokens and substitute them with projected inter-modal features, employing residual positional alignment to explicitly utilize inter-modal correlations within a standard transformer architecture [63]. Such dynamic fusion strategies ensure that the model remains sensitive to the most salient cross-modal cues required for precise physical interaction.

Finally, the robustness of these alignment strategies under varying conditions is paramount for

deployment in dynamic physical worlds. Position-aware contrastive alignment networks have been developed to guide vision-language interaction through prior position information, enhancing the ability to locate referred objects accurately among instances, a capability directly transferable to object manipulation and navigation [64]. By integrating these diverse mechanisms—ranging from discrete token clustering and hierarchical semantic decomposition to dynamic token fusion and modality-agnostic unification—modern embodied agents are increasingly capable of harmonizing perception, cognition, and action within a single, coherent feature space. These strategies not only refine the unified foundations described in Section 2.1 but also provide the necessary structured representations that enable the radical shift toward fully discrete, homogeneous token sequences explored in the following section.

2.3 Discrete Tokenization and Unified Representation Learning

2.3 Discrete Tokenization and Unified Representation Learning

Building upon the alignment strategies discussed in the previous section, which focus on harmonizing heterogeneous modalities through projection and clustering, a more radical architectural evolution has emerged: treating all modalities as homogeneous token sequences. This paradigm shift moves beyond merely aligning distinct feature spaces to fully discretizing continuous sensory inputs—such as images, audio, 3D point clouds, and even continuous action trajectories—into finite sets of discrete tokens. By doing so, researchers can leverage the powerful autoregressive capabilities of frozen Large Language Models (LLMs) to process and generate multimodal data without the need for modality-specific decoders. This approach effectively transforms the complex challenge of multimodal alignment into a unified sequence modeling problem, where the distinction between perception, language, and action is reduced to differences in token vocabulary and embedding spaces, setting the stage for the parameter-efficient adaptations detailed in the subsequent section.

A primary advantage of this discrete tokenization is the creation of a shared semantic space where diverse modalities interact seamlessly, overcoming the modality entanglement often seen in continuous feature embedding approaches. Instead of relying on complex projection layers, discrete tokenization methods map inputs from any modality into a sequence of tokens compatible with standard LLM tokenizers. For instance, the TEAL framework proposes to tokenize and embed all modalities, allowing a frozen LLM to predict multi-modal tokens autoregressively just as it would predict text tokens [65]. This strategy enables the textual LLM to serve as a universal interface, maintaining its high performance in textual understanding while extending its capabilities to non-textual generation tasks. Similarly, the concept of treating different modalities as different “languages” has been successfully applied in tri-modal translation systems, where speech and image data are tokenized into discrete units to facilitate translation across arbitrary modality pairs [66].

The efficacy of this unified approach relies heavily on the design of the visual or sensory tokenizer, which must compress high-dimensional sensory data into a compact sequence of tokens that retain sufficient semantic richness for downstream reasoning. The LaVIT model introduces a dynamic discrete visual tokenization mechanism that translates images into sequences of discrete tokens akin to a foreign language, enabling the model to handle image and text indiscriminately under the same generative learning paradigm [67]. Furthermore, the SEED tokenizer emphasizes that image tokens should be independent of 2D physical patch positions and instead exhibit intrinsic interdependence aligned with the left-to-right autoregressive prediction mechanism of LLMs [68]. This alignment is crucial for enabling emergent abilities such as multi-turn in-context multimodal generation, where the model can “see” and “draw” simultaneously within a unified autoregressive

transformer.

In the context of embodied reasoning, discrete tokenization extends beyond static perception to include dynamic actions and state representations, directly addressing the need to close the perception-action loop. Embodied agents must not only perceive the world but also act within it, necessitating the inclusion of action and state tokens within the unified sequence. The MultiPLY model exemplifies this by encoding 3D scenes as abstracted object-centric representations and introducing specific action tokens to denote agent interactions, alongside state tokens representing multisensory observations [69]. This allows the LLM to generate action tokens that instruct the agent to perform specific movements, with the resulting environmental feedback appended back as state tokens to close the perception-action loop. Similarly, in the domain of human motion synthesis, motions of diverse body parts are quantized into separate codebooks, framing multimodal motion generation as a token prediction task that draws from specialized codebooks based on the control signal modality [70]. The TM2T model further reinforces this by using discrete motion tokens to enable the reciprocal generation of 3D human motions and texts, treating motion and text signals on a level playing ground [71].

The adoption of discrete tokenization also facilitates efficient compression and abstraction of temporal data, which is critical for long-horizon planning in embodied tasks. By viewing temporal action abstractions as a sequence compression problem, techniques like Byte Pair Encoding (BPE)—originally designed for text compression—can be adapted to learn powerful action primitives from continuous control domains. The PRISE approach combines continuous action quantization with BPE to discover high-level skills that significantly boost performance in multitask imitation learning [72]. This mirrors findings in other domains where BPE significantly decreases sequence length while increasing vocabulary expressiveness, leading to faster inference and better generation results [73]. Moreover, representing continuous trajectories as sequences of discrete motion tokens allows models like MotionLM to cast multi-agent motion forecasting as a language modeling task, bypassing the need for explicit latent variable optimization and producing joint distributions over interactive agent futures in a single autoregressive process [74].

Despite the promise of unified token sequences, challenges remain in ensuring that the discrete representations capture the necessary granularity for fine-grained control and physical commonsense. The quality of the unified representation depends on the tokenizer’s ability to balance compression with information retention. Approaches like Finite Discrete Tokens (FDT) aim to reduce the granularity gap between visual patches and text tokens by grounding multimodal inputs to a set of learnable tokens representing specific visual-semantic concepts [56]. Additionally, the integration of action and state tokens requires careful design to ensure that the LLM can correctly interpret the causal relationships between actions and environmental changes. The Unified-IO 2 model demonstrates the scalability of this approach by tokenizing inputs and outputs—including images, text, audio, action, and bounding boxes—into a shared semantic space, achieving state-of-the-art performance across more than 35 benchmarks including robotic manipulation [75].

Ultimately, discrete tokenization and unified representation learning provide a robust foundation for building generalist embodied agents by reducing all modalities to a common token stream. This architectural simplification enables the direct transfer of reasoning capabilities from language to vision and action, facilitating complex tasks such as tool use, navigation, and interactive dialogue. However, while this unified approach streamlines the modeling process, the sheer scale of these models necessitates efficient adaptation strategies for downstream tasks. As research progresses, the focus is shifting towards optimizing tokenizer efficiency and refining the alignment between discrete tokens and continuous physical dynamics, naturally leading to the exploration of parameter-

efficient fine-tuning and modular designs that allow these massive unified models to be specialized for specific embodied environments without prohibitive computational costs.

2.4 Parameter-Efficient Adaptation and Modular Design

2.4 Parameter-Efficient Adaptation and Modular Design

While the previous section established how discrete tokenization unifies heterogeneous modalities into a common sequence space, the sheer scale of these unified foundation models presents a new challenge: the prohibitive computational cost of full fine-tuning for diverse embodied reasoning tasks. Consequently, the field has shifted toward Parameter-Efficient Fine-Tuning (PEFT) strategies that preserve the rich semantic knowledge embedded in pre-trained models while enabling rapid, resource-conscious adaptation to specific sensory-motor loops. This paradigm is critical for embodied agents, which must often operate on edge devices and adapt to dynamic environments without catastrophic forgetting of general world knowledge. By freezing the core backbone and updating only a small subset of parameters, PEFT serves as the essential bridge between the unified representations discussed earlier and the sparse, dynamic routing mechanisms explored in the subsequent section.

A primary strategy in this domain involves integrating lightweight adapter modules into frozen backbones. Traditional architectures typically employ feed-forward neural networks inserted between transformer layers; however, recent advancements demonstrate that incorporating attention mechanisms within these adapters significantly enhances expressive power. For instance, the introduction of tiny-attention adapters, which utilize extremely small per-head dimensionalities, allows the model to modify hidden states by conditioning on global context, effectively acting as a mixture of experts that can be averaged during deployment to reduce inference costs [76]. Furthermore, to address the specific challenges of multimodal instruction following, researchers have proposed the Mixture-of-Modality Adaptation (MMA) framework. MMA replaces heavy cross-modal projection layers with lightweight adapters and employs a routing algorithm to automatically shift between single-modal and multi-modal instructions, thereby maintaining natural language understanding capabilities while efficiently bridging the gap between vision encoders and Large Language Models (LLMs) [77].

Building on these adapter concepts, modular design principles have emerged to decouple the handling of diverse modalities, allowing for more flexible and robust system architectures. A notable example is the PILL architecture, which introduces a “Plug Into LLM” approach featuring a Mixture-of-Modality-Adapter-Expert and a Modality-Attention-Gating mechanism. This design explicitly decouples complex modalities by assigning independent adapter experts to handle specific input types, while the attention gating mechanism adaptively controls the contribution of each modality token to the overall representation. This separation ensures that the addition of new sensory inputs, such as speech or video, does not interfere with the stable processing of existing modalities, a crucial requirement for embodied agents operating in heterogeneous environments [78]. Similarly, the concept of modularity has been extended to prompt tuning, where Conditional Prompt Tuning utilizes a Mixture of Prompt Experts (MoPE) to dynamically route instances to the most suitable prompt experts. By disentangling prompt vectors into specialized types that capture global and instance-level features, this approach effectively transfers pre-trained unimodal knowledge to downstream multimodal tasks with minimal trainable parameters [79].

The challenge of maintaining performance across varying domains and task distributions is further addressed through mixture-of-adapters frameworks. In scenarios involving significant distribution

shifts, such as those encountered in sim-to-real transfer for robotics, a static adapter may be insufficient. The Mixture-of-Adapters (MoA) approach employs multiple adapters with varying capacities and utilizes learnable routers to allocate tokens to the most appropriate adapter, effectively regularizing the model against domain shifts [80]. This concept of dynamic allocation is also evident in the Soft Mixture of Adapters (Soft-MoA), which leverages soft assignment between input tokens and adapter experts to ensure balanced utilization and improved scaling, particularly in audio and speech processing tasks that are increasingly relevant for multimodal embodied interaction [81]. Moreover, the flexibility of modular designs allows for the automatic discovery of optimal adapter structures and placement via Neural Architecture Search (NAS), eliminating the need for handcrafted, fixed adapter configurations and enhancing learning efficiency across multiple domains [82].

Preserving pre-trained knowledge while adapting to new tasks remains a central concern, particularly to avoid catastrophic forgetting. Strategies such as Model Tailor address this by identifying a sparse “model patch” through salience analysis and replacing only a small fraction of fine-tuned parameters, thereby maintaining nearly all original task effectiveness while acquiring new capabilities [83]. Complementing this, self-expansion frameworks like SEMA automatically decide whether to reuse existing adapter modules or add new ones based on detected distribution shifts, utilizing representation descriptors to trigger expansion only when necessary [84]. Additionally, modular approaches facilitate robust learning even when modalities are missing or corrupted. Frameworks like MMANet employ modality-aware regularization to guide the deployment network in balancing weak modality combinations, while others utilize reconstruction-based methods to impute missing sensory data before querying the model, ensuring reliable operation in imperfect sensing conditions typical of real-world robotics [85; 86].

Ultimately, the transition toward parameter-efficient and modular designs represents a fundamental shift in how embodied agents are constructed. By decoupling modality handling through expert adapters, attention gates, and dynamic routing mechanisms, these architectures enable scalable, adaptable, and robust intelligence. They allow agents to leverage the vast knowledge of frozen foundation models derived from unified tokenization while efficiently specializing for the intricate perception-cognition-action loops required in physical environments. This modular foundation not only strikes a critical balance between computational feasibility and functional versatility but also naturally sets the stage for the next evolutionary step: moving from static or semi-dynamic adapter configurations to fully dynamic Mixture-of-Experts (MoE) architectures that actively route computational resources based on input complexity.

2.5 Dynamic Modality Routing and Mixture-of-Experts

2.5 Dynamic Modality Routing and Mixture-of-Experts

Building upon the parameter-efficient and modular design principles discussed in the previous section, the pursuit of scalable embodied intelligence necessitates a further paradigm shift from static adapter configurations to sparse, dynamic frameworks. While Section 2.4 established how lightweight modules can be inserted or expanded to handle diverse modalities, this subsection explores how Mixture-of-Experts (MoE) architectures actively route computational resources to these specialized components based on input complexity. By decoupling model capacity from inference cost, MoE mechanisms allow Vision-Language Models (VLMs) to activate only a subset of parameters for any given token. In the context of embodied reasoning, where agents must process heterogeneous streams of visual, linguistic, and proprioceptive data in real-time, dynamic modal-

ity routing serves as a critical enabler for efficient deployment, optimizing the trade-off between performance and latency by selectively engaging sub-networks tailored to specific sensory demands.

A fundamental challenge in adapting MoE to embodied vision tasks lies in the design of the routing mechanism itself. Unlike natural language processing where token sequences are relatively uniform, visual data exhibits high variability in spatial redundancy and semantic density. Research into router formulations has revealed that the strategy for assigning tokens to experts significantly impacts model efficacy. For instance, empirical studies on vision MoE models indicate that “Expert Choice” routers, which match tokens to experts, generally outperform “Token Choice” routers in sparse settings, while soft MoE assignments often yield superior results under fixed compute budgets compared to hard gating mechanisms [87]. This insight is pivotal for embodied agents that must process high-resolution egocentric video feeds; a soft routing strategy allows the model to integrate nuanced visual features from multiple experts without the instability associated with hard assignments, ensuring robust perception in dynamic environments.

To further enhance scalability within these dynamic frameworks, recent advancements have focused on integrating MoE with low-rank adaptation techniques, creating architectures that expand parameter counts without proportional increases in inference costs. The Omni-SMoLA framework exemplifies this by utilizing a “Soft Mixture of Low-rank Experts” to boost generalist multimodal models. By employing lightweight experts that residually learn specialized knowledge on top of a frozen foundational backbone, Omni-SMoLA achieves state-of-the-art performance across diverse generative vision-and-language tasks while avoiding the prohibitive storage costs of replicating full expert models [88]. This architecture is particularly relevant for embodied agents that require broad generalization capabilities across unseen environments and tasks, as it allows the model to accumulate specialized knowledge for distinct manipulation or navigation scenarios without catastrophic forgetting or excessive memory overhead.

Beyond general scaling, dynamic routing enables the creation of modality-aware systems that refine the modular concepts introduced earlier by actively managing mixed inputs. The PILL architecture, previously noted for its modular adapter experts, leverages this dynamic capability through its “Modality-Attention-Gating” mechanism. This design allows the model to independently process different modalities—such as speech, video, and text—using dedicated adapter experts, while the attention gate adaptively controls the contribution of each modality to the final representation [78]. Such modularity is essential for embodied agents operating in noisy real-world settings where certain sensory channels may be unreliable or irrelevant; the system can dynamically down-weight corrupted inputs and rely more heavily on robust modalities, thereby enhancing overall system resilience.

Furthermore, the specialization of experts can be driven by the specific characteristics of the visual content itself, extending the concept of task-specific adaptation. The MoVA model addresses the limitations of single vision encoders by adaptively routing and fusing task-specific vision experts based on a coarse-to-fine mechanism. In this framework, a context-aware routing strategy leverages the large language model’s understanding of instructions to select the most suitable vision experts, while a mixture-of-vision-expert adapter extracts and fuses task-specific knowledge [89]. This capability is crucial for embodied reasoning, where an agent might need to switch between experts specialized in document reading, object detection, or depth estimation depending on whether the task involves reading a label, grasping an object, or navigating a room.

Efficiency in dynamic routing also extends to the handling of varying input difficulties, addressing the limitation of static resource allocation. Traditional MoE models often activate a fixed number

of experts regardless of input complexity, which can be computationally wasteful for simple instances. To address this, dynamic expert selection frameworks have been proposed that adjust the number of activated experts based on the confidence level of the router or the perceived difficulty of the task. Such methods ensure that harder tasks requiring complex reasoning activate more experts, while simpler perceptual tasks utilize fewer resources, leading to significant improvements in computational efficiency without sacrificing accuracy [90]. This adaptive compute paradigm aligns perfectly with the needs of embodied agents, which must operate under strict energy and latency constraints while performing a wide spectrum of tasks ranging from simple localization to complex long-horizon planning.

Finally, the integration of MoE into efficient inference systems addresses the practical challenges of deploying these large-scale dynamic models on resource-constrained hardware. Systems like SiDA and Pre-gated MoE leverage the inherent sparsity of expert activation to optimize memory usage and reduce latency, enabling the deployment of massive models on single GPUs or edge devices [91; 92]. For embodied agents, which often rely on onboard computation for real-time control, these system-level optimizations are as critical as the architectural innovations themselves. While the strategies discussed here focus on optimizing parameter activation and routing logic, the following section will examine complementary approaches that optimize the visual encoding process itself, specifically targeting token reduction and resolution flexibility to further alleviate computational bottlenecks.

2.6 Optimizing Visual Encoding for Efficiency and Resolution

2.6 Optimizing Visual Encoding for Efficiency and Resolution

While Section 2.5 demonstrated how dynamic modality routing and Mixture-of-Experts (MoE) architectures optimize the *activation* of parameters based on input complexity, a complementary bottleneck remains in the *generation* of those inputs: the visual encoding process itself. Even with sparse expert activation, the quadratic computational complexity of Transformer-based self-attention relative to the number of input tokens can prohibit real-time performance in embodied agents. These agents must process high-resolution, multi-aspect-ratio sensory streams continuously to support navigation and manipulation, making the sheer volume of visual tokens a critical constraint. Consequently, this subsection shifts focus from routing logic to the foundational optimization of visual encoders, exploring strategies for adaptive token reduction, flexible resolution handling, and advanced tokenizer pre-training to balance semantic fidelity with computational efficiency.

A dominant strategy for mitigating token overload involves adaptive reduction mechanisms that dynamically prune or merge redundant visual tokens based on their semantic importance. Traditional Vision Transformers (ViTs) process fixed-length sequences of patch tokens, often allocating equal resources to homogeneous background regions and salient objects. To address this inefficiency, recent works propose dynamic frameworks that operate without extensive retraining. For instance, the concept of “Token Fusion” amalgamates the benefits of token pruning and merging, utilizing techniques like MLERP merging to preserve feature norm distributions while reducing sequence length [93]. Similarly, adaptive sparse token pruning frameworks leverage learnable thresholds to hierarchically discard unimportant tokens, achieving substantial throughput improvements with negligible accuracy loss [94]. In the context of dense prediction tasks critical for robotics, methods such as AiluRus utilize spatial-aware density-based clustering to merge semantically similar tokens, effectively accelerating inference by nearly 50% while maintaining performance [95]. Furthermore,

approaches like ELIP introduce language-guided token pruning, removing less influential visual tokens based on supervision from language outputs, thereby aligning visual compression directly with multimodal task objectives [96]. These mechanisms are particularly vital for embodied agents that must prioritize salient environmental features over static backgrounds, ensuring that the sparse experts discussed in the previous section receive only the most informative inputs.

Parallel to token reduction, addressing the limitations of fixed-resolution inputs is crucial for robust embodied perception. Real-world robotic scenarios involve objects and environments at varying scales and aspect ratios, which fixed-size resizing strategies often distort, leading to a loss of fine-grained details necessary for manipulation. The NaViT architecture challenges the standard pipeline by employing sequence packing to process inputs of arbitrary resolutions and aspect ratios natively, demonstrating improved training efficiency and robustness [97]. Building on this, ViTAR introduces dynamic resolution adjustment modules and fuzzy positional encodings, enabling a single model to adapt to resolutions ranging from standard classification sizes to ultra-high resolutions suitable for fine-grained manipulation tasks [98]. For Large Multimodal Models (LMMs), LLaVA-UHD proposes a modularization strategy that divides native-resolution images into variable-sized slices, coupled with a compression module to condense tokens, allowing the model to perceive high-resolution details efficiently [99]. Additionally, the “Make A Long Image Short” approach employs a lightweight Token-Length Assigner to dynamically determine the minimum sufficient number of tokens required for accurate prediction per image, optimizing the trade-off between computational cost and visual fidelity [100]. Such flexibility ensures that embodied agents can allocate computational resources dynamically, focusing high-resolution processing on complex interactions while utilizing coarser representations for open-space navigation.

The efficacy of these encoding strategies is fundamentally dependent on the quality of the visual tokenizer itself. The choice of pre-training methodology significantly influences the tokenizer’s ability to capture both global semantics and fine-grained details necessary for grounding language in physical actions. Empirical investigations reveal that while fully supervised models capture richer semantics, self-supervised models often excel at fine-grained perception, suggesting a need for hybrid approaches [101]. To bridge this gap, researchers have explored discrete tokenization methods that align visual representations with the autoregressive nature of LLMs. The SEED framework, for example, designs an image tokenizer that produces tokens with 1D causal dependency and high-level semantic abstraction, enabling unified image-to-text and text-to-image capabilities within a single LLM backbone [102]. Moreover, the investigation into homogeneous visual tokenizers, such as HOOK, demonstrates that splitting images into semantically independent regions rather than fixed patches can drastically reduce token counts while improving performance on sparse and dense tasks [103]. Further advancements include bottom-up patch summarization techniques like BUS, which coordinate coarse extraction and top-level abstraction to learn concise visual summaries, boosting training efficiency by 50% [104].

Finally, the optimization of visual encoding extends to the alignment of spatial and temporal dimensions, serving as a critical bridge to the specialized temporal and 3D architectures discussed in the following section. While static image optimization reduces per-frame latency, embodied agents operating in dynamic environments must also handle redundancy across time. Methods like CenterCLIP utilize multi-segment token clustering to eliminate redundancy in consecutive video frames, significantly reducing memory costs while enhancing spatiotemporal alignment [55]. Similarly, semantic-aware temporal accumulation scores allow for the integral pruning of spatiotemporal tokens by evaluating both temporal redundancy and semantic importance, achieving over 30% computation reduction in video recognition [105]. Collectively, these advancements in adap-

tive token management, resolution flexibility, and tokenizer design form the bedrock of efficient multimodal architectures. By moving beyond rigid, fixed-resolution pipelines, modern VLMs not only achieve the speed required for real-world interaction but also establish the efficient data representation necessary for the complex temporal modeling and 3D geometric reasoning explored next.

2.7 Specialized Architectures for Temporal and 3D Inputs

2.7 Specialized Architectures for Temporal and 3D Inputs

Building upon the efficient visual encoding strategies discussed in Section 2.6, which optimize token usage for static and high-resolution images, embodied reasoning further demands architectures capable of processing continuous temporal streams and complex three-dimensional geometries. While adaptive token reduction and flexible resolution handling address computational bottlenecks in single-frame perception, standard Vision-Language Models (VLMs) trained on static image-text pairs often lack the intrinsic mechanisms to model the spatiotemporal evolution of scenes or the geometric constraints inherent in physical environments. To bridge this gap, recent research has focused on developing specialized architectural extensions that integrate robust temporal modeling modules for video understanding and frameworks that unify 3D point clouds, depth maps, and multisensory data within language-model-centric paradigms.

In the domain of temporal modeling, the primary challenge lies in efficiently capturing long-range dependencies and motion dynamics without incurring prohibitive computational costs, a concern that echoes the efficiency goals of token pruning but extends them across the time dimension. Traditional approaches utilizing 3D Convolutional Neural Networks (CNNs) have demonstrated efficacy in action recognition but often suffer from high parameter counts and computational overhead compared to their 2D counterparts [106]. To address these efficiency concerns, architectures like TEINet introduce temporal modules that decouple channel correlation from temporal interaction, allowing for flexible and effective temporal feature learning while maintaining high inference speeds. Furthermore, the interpretation of spatiotemporal features remains a critical area of investigation; comparative studies between 3D CNNs and Convolutional LSTMs reveal distinct learning behaviors, where 3D convolutions tend to focus on shorter, contiguous spatial events, whereas recurrent models may capture different temporal structures [107].

Beyond convolutional approaches, transformer-based architectures have emerged as powerful tools for grounding spatio-temporal language, leveraging the attention mechanisms refined in earlier sections to operate across both space and time. These models can learn the meaning of behavioral traces described in natural language, effectively mapping time-extended predicates to visual observations [108]. A notable advancement in this direction is the integration of language-based context summaries to enhance object interaction anticipation. The TransFusion architecture, for instance, utilizes pre-trained vision-language models to summarize past action contexts into textual descriptions, which are then fused with current visual inputs to forecast future interactions, demonstrating that language can serve as an efficient compressor of spatiotemporal history [109]. Additionally, unified autoregressive models like Unified-IO 2 have expanded the scope of temporal processing by tokenizing video and audio inputs alongside text and actions, enabling a single encoder-decoder transformer to handle diverse multimodal sequences seamlessly [75].

Parallel to temporal advancements, the integration of 3D geometric data poses unique challenges due to the unstructured nature of point clouds and the need for viewpoint invariance, requiring a shift from the 2D patch-based tokenization discussed previously to more sophisticated 3D rep-

resentation learning. Early unified models often relied on projected multi-view images, limiting their ability to reason directly about 3D geometry. In contrast, recent frameworks such as Uni3DL operate directly on point clouds, employing a query transformer to learn task-agnostic semantic outputs and a task router to generate specific predictions for detection, segmentation, and visual grounding [110]. This direct processing of 3D data is further enhanced by specialized encoders like ReCon++, which leverage multi-view image distillation to improve geometric understanding before feeding features into large language models, as seen in ShapeLLM [111]. Such architectures enable agents to perform embodied visual grounding and interact with 3D objects with a level of precision that 2D-only models cannot achieve.

The necessity for holistic 3D perception is further underscored by the introduction of EmbodiedScan, which advocates for ego-centric 3D perception suites that process arbitrary numbers of multi-modal inputs to understand scenes from a first-person perspective [112]. This shift towards ego-centricity is crucial for robots that must transform exocentric knowledge into actionable affordances. Moreover, the complexity of embodied tasks often requires more than just visual and geometric data; it demands the integration of multisensory cues. The MultiPLY framework exemplifies this by incorporating visual, audio, tactile, and thermal information into a large language model via object-centric representations and action tokens, allowing agents to actively explore and correlate multisensory states with linguistic concepts [69].

To further unify these capabilities, generative world models like 3D-VLA have been proposed to link 3D perception, reasoning, and action. By aligning embodied diffusion models with a 3D-based LLM, these systems can predict future goal images and point clouds, effectively simulating physical dynamics for planning purposes [24]. This generative capacity is complemented by approaches that utilize implicit object memory to aggregate features across long temporal horizons using projective geometry, significantly enhancing embodied object detection in complex, dynamic streams [113]. Additionally, the development of generalist agents like LEO demonstrates the viability of training shared LLM-based architectures on extensive 3D vision-language-action datasets, enabling proficiency across a spectrum of tasks from 3D captioning to robotic manipulation [114].

Finally, the efficiency of processing these high-dimensional inputs is paramount for real-time embodied interaction, necessitating the adaptation of the efficient encoding principles from Section 2.6 to omni-modal contexts. Techniques such as ViT-Lens facilitate omni-modal representation learning by projecting novel modalities, including 3D point clouds and depth, into a shared embedding space compatible with pre-trained Vision Transformers, thereby unlocking powerful pretrained knowledge for diverse sensory inputs without exhaustive retraining [115]. Collectively, these specialized architectures represent a significant leap forward, moving beyond passive observation to active, spatiotemporally aware, and geometrically grounded understanding essential for true embodied intelligence.

3 Spatial Understanding and 3D World Modeling

3.1 Explicit Geometric Representations and Cross-Modal Transfer

3.1 Explicit Geometric Representations and Cross-Modal Transfer

The transition from passive 2D visual understanding to active 3D embodied reasoning necessitates a robust handling of explicit geometric representations. Unlike the regular grid structure of images, the physical world perceived by embodied agents is inherently three-dimensional, irregular, and sparse. Consequently, the field has converged on three primary discrete representations: point

clouds, voxel grids, and meshes, each offering distinct trade-offs between geometric fidelity, computational efficiency, and suitability for downstream robotic tasks. Point clouds, as the native output of LiDAR and depth sensors, provide an unstructured yet precise description of surface geometry but suffer from permutation invariance challenges and a lack of explicit connectivity. Voxel grids impose a regular Euclidean topology that facilitates the application of standard convolutional operations, yet they incur cubic computational complexity relative to resolution, often limiting their application to coarse-grained scene understanding. Meshes, defined by vertices and faces, offer a compact and continuous surface representation ideal for physics simulation and collision avoidance, though their variable topology complicates direct deep learning processing. A central challenge in leveraging these representations for embodied AI is the scarcity of large-scale, annotated 3D datasets compared to the abundance of 2D image data. To bridge this domain gap, recent research has heavily invested in cross-modal transfer techniques, specifically utilizing knowledge distillation and weight inflation to transplant rich semantic priors from pretrained 2D models into 3D architectures. This focus on discrete, explicit structures serves as a critical foundation before exploring the paradigm shift toward continuous neural implicit fields discussed in the subsequent section.

One of the most effective strategies for enhancing data efficiency in 3D perception is the direct transfer of 2D model architectures and weights to 3D domains. The feasibility of this approach relies on the observation that low-level geometric features learned from 2D images can be generalized to 3D structures through appropriate transformation. For instance, the concept of weight inflation allows researchers to adapt 2D convolutional kernels for 3D processing by copying or expanding pretrained weights, thereby initializing 3D networks with powerful feature extractors derived from massive image corpora. This methodology has been empirically validated by approaches such as [116], which demonstrated that finetuning transformed image-pretrained models requires minimal architectural changes—often limited to input, output, and normalization layers—while achieving competitive performance on 3D classification tasks. This transfer not only accelerates convergence but also significantly boosts data efficiency, enabling high accuracy even in few-shot learning scenarios where 3D labeled data is extremely scarce. Furthermore, the theoretical underpinning of this success has been linked to the phenomenon of neural collapse, suggesting that the feature spaces of 2D and 3D models converge to similar structures despite the dimensional disparity.

Beyond simple weight initialization, structured knowledge distillation has emerged as a sophisticated mechanism for aligning 2D and 3D feature spaces. This paradigm typically employs a powerful 2D teacher network to guide the training of a 3D student network, forcing the latter to mimic the semantic richness of the former. However, direct application of 2D distillation methods to point clouds often yields suboptimal results due to the sparsity and irregularity of 3D data. To address this, specialized frameworks have been developed to extract and distill local geometric structures specifically tailored for point clouds. [117] introduces a local distillation strategy that utilizes dynamic graph convolution to highlight crucial points or voxels, ensuring that the student network learns efficient geometric representations that outperform even their teachers in certain detection metrics. Similarly, [118] addresses the issue of representation disparity between over-parameterized teachers and compact students by employing an information bottleneck perspective, effectively minimizing the feature gap in proposal regions and enabling the deployment of extremely compact 3D detectors suitable for resource-constrained embodied agents.

Cross-modal transfer is further enhanced by leveraging multi-view consistency, which serves as a bridge between the 2D and 3D domains. By projecting 3D points into multiple 2D views, models can exploit the vast knowledge embedded in 2D vision foundation models. [119] proposes a novel loss function that enforces consistency between the 2D projections of 3D features and the outputs

of pretrained 2D networks, effectively retaining 3D geometric signals while absorbing 2D semantic knowledge. This approach avoids the pitfalls of naive dimensionality lifting by directly training 3D networks with 2D guidance. Complementing this, [120] utilizes a visibility-aware feature projection to aggregate point-wise embeddings into view-specific descriptors, aligning them with multi-view visual features from a 2D teacher. This heterogeneous feature alignment allows 3D encoders to overcome their limited learning capacity and achieve state-of-the-art performance in shape recognition without exhaustive network modifications.

The efficacy of these transfer learning techniques extends to unsupervised and self-supervised settings, which are critical for embodied agents operating in open worlds where labeled data is unavailable. Methods like [121] exploit the geometric relationship between RGB cameras and LiDAR sensors to establish correspondences in both image-plane and bird’s-eye views. By using a self-supervised 2D network as a teacher, this framework guides the feature learning of 3D student networks without requiring manual annotations, significantly improving performance on downstream tasks such as semantic segmentation and object detection. Additionally, [122] revisits masked modeling through the lens of knowledge distillation, where frozen 2D Transformers act as cross-modal teachers for 3D autoencoders. This approach distills “dark knowledge” regarding foundational geometry into 3D learners, resulting in superior generalization capabilities across various benchmarks.

Finally, the integration of these representations supports more complex geometric reasoning required for manipulation and navigation. While point clouds excel in perception, meshes and voxels are often preferred for planning and interaction. Techniques such as [123] demonstrate how learning to deform points sampled from high-quality meshes can produce arbitrarily dense representations with fine-grained geometry, facilitating tasks like label transferring for semantic segmentation. Moreover, hybrid approaches like [124] unify point-based, voxel-based, and tree-based representations under a single learning framework, allowing agents to dynamically switch between representations based on bit-depth requirements and task constraints. These advancements in explicit geometric modeling and cross-modal transfer collectively form the bedrock of modern 3D world modeling, enabling embodied agents to perceive, reason, and interact with the physical environment with unprecedented efficiency and accuracy. While these discrete methods provide essential structural grounding, the limitations in resolution and memory inherent to voxelization and point sampling motivate the exploration of continuous neural implicit fields, which offer a seamless alternative for high-fidelity scene reconstruction.

3.2 Neural Implicit Fields and Continuous Scene Modeling

3.2 Neural Implicit Fields and Continuous Scene Modeling

Building upon the discrete, explicit geometric foundations discussed in Section 3.1, the field has witnessed a paradigm shift toward continuous neural implicit fields. While point clouds, voxels, and meshes provide essential structural grounding, they are inherently limited by resolution constraints, memory discretization, and the trade-offs between geometric fidelity and computational cost. Neural implicit representations address these limitations by encoding scene geometry and appearance as continuous functions parameterized by neural networks, offering a seamless alternative for high-fidelity scene reconstruction. This subsection focuses primarily on Neural Radiance Fields (NeRF) and their variants, which have emerged as the de facto standard for novel view synthesis and dense 3D modeling [125]. Unlike their discrete counterparts, these models represent a scene as a continuous volume where a multilayer perceptron (MLP) maps spatial coordinates and viewing directions to volume density and view-dependent radiance, enabling the synthesis of photorealistic

images from arbitrary viewpoints through differentiable volume rendering.

A defining characteristic of neural implicit fields is their capacity to capture fine-grained geometry and appearance details even when trained on sparse input views, a capability critical for embodied agents operating in data-scarce environments. However, standard NeRF formulations often struggle with overfitting and geometric artifacts under such constraints. To mitigate this, recent research has introduced sophisticated regularization strategies and architectural innovations. For instance, [126] reduces artifacts by regularizing the geometry and appearance of patches rendered from unobserved viewpoints and employing annealing strategies for ray sampling spaces. Similarly, [127] proposes augmented models that encourage simpler solutions by exploring positional encoding and view-dependent radiance, using the depth estimated by these simpler models to supervise the primary NeRF training. Other approaches, such as [128], leverage visibility priors computed via plane sweep volumes to provide dense supervision without relying on potentially inaccurate pre-trained depth networks. Furthermore, [129] introduces feature matching-based sparse geometry regularization to pinpoint high-frequency keypoints, safeguarding fine details that are often lost in few-shot scenarios.

Beyond handling sparse inputs, scalability remains a pivotal challenge for deploying neural implicit fields in the large-scale, unbounded environments typical of embodied AI. The original NeRF architecture, reliant on a single large MLP, faces prohibitive computational costs for training and rendering in complex urban settings. To overcome these limitations, researchers have developed adaptive and hierarchical structures that partition the scene into manageable sub-regions. [130] proposes subdividing scenes into axis-aligned bounding boxes using a tree hierarchy, assigning smaller NeRFs to different subspaces based on scene complexity to balance workloads and accelerate rendering. Similarly, [131] employs a level-of-detail approach where query coordinates with high uncertainties are forwarded to larger, more capable neural networks, ensuring efficient resource allocation between simple and complex scene regions. For large-scale urban scenes, [132] integrates a compact multiresolution ground feature plane representation with a lightweight NeRF branch, achieving high-fidelity rendering while maintaining computational efficiency. Additionally, [133] utilizes a two-stage global-focal architecture to maintain scene-wide consistency while allowing sub-encoders to focus on capturing local details, effectively addressing the blurring issues common in large-scale reconstructions.

To further enhance robustness and utility for robotic tasks, the evolution of neural implicit fields increasingly encompasses the integration of explicit geometric priors and hybrid representations. [134] decouples occupancy and color learning by leveraging LiDAR data to build differentiable 3D occupancy grid maps, significantly improving performance in large outdoor driving scenes with sparse views. This fusion of sensor modalities allows for more accurate depth prediction and novel view synthesis in challenging conditions. Moreover, the move towards explicit neural fields, such as [135], enables efficient manipulation and composition of 3D models by learning hybrid tensor rank decompositions, bridging the gap between the flexibility of implicit representations and the manipulability of explicit meshes. [136] further advances this by utilizing adaptive tetrahedral grids derived from Delaunay triangulation, offering superior detail around surfaces compared to uniform voxel subdivisions. These hybrid approaches effectively synthesize the strengths of the discrete methods from Section 3.1 with the continuity of implicit fields.

Continuous scene modeling is not limited to static environments; it extends dynamically to capture humans in motion and temporal changes, laying the groundwork for the dynamic physical understanding explored in Section 3.3. [137] combines neural scene representation with implicit statistical geometric human models to robustly reconstruct moving subjects from sparse camera views. For dynamic scene updates, [138] introduces a pipeline to identify scene changes and update

pre-trained NeRFs using only a few new images, bypassing the need for time-consuming re-training. Additionally, [139] addresses the challenges of unconstrained image collections by leveraging interactive information across multiple rays to handle varying appearances and transient objects, mimicking human perceptual processes.

In summary, neural implicit fields have evolved from simple MLP-based volume renderers into sophisticated, scalable, and robust frameworks capable of modeling complex 3D worlds from sparse and noisy data. By addressing challenges related to sparse inputs, large-scale scalability, and dynamic changes through architectural innovations and hybrid representations, these continuous models provide a foundational layer for embodied agents to achieve precise spatial understanding. The ongoing refinement of these techniques, including efforts to improve rendering speeds via distillation into explicit structures as seen in [140] and [141], ensures their increasing viability for real-time embodied reasoning. Crucially, by providing high-fidelity, continuous geometric representations, these models serve as the essential static backbone upon which generative world models can be built to simulate physical dynamics and future states, as will be discussed in the following section.

3.3 Generative World Models for Physical Dynamics

3.3 Generative World Models for Physical Dynamics

Building upon the continuous scene representations established in neural implicit fields, the next critical leap for embodied reasoning involves transitioning from static 3D perception to dynamic physical understanding. While Section 3.2 detailed how agents can reconstruct high-fidelity geometries from sparse data, effective interaction requires anticipating how these geometries evolve under action constraints. Generative world models serve as the computational engine for this capability, enabling agents to perform “hindsight for foresight” reasoning by simulating the consequences of potential actions before execution. These models synthesize 3D structures and predict future physical states, effectively bridging the gap between the high-level semantic planning discussed in upcoming sections and low-level physical dynamics. The landscape of generative world models is currently dominated by two primary paradigms: Generative Adversarial Networks (GANs) and diffusion models, each offering distinct advantages in modeling complex, stochastic, and high-dimensional physical environments.

Generative Adversarial Networks have established a strong foundation for predicting future states in dynamic systems by learning the underlying distribution of physical trajectories. In the context of embodied agents, GANs are particularly effective for modeling the multi-modal nature of future outcomes, where a single current state can lead to diverse valid futures depending on agent interactions and environmental stochasticity. For instance, frameworks like SoPhie leverage GANs to predict paths that are compliant with both social and physical constraints, utilizing attention mechanisms to extract salient scene context and agent interactions [142]. This approach demonstrates that generative models can successfully integrate visual scene information with trajectory history to produce physically plausible predictions. Similarly, the Trajectron model employs graph-structured variational deep generative modeling to forecast multiple potential futures for interacting agents, highlighting the capacity of GAN-based architectures to handle dynamic scenarios with time-varying numbers of agents [143].

Beyond trajectory prediction, GANs have been instrumental in learning latent dynamics directly from high-dimensional sensory inputs, such as images, which is crucial for model-based planning in unknown environments. The Deep Planning Network (PlaNet) exemplifies this by learning a

latent dynamics model that combines deterministic and stochastic components to predict rewards and state transitions over multiple time steps, enabling fast online planning in latent space [144]. Furthermore, recent advancements have explored hierarchical hybrid frameworks that combine deep learning with reinforcement learning to capture multi-scale interactions, ensuring that predicted trajectories are not only accurate but also kinematically feasible [145]. The ability of GANs to model complex distributions is further evidenced in works like HP-GAN, which predicts probabilistic 3D human motion sequences, generating multiple plausible futures from a single input sequence [146]. These models often utilize specialized loss functions and architectural modifications, such as difference-guided generation, to enhance the accuracy of future frame prediction in video sequences, thereby improving the fidelity of the simulated physical dynamics [147].

Parallel to the advancement of GANs, diffusion models have emerged as a powerful alternative for generative world modeling, offering superior stability and sample quality in high-dimensional spaces. Diffusion-based approaches are increasingly being adopted for task-agnostic planning and trajectory generation due to their flexibility in sampling diverse outcomes. The Diffused Task-Agnostic Milestone Planner illustrates this potential by utilizing a diffusion-based sequence model to plan milestones in a latent space, allowing agents to solve long-horizon, sparse-reward tasks that are challenging for traditional reinforcement learning methods [148]. In the domain of autonomous driving, diffusion-based models have demonstrated state-of-the-art performance in multi-agent trajectory prediction by capturing complex interactions and the multimodal nature of traffic scenarios through interaction-aware guidance signals [149]. The theoretical underpinnings of these models are comprehensively surveyed in recent literature, which highlights their applicability across diverse domains including biology and healthcare, suggesting a broad potential for embodied AI applications [150].

A critical frontier in this domain is the integration of explicit 3D reasoning into generative world models, extending the continuous representations from Section 3.2 into the temporal dimension. Traditional approaches often operate in 2D image space, failing to capture the full geometric and physical consistency required for robust manipulation and navigation. However, emerging frameworks like 3D-VLA represent a significant leap forward by linking 3D perception, reasoning, and action through a generative world model. 3D-VLA utilizes embodied diffusion models aligned with a large language model to predict goal images and point clouds, effectively injecting generation abilities into the planning process [24]. This aligns with broader efforts to synthesize immersive 3D indoor scenes from limited inputs, where image-to-image GANs map reprojections of incomplete point clouds to full high-resolution RGB-D images, facilitating data augmentation for downstream robotics tasks [151]. Additionally, research into 3D-aware video generation combines neural implicit representations with time-aware discriminators to synthesize 3D videos supervised only by monocular inputs, learning rich embeddings of decomposable 3D structures and motions [152].

The utility of these generative models extends to specific physical domains, such as molecular conformation and material science, where preserving physical laws is paramount. Diffusion-driven frameworks have been developed to deduce three-dimensional molecular configurations from graph representations, addressing the challenge of preserving roto-translational invariance to ensure generated structures adhere to physical laws [153]. Similarly, physics-guided deep learning models incorporate physically oriented data augmentation and loss terms to generate crystal structures with higher physical feasibility, demonstrating the necessity of embedding physical constraints directly into the generative process [154]. The concept of learning effective physical laws is further explored in Lagrangian Deep Learning, which uses layers of Lagrangian displacements to learn effective descriptions of underlying physics, enabling the scaling of generative models to very high

dimensions with significantly reduced computational costs [155].

Moreover, the integration of causal reasoning into generative models allows agents to disentangle the causes of events and reason backward to achieve goals. Schema Networks, for example, function as object-oriented generative physics simulators capable of zero-shot transfer by learning the dynamics of an environment directly from data and reasoning through causal relationships [156]. This causal understanding is complemented by models that learn intuitive physics through self-organizing maps, enabling agents to execute kinematic simulations of the environment’s future for action planning after a phase of playful exploration [157]. The combination of visual grounding with dynamics priors allows neural models to infer physical properties within few observations, adapting quickly to unseen scenarios involving rigid objects, deformable materials, and fluids [158].

In summary, generative world models, powered by both GAN and diffusion architectures, are fundamentally reshaping how embodied agents understand and interact with physical dynamics. By synthesizing 3D structures and predicting future states with high fidelity, these models enable agents to plan actions within simulated environments, reducing the reliance on costly real-world trial and error. Crucially, by providing a mechanism to simulate the outcomes of actions in a geometrically consistent manner, these dynamic models lay the necessary groundwork for the semantic grounding discussed in Section 3.4, where abstract linguistic concepts must be mapped onto not just static geometry, but also the evolving physical states of the world. The convergence of 3D representation learning, causal reasoning, and advanced generative techniques promises to yield increasingly sophisticated agents capable of navigating and manipulating the complex, dynamic physical world with human-like foresight.

3.4 Semantic Grounding in 3D Spaces

3.4 Semantic Grounding in 3D Spaces

Building upon the dynamic physical simulations established in Section 3.3, the next critical step for embodied reasoning is bridging the gap between these geometric and temporal representations and high-level human intent. While generative world models enable agents to anticipate physical outcomes, effective interaction requires a robust mechanism for semantic grounding: precisely mapping abstract linguistic concepts onto the continuous 3D structures of the physical world. Traditional approaches to 3D understanding have largely relied on closed-set classification, limiting agents to recognizing only those object categories present in their training data. However, for an embodied agent to operate effectively in open-world environments, it must possess the capability to ground free-form language descriptions onto 3D structures, facilitating open-vocabulary semantic segmentation and complex visual reasoning tasks without the burden of exhaustive manual annotations. Recent advancements have shifted the paradigm towards leveraging the synergy between large-scale vision-language foundation models and 3D geometric representations, enabling agents to interpret and interact with novel objects and scenes through natural language.

A primary challenge in this domain is bridging the modality gap between the discrete, token-based nature of language and the continuous, often sparse, structure of 3D data such as point clouds and neural fields. Early efforts attempted to address this by projecting 3D data into 2D views to utilize established 2D vision-language models, yet this often resulted in information loss and occlusion issues. To overcome these limitations, researchers have developed methods that directly align 3D features with linguistic embeddings. For instance, the integration of pre-trained models like CLIP into 3D pipelines has proven effective. The work [159] demonstrates a training-free approach where dense pixel features from CLIP are projected onto point clouds, allowing 3D scene understanding

models to inherit rich open-world semantics without requiring additional 3D annotations. This distillation process enables the recognition of long-tailed concepts and open-vocabulary semantics that were previously inaccessible to standard 3D encoders.

Beyond simple feature projection, more sophisticated architectures have emerged to handle the granularity mismatch between global image-text pairs and local 3D regions. The framework [160] addresses the alignment granularity gap by explicitly learning pixel-level alignment alongside object- and region-level alignment. By constructing pseudo multi-granular semantic correspondences, this approach facilitates fine-grained cross-modal contrastive learning, significantly improving zero-shot segmentation results without dense annotations. Similarly, in the context of 3D point clouds, [161] proposes a unified network that aligns point clouds with images, language, and depth. By utilizing hierarchical 3D caption pairs and geometric constraints, it achieves state-of-the-art performance in open-vocabulary semantic and instance segmentation, demonstrating the importance of fine-grained feature representation in grounding tasks.

The emergence of Neural Semantic Fields has further revolutionized semantic grounding by offering a continuous, differentiable medium for mapping language to geometry, effectively extending the neural implicit representations discussed in Section 3.2 into the semantic domain. Unlike discrete voxel grids or meshes, neural fields allow for the segmentation and learning of concepts directly on the underlying 3D representation. The paper [162] pioneers this direction by representing each 3D coordinate as a high-dimensional descriptor, enabling concept grounding through similarity computation between 3D descriptors and language concept embeddings. This allows both semantic and instance segmentations to emerge directly from question-answering supervision, showcasing superior generalization to unseen shape categories. Expanding on this, [163] introduces a generalizable implicit representation that aggregates geometry-aware features via a cost volume and multi-view joint fusion. This method eliminates the need for ground truth semantic labels or depth priors, achieving state-of-the-art performance in both 2D and 3D open-vocabulary tasks while generalizing across diverse scenes.

For applications requiring real-time interaction, such as robotic navigation and manipulation, efficiency is paramount. [164] presents a groundbreaking approach that combines pre-trained vision-language foundation models with Truncated Signed Distance Functions (TSDF) for real-time mapping. By extracting region-based embeddings and integrating them with 3D knowledge via an enhanced Hungarian-based feature-matching mechanism, Open-Fusion delivers annotation-free 3D segmentation capable of handling open-world semantics in dynamic environments. Furthermore, weakly supervised approaches have shown promise in reducing reliance on labeled data. The method described in [165] exploits the reasoning capabilities of CLIP and DINO, distilling their knowledge into a Neural Radiance Field (NeRF) using only open-vocabulary text descriptions. This approach effectively lifts 2D features into view-consistent 3D segmentation, outperforming fully supervised models in certain scenarios and highlighting the potential of foundation models to drive data-efficient learning.

Grounding also extends to complex relational reasoning, where agents must understand not just objects but their spatial and functional relationships—a prerequisite for the articulated object modeling discussed in Section 3.5. [166] addresses this by translating language into executable programs with hierarchical structures, leveraging large language-to-code models. This neuro-symbolic framework introduces functional modules capable of reasoning about high-arity relations among multiple objects, which is crucial for disambiguating targets in cluttered 3D scenes. Additionally, [167] strengthens relation-oriented mapping by replacing language-independent encoding with cross-modal encoding guided by global semantics. This graph-based approach significantly improves

localization performance in multi-relation challenges, ensuring that agents can correctly identify objects based on complex referential descriptions.

Finally, the transfer of knowledge from 2D to 3D domains remains a critical strategy for overcoming data scarcity. [168] proposes a baseline that aligns 2D and 3D encoders at both feature and output levels, utilizing multi-granularity cross-modal feature alignment to transfer visual-linguistic knowledge. Similarly, [169] explores unsupervised methodologies by transferring knowledge from 2D image datasets to 3D point clouds, achieving state-of-the-art performance without any 3D labels. These cross-modal transfer techniques, combined with the advent of unified models like [110] which operates directly on point clouds to support diverse vision-language tasks, are paving the way for embodied agents that can seamlessly ground language in 3D space. By establishing this semantic bridge, agents are equipped to not only perceive static geometry but also to engage with the dynamic, articulated objects that define human-centric environments, setting the stage for the interactive perception capabilities explored in the following section.

3.5 Interactive Perception and Articulated Object Modeling

3.5 Interactive Perception and Articulated Object Modeling

Building upon the foundational capabilities of semantic grounding in static 3D spaces discussed previously, the next critical frontier for embodied agents is the transition to dynamic, interactive perception. While Section 3.4 established how agents map language to stationary geometric structures, real-world manipulation demands an understanding of objects that change state through interaction. Specifically, embodied agents operating in human-centric environments must comprehend and manipulate articulated objects—entities composed of multiple rigid parts connected by movable joints, such as doors, drawers, laptops, and tools. Constructing accurate digital twins of these objects requires more than visual reconstruction; it demands the inference of underlying kinematic structures and the ability to predict how these objects deform under interaction. This subsection investigates approaches that leverage interaction data to build comprehensive models of articulated objects, emphasizing the integration of explicit kinematic constraints with implicit neural representations to facilitate robust manipulation and collision avoidance.

A primary challenge in articulated object modeling is the unsupervised discovery of joint structures and part segmentation from visual observations. Early methods often relied on predefined templates or extensive manual annotations, limiting their applicability to unseen categories. Recent advances have shifted toward learning both appearance and structure simultaneously by observing objects in motion. For instance, the work [Watch It Move Unsupervised Discovery of 3D Joints for Re-Posing of Articulated Objects] proposes a novel approach to learn the structure of previously unseen articulated objects by observing them move from multiple views without joint annotation supervision. By modeling object parts as 3D ellipsoids and identifying joints where adjacent parts move relative to one another, this method effectively combines explicit geometric representations with implicit compensations to handle diverse structures ranging from quadrupeds to robotic arms. Similarly, [170] introduces a framework that reconstructs part-level geometry and estimates articulation models through interactive perception. By utilizing implicit neural representations for joint geometry and articulation modeling, Ditto can recreate interactive digital twins from pairs of visual observations before and after interaction, enabling category-agnostic reconstruction that can be directly deployed in physical simulations.

The fidelity of these digital twins is further enhanced by integrating kinematic and dynamic properties derived from video data. Accurate simulation requires not only the correct kinematic tree but

also precise dynamic parameters such as mass, friction, and joint limits. The pipeline introduced in [171] combines inverse rendering with differentiable simulation to automatically discover joint types and tune dynamic properties from depth or RGB videos. This approach ensures that control policies optimized in the derived simulation can successfully transfer back to the original physical system, bridging the gap between perception and control. Furthermore, self-supervised methods like [SM³ Self-Supervised Multi-task Modeling with Multi-view 2D Images for Articulated Objects] leverage multi-view images captured before and after interaction to model articulated objects and infer joint parameters without annotations. By constructing 3D geometries and textures from 2D inputs, SM³ achieves integrated optimization of movable parts and joint parameters, demonstrating superior adaptability in real-world scenarios compared to supervised baselines.

Beyond kinematic structure, understanding the physical contact and interaction priors between agents and articulated objects is crucial for dexterous manipulation. The complexity of hand-object interactions, particularly with articulated mechanisms, necessitates models that can predict contact regions and valid pose configurations. [ContactArt Learning 3D Interaction Priors for Category-level Articulated Object and Hand Poses Estimation] addresses this by learning 3D interaction priors, including a discriminator capturing part arrangement distributions and a diffusion model generating contact regions. These priors, learned from a dataset collected via visual teleoperation, significantly improve joint hand and articulated object pose estimation by transferring structural knowledge to real-world data with minimal domain gap. Complementing this, [V-MAO Generative Modeling for Multi-Arm Manipulation of Articulated Objects] presents a variational generative model that learns contact point distributions over object rigid parts for multi-arm manipulation. By formulating an object-centric control strategy, V-MAO enables collaborative robot arms to complete complex manipulation tasks on articulated objects with high success rates in simulation.

The integration of these perceptual models into closed-loop control systems allows for real-time collision avoidance and adaptive planning. High-performance whole-body collision avoidance systems, such as the one described in [Towards Connecting Control to Perception High-Performance Whole-Body Collision Avoidance Using Control-Compatible Obstacles], utilize digital twin systems to provide millisecond-real-time control stages with accurate world descriptions. By compressing ultra-realistic virtual replicas into geometric primitives known as “primitive skeletons,” this method closes the loop between high-resolution environmental perception and low-latency robot control, ensuring safety in collaborative scenarios. Additionally, for deformable objects which pose infinite shape variations, [ACID Action-Conditional Implicit Visual Dynamics for Deformable Object Manipulation] introduces action-conditional visual dynamics models based on structured implicit neural representations. ACID learns implicit representations of occupancy and flow-based forward dynamics from partial RGB-D observations, enabling goal-conditioned manipulation of volumetric deformable objects with significant improvements in task success rates.

Finally, the synthesis of realistic human-object interactions serves as both a validation of these models and a vital source of training data for embodied agents. Frameworks like [172] and [173] utilize generative models to synthesize natural interactions controlled by high-level semantic specifications. By explicitly modeling contact between human body surfaces and object geometry, these methods generate physically plausible interaction sequences that respect the kinematic constraints of articulated objects. Such synthetic data is invaluable for training agents to understand the affordances of complex objects, as demonstrated by [174], which leverages large language models to infer diverse 3D interaction models from single images. Together, these approaches illustrate a converging trend where interactive perception, implicit representation, and kinematic reasoning combine to create robust, generalizable digital twins capable of supporting sophisticated embodied

reasoning and manipulation in dynamic, unstructured environments.

4 Reasoning Mechanisms, Planning, and Cognitive Control

4.1 Chain-of-Thought Reasoning and Prompting Strategies

4.1 Chain-of-Thought Reasoning and Prompting Strategies

The integration of Chain-of-Thought (CoT) reasoning into embodied artificial intelligence marks a fundamental paradigm shift from reactive perception-action loops to deliberative cognitive control. For embodied agents navigating complex physical environments and executing multi-step manipulation tasks, the ability to decompose high-level instructions into interpretable intermediate steps is paramount. While traditional CoT prompting has demonstrated remarkable efficacy in guiding Large Language Models (LLMs) to generate step-by-step rationales for static natural language processing tasks, its direct application to embodied settings faces unique hurdles. These include the necessity for precise multimodal alignment, robustness against environmental dynamism, and the critical mitigation of error accumulation over long horizons. Recent research has therefore focused on adapting CoT techniques through specialized prompting strategies, duty-distinct architectures, and logical self-correction frameworks to bridge the gap between abstract reasoning and physical execution.

A primary limitation of standard linear CoT in embodied scenarios is its susceptibility to early-stage errors, which can propagate catastrophically through subsequent action steps, leading to task failure. To mitigate this, researchers have developed frameworks that incorporate self-verification and logical self-correction mechanisms directly into the reasoning loop. For instance, the concept of self-verification allows models to treat their generated conclusions as conditions for re-evaluating the original problem, effectively performing a backward check to validate the reasoning chain [175]. This capability is particularly critical for embodied agents, where a single incorrect grasp or navigation decision can render a task impossible. Furthermore, the reliability of reasoning chains can be enhanced by assessing the entailment relationship between the query and the candidate reasoning path; if confidence is low, the agent can opt to predict directly or seek clarification rather than proceeding with a flawed plan [176]. Such selective filtering mechanisms ensure that agents do not blindly follow hallucinated or logically unsound intermediate steps, thereby improving safety and success rates in physical interactions.

Beyond simple linear decomposition, advanced CoT strategies for embodied reasoning often require a strict division of labor between different cognitive capabilities to handle the complexity of multimodal inputs. The “Duty-Distinct Chain-of-Thought” (DDCoT) approach exemplifies this by explicitly separating the responsibilities of recognition and reasoning [DDCoT Duty-Distinct Chain-of-Thought Prompting for Multimodal Reasoning in Language Models]. In multimodal embodied tasks, visual perception and logical planning demand distinct processing strengths. DDCoT leverages specialized visual models for accurate scene understanding while employing LLMs for high-level strategic reasoning, integrating these outputs through a joint process that maintains a critical attitude via negative-space prompting. This architectural separation prevents the conflation of perceptual noise with logical errors, a common failure mode in unified models attempting to process raw sensory data and abstract goals simultaneously. Similarly, the Cantor framework adopts a perception-decision architecture where the model first acts as a decision generator to analyze visual context before leveraging cognitive functions to derive higher-level information, ensuring that the generated CoT is grounded in actual environmental states rather than hallucinated premises [177].

The structure of the reasoning process itself has also evolved from simple sequential chains to more complex graph-based and recursive structures, better mimicking the non-linear nature of human expert thinking in dynamic environments. While standard CoT follows a rigid path, human reasoning often involves backtracking and the evaluation of multiple hypotheses. Graph-of-Thought (GoT) and Hypergraph-of-Thought (HoT) paradigms model thought units as nodes in a graph, allowing for the exploration of multiple reasoning paths and the aggregation of diverse information sources [Beyond Chain-of-Thought, Effective Graph-of-Thought Reasoning in Language Models] [Thinking Like an Expert Multimodal Hypergraph-of-Thought (HoT) Reasoning to boost Foundation Models]. These approaches enable embodied agents to perform high-order multi-hop reasoning and comparative judgments, which are essential for tasks requiring the manipulation of articulated objects or navigation in cluttered spaces. Additionally, recursive strategies like Socratic Questioning break down complex problems into approachable sub-questions, navigating the thinking space more robustly than single-pass generation [The Art of SOCRATIC QUESTIONING Recursive Thinking with Large Language Models]. This divide-and-conquer approach is particularly beneficial for long-horizon embodied tasks where the context window might otherwise be exceeded or where the dependency between distant steps is critical.

Furthermore, ensuring the faithfulness and interpretability of CoT in embodied agents remains an active area of investigation, serving as a prerequisite for the rigorous grounding discussed in subsequent planning frameworks. Studies have shown that CoT explanations can sometimes be unfaithful, rationalizing incorrect actions or failing to reflect the true decision-making process [Language Models Don’t Always Say What They Think Unfaithful Explanations in Chain-of-Thought Prompting]. To address this, frameworks like Verify-and-Edit and Knowledge-Driven CoT integrate external knowledge sources to verify and modify reasoning traces, ensuring that the agent’s internal monologue aligns with factual reality and physical constraints [178] [Knowledge-Driven CoT Exploring Faithful Reasoning in LLMs for Knowledge-intensive Question Answering]. The length and structure of the reasoning steps also play a crucial role; research indicates that lengthening reasoning steps can significantly enhance performance on complex tasks, even if the additional tokens do not introduce new information, suggesting that the act of elaborating the thought process itself aids in focusing the model’s attention [179].

In summary, the adaptation of CoT for embodied reasoning extends far beyond simple prompt engineering. It involves sophisticated architectural designs that separate perception from planning, robust self-correction loops to handle physical uncertainty, and non-linear reasoning structures capable of managing complex state spaces. By integrating duty-distinct processing, logical verification, and graph-based exploration, modern embodied agents are becoming increasingly capable of executing interpretable, reliable, and multi-step strategies. These advancements in generating coherent and validated reasoning traces lay the essential groundwork for the next generation of generalist agents, providing the semantic flexibility required for the neuro-symbolic integration and grounded planning strategies explored in the following section.

4.2 Neuro-Symbolic Integration for Grounded Planning

4.2 Neuro-Symbolic Integration for Grounded Planning

While the advanced prompting strategies and self-correction mechanisms discussed in Section 4.1 significantly enhance the coherence of reasoning traces, they often remain insufficient for guaranteeing physical feasibility in complex environments. Purely neural approaches, even with sophisticated Chain-of-Thought structures, frequently suffer from hallucinations and a lack of explicit understand-

ing regarding geometric constraints, leading to action sequences that are semantically plausible but physically infeasible. To bridge this gap between abstract reasoning and executable reality, recent research has increasingly turned to neuro-symbolic architectures. This paradigm represents a pivotal shift aimed at reconciling the semantic flexibility and commonsense knowledge of Large Language Models (LLMs) with the rigorous logical guarantees and state-tracking capabilities of classical planning systems. By hybridizing the generative power of LLMs with structured symbolic planners, scene graphs, and inverse planning engines, these systems ensure that generated plans are not only aligned with human intent but are also strictly grounded in the geometric and topological realities of the environment.

A fundamental challenge in this domain is grounding high-level linguistic commands into the specific topology and object states of a robot’s environment. Purely neural approaches often struggle to maintain consistency over long horizons or in large-scale environments due to the combinatorial explosion of the action space. To mitigate this, researchers have adopted structured environmental representations, such as scene graphs, to serve as an intermediate bridge between perception and planning. For instance, the [180] framework introduces a scalable approach where LLMs perform semantic searches over hierarchical 3D scene graphs. By collapsing the full graph into manageable subgraphs, the system allows the LLM to generate high-level plans that are subsequently validated and refined through an iterative replanning pipeline involving a scene graph simulator. This method effectively reduces the planning horizon for the LLM while ensuring that every proposed action corresponds to a valid state transition within the explicit environmental model. Similarly, [181] leverages scene graphs to translate environmental topology into actionable knowledge, enabling the fast generation of precise planning problem descriptions. By decomposing long-term goals into autoregressive sub-goals guided by the scene graph, DELTA achieves higher planning success rates and significantly shorter computation times compared to state-of-the-art methods that rely solely on LLM inference.

Beyond static scene representations, neuro-symbolic integration frequently involves coupling LLMs with classical planners that operate on formal logic specifications, such as Linear Temporal Logic (LTL) or the Planning Domain Definition Language (PDDL). This combination allows systems to benefit from the optimality guarantees of symbolic solvers while utilizing LLMs as heuristic guides or translators. The work presented in [182] exemplifies this by converting natural language tasks into LTL automata using an LLM, which then guides a hierarchical planner over a scene graph. Crucially, this approach employs a provably consistent LTL heuristic to supplement the potentially inadmissible guidance provided by the LLM, ensuring that the final plan is optimal even if the language model’s initial suggestions are imperfect. In a similar vein, [183] proposes a framework where LLMs act as translators, converting natural language into an intermediate task representation consumable by a Task and Motion Planning (TAMP) algorithm. The system employs autoregressive re-prompting to detect and correct syntactic and semantic errors in the translation process, effectively acting as a self-correcting mechanism that bridges the gap between vague human instructions and rigorous motion planning constraints.

The necessity of such hybridization is further underscored by the limitations of end-to-end neural planners in handling complex physical constraints and multi-agent coordination. While LLMs can approximate human intuition for goal decomposition—a capability enhanced by the duty-distinct and graph-based reasoning discussed previously—they lack the inherent ability to verify temporal interference or geometric feasibility without external validation. The [184] study addresses this by combining LLM-based goal decomposition with classical PDDL planners. Here, the LLM approximates human intuition to break down complex multi-agent goals into independent subgoals, which

are then solved efficiently by single-agent classical planners. This division of labor preserves the execution success guarantees of symbolic planning while leveraging the LLM’s ability to handle ambiguity and decompose tasks dynamically. Furthermore, [185] demonstrates a modular architecture where a vision-and-language network predicts a symbolic goal representation, which is then passed to a symbolic planner to generate the action sequence. This separation allows the neural component to focus on perceptual grounding and intent inference, while the symbolic component ensures the logical consistency of the resulting plan.

Addressing the issue of dynamic and partially observable environments, advanced frameworks incorporate feedback loops where the symbolic planner informs the neural model about feasibility, creating a closed-loop reasoning process. [186] highlights the importance of environment feedback, proposing that LLMs can form an “inner monologue” by processing success detection and scene descriptions to refine their plans. Although primarily neural, this concept aligns with neuro-symbolic principles by emphasizing the need for continuous grounding in state changes, echoing the self-verification themes of Section 4.1. More explicitly, [187] introduces a replanning strategy that orchestrates interplay between cognitive levels to handle physically grounded, logical, and semantic errors. By detecting failures in executability, the system triggers a corrective planning phase that revises the action sequence, ensuring robustness in open-world scenarios. Additionally, [188] illustrates how smaller language models can be fine-tuned to reason directly over scene graph structures, effectively learning to decompose tasks into subgoal specifications that a downstream planner can execute. This approach demonstrates that even without massive scale, grounding the input of language models in structured domain representations enables effective long-horizon task planning.

Finally, the integration of affordance reasoning into symbolic frameworks further enhances the physical groundedness of plans. [189] combines an affordance-based scene representation with a planning system, allowing symbolic planning with arbitrary objects by deriving the planning domain from action possibilities rather than predefined object classes. This approach enables the system to handle incomplete information and missing objects by exploring the scene and suggesting alternatives, a capability often lacking in rigid symbolic systems. Similarly, [190] utilizes a two-level scene graph representation—geometric and symbolic—to jointly predict high-level task plans and low-level motions. This hierarchical abstraction allows the system to scale to long-horizon tasks while maintaining computational efficiency, achieving high success rates in real-world manipulation scenarios. Collectively, these works illustrate that neuro-symbolic integration is not merely a combination of tools but a necessary architectural paradigm for achieving robust, explainable, and physically feasible embodied reasoning. By anchoring the semantic generalization of LLMs in the rigorous logic of symbolic planners and the structural fidelity of scene graphs, these systems overcome the hallucinations and physical inconsistencies that plague purely neural approaches. This establishment of grounded, feasible plans provides the essential foundation for the subsequent paradigm discussed in Section 4.3, where LLMs act as central controllers to manage the temporal decomposition of these validated actions over extended horizons.

4.3 LLMs as Central Controllers for Long-Horizon Decomposition

4.3 LLMs as Central Controllers for Long-Horizon Decomposition

While neuro-symbolic integration (Section 4.2) addresses the grounding of plans in physical and logical constraints, a parallel and complementary paradigm focuses on leveraging Large Language Models (LLMs) as central cognitive controllers to manage the temporal complexity of long-horizon tasks. This approach represents a shift from monolithic policy learning toward hierarchical archi-

tectures where the LLM acts as a high-level “brain.” Its primary function is to decompose complex, open-ended natural language instructions into manageable, executable sub-goals. This decomposition is critical because embodied tasks often involve intricate temporal dependencies and spatial constraints that exceed the immediate reasoning capacity of low-level controllers or the context limits of standard prompting strategies. By utilizing hierarchical structures and few-shot learning, these systems ensure that agents maintain coherence over extended interaction sequences, serving as a foundational layer upon which adaptive replanning mechanisms (discussed in Section 4.4) can operate.

A primary mechanism employed by LLM-based controllers is the recursive decomposition of tasks. Traditional approaches often struggle when a single instruction requires a sequence of distinct actions, where the failure of one sub-task leads to total task collapse. To address this, frameworks like ADaPT (As-Needed Decomposition and Planning) introduce a dynamic planning strategy where the LLM explicitly decomposes complex sub-tasks only when necessary, such as when the model detects an inability to execute a current step directly [191]. This approach allows the system to adaptively adjust to both the complexity of the task and the specific capabilities of the executor LLM, recursively breaking down problems until they reach a level of solvability. Empirical evaluations have demonstrated that such as-needed decomposition significantly outperforms static planning baselines in environments requiring multi-step reasoning, achieving substantial improvements in success rates across diverse benchmarks. Similarly, the concept of Decomposed Prompting advocates for a modular approach where complex tasks are split into simpler sub-tasks delegated to a library of specialized prompts or models [192]. This modularity not only enhances performance on symbolic reasoning and long-context multi-hop questions but also facilitates the replacement of specific reasoning modules with more effective solvers, thereby increasing the robustness of the overall system.

Beyond simple linear decomposition, advanced architectures employ hierarchical policies to unleash the creative potential of LLMs in exploring diverse solution spaces. The “LLM as Hierarchical Policy” framework posits a leader-follower structure, where a visionary leader LLM proposes multiple high-level problem-solving tactics or hints, and a follower LLM executes detailed reasoning chains based on these directives [193]. This separation of concerns allows the system to generate a group of solutions for each high-level strategy, subsequently selecting the optimal path through a tournament-based evaluation. This method effectively broadens the exploration capability of the agent, preventing it from getting trapped in local optima—a common failure mode in challenging problem-solving scenarios. Furthermore, the integration of tree-based hard attention mechanisms, as seen in frameworks like TEAROOM, enables LLMs to process hierarchically structured text inputs more effectively [194]. By selectively focusing on relevant leaves in relation to a root node, these models can generate tailored symbolic representations that align more closely with the underlying logic of complex tasks, iteratively refining their outputs to approximate the ground truth.

The efficacy of LLMs as central controllers is further amplified by their ability to learn and reuse skills in a lifelong manner, transforming static decomposition into a cumulative learning process. The “Self-driven Grounding” (SDG) framework exemplifies this by enabling LLMs to autonomously propose hypotheses for sub-goals and verify their feasibility through environment interaction [195]. Once a sub-goal is verified, the system learns generalized skills that can be reused for future, more complex tasks, effectively building a library of executable capabilities without extensive human demonstration. This aligns with the vision of lifelong robot learning, where LLM-based planners query new skills and teach robots in a data-efficient manner, allowing for the accumulation of knowledge over time [196]. Additionally, frameworks like O3D (Offline Data-driven Discovery and

Distillation) leverage offline interaction data to automatically discover reusable skills and distill generalizable knowledge, enhancing the decision-making capabilities of LLMs without the need for fine-tuning [197]. These approaches underscore the transition from static, one-off planning to dynamic, skill-acquiring systems capable of handling open-world scenarios.

However, effective long-horizon decomposition relies heavily on the agent’s ability to retain context across extended timelines. Memory mechanisms play an indispensable role in sustaining the coherence of these decomposed plans; without persistent memory, an LLM controller may lose track of previously achieved sub-goals or fail to recognize the context required for subsequent steps. The Large Decision Model with Memory (LDM²) addresses this by constructing dynamic prompts based on a memory of state-action tuples [198]. Through a process of memory formation and refinement, often aided by tree exploration, the model enriches its understanding of the global environment, allowing it to make more informed decisions in complex interactive settings. Similarly, the “Empowering Working Memory” concept draws from cognitive psychology to propose architectures with centralized working memory hubs and episodic buffers, enabling agents to retain memories across distinct dialog episodes and maintain continuity during intricate tasks [199]. The integration of symbolic memory, such as SQL databases in ChatDB, further augments this capability by providing a precise, error-resistant storage mechanism for complex multi-hop reasoning, overcoming the approximate nature of neural memory [200].

Finally, the interaction between the central controller and the execution layer is increasingly being optimized through reinforcement learning and active inquiry to balance computational efficiency with reasoning depth. The “When2Ask” framework formulates the interaction between an agent and an LLM as a Markov decision process, learning precisely when it is necessary to query the LLM for high-level instructions [201]. This reduces computational costs and latency while maintaining robustness against partial observability. Moreover, frameworks that incorporate active inquiry, such as those learning to ask relevant prompt questions via reinforcement learning, enable the controller to resolve ambiguities before committing to a plan [202]. By employing a bilevel leader-follower structure, these systems can introspectively revise prompts based on historical findings, guiding the chain-of-thought process toward decisive, high-performing actions. Collectively, these advancements illustrate that LLMs, when equipped with hierarchical decomposition, dynamic memory, and adaptive interaction strategies, serve as powerful central controllers. While this section focuses on the architectural design of the controller itself, the following section (4.4) will explore how these controllers engage in closed-loop planning and adaptive replanning to handle the inevitable discrepancies between generated plans and real-world execution.

4.4 Closed-Loop Planning and Adaptive Replanning

4.4 Closed-Loop Planning and Adaptive Replanning

While Section 4.3 established LLMs as central controllers capable of hierarchically decomposing long-horizon tasks, such high-level planning inherently assumes a degree of environmental stability that rarely exists in real-world embodied scenarios. In practice, agents operate within stochastic environments where actions may fail, objects shift unexpectedly, and sensory inputs are often noisy or occluded. Consequently, the transition from the static, open-loop decomposition strategies discussed previously to dynamic, closed-loop execution represents a pivotal advancement. This paradigm shift moves beyond the generation of a single, rigid sequence of sub-goals toward a reactive and iterative process. By integrating continuous feedback loops, agents can detect discrepancies between their internal plans and physical reality, engaging in an “inner monologue” to self-correct

and adapt without external intervention. This section explores how these closed-loop mechanisms bridge the gap between the abstract cognitive planning of Section 4.3 and the physically grounded trajectory generation detailed in Section 4.5.

A foundational mechanism in this domain is the integration of language models as high-level controllers that utilize environmental feedback to sustain an internal dialogue. The concept of an “inner monologue” has been effectively demonstrated in frameworks where Large Language Models (LLMs) are not merely prompted once but are engaged in a continuous loop of perception, planning, and reflection. By leveraging environment feedback sources such as success detection, scene descriptions, and human interaction, LLMs can richly process robotic control scenarios [186]. This approach allows the agent to verbalize its reasoning steps, effectively turning the planning process into a transparent, interpretable stream of consciousness that adapts as the physical world changes. For instance, if a manipulation attempt fails, the agent can describe the failure in natural language, re-evaluate its preconditions, and generate a new plan, significantly improving high-level instruction completion in long-horizon tasks.

Building on this feedback-driven dialogue, recent advancements have introduced specific architectural strategies to mitigate the hallucinations and rigidities often associated with static LLM planning. The AdaPlanner framework exemplifies this by proposing a closed-loop approach where the agent adaptively refines its self-generated plan in response to environmental feedback [203]. Unlike traditional methods that execute greedy actions or stick to a pre-computed trajectory regardless of outcome, AdaPlanner employs both in-plan and out-of-plan refinement strategies. This dual capability enables the agent to adjust parameters within an existing plan or completely restructure its approach when faced with unexpected obstacles. Furthermore, by utilizing a code-style prompt structure and a skill discovery mechanism that treats successful plans as few-shot exemplars, such systems achieve superior sample efficiency and performance in complex sequential decision-making tasks [203].

The necessity of closed-loop mechanisms is further underscored by the limitations of open-loop perception-decision systems, which often struggle with generalization in dynamic settings. Research into human-like navigation has highlighted that converting actions back into internal latent states to form a two-stage closed loop of perception, decision-making, and reasoning substantially promotes robustness [204]. In these systems, the reasoning model acts as a corrective filter, allowing the agent to recover from deviations and adapt to novel scenarios that were not present in the training distribution. This is particularly critical in mobile manipulation, where the COME-robot framework demonstrates that utilizing vision-language foundation models for open-ended reasoning and adaptive planning enables agents to actively perceive environments, perform situated reasoning, and recover from failures in real-world settings [205]. By treating action primitives as callable modules verified by a multimodal “brain,” these agents can detect task progress and trigger replanning only when necessary, optimizing computational resources while maintaining high success rates.

Beyond language-centric approaches, model-based reinforcement learning offers complementary techniques for adaptive control through the lens of internal world models. Agents equipped with the ability to simulate future trajectories can engage in “visual imagination” to plan novel actions before execution [206]. When the predicted outcomes diverge from actual observations, the agent can iteratively refine its inferred states. A training-free approach known as iterative inference at decision-time allows agents to fine-tune their state estimates based on the coherence of future representations, proving especially beneficial in partially observable environments [207]. Similarly, the Deep Planning Network (PlaNet) illustrates how learning latent dynamics from pixels enables fast online planning, where the agent constantly updates its belief state to handle contact dynamics

and sparse rewards [144]. These latent imagination strategies provide a crucial middle ground, offering the predictive foresight required for robust replanning before handing off precise trajectory execution to the diffusion-based methods discussed next.

Furthermore, the integration of introspection and anomaly detection serves as a critical layer for long-term autonomy. Systems that incorporate a “recovery critic” atop an introspection module can classify anomalies and execute specific recovery policies, such as re-enactment or adaptation, to resolve accidental disturbances [208]. This level of self-monitoring ensures that agents do not merely fail silently but actively diagnose the root cause of execution errors. The interplay between deliberate planning and habitual control also plays a role; an arbitrator can switch between computationally expensive planning and fast, habitual responses based on environmental uncertainty, ensuring adaptability when visual occlusions or dynamic changes occur [209].

In summary, closed-loop planning and adaptive replanning constitute the operational backbone that breathes life into the hierarchical decompositions proposed in Section 4.3. By synthesizing inner monologue capabilities, iterative state refinement, and explicit error recovery mechanisms, modern agents can transcend the limitations of static policies. Whether through the linguistic flexibility of LLMs refining plans via feedback [186] or the predictive power of latent world models correcting trajectory deviations [207], the overarching goal remains the same: to create agents that are not only capable of acting but also of thinking about their actions in real-time. However, while these methods excel at high-level strategy adjustment and state estimation, they often rely on simpler motion primitives for execution. The following section (4.5) addresses this limitation by exploring how diffusion-based models provide the necessary granular, physically feasible trajectory generation to fully realize these adaptive plans in continuous, high-dimensional spaces.

4.5 Diffusion-Based Planning and Generative Simulators

4.5 Diffusion-Based Planning and Generative Simulators

While the closed-loop mechanisms and inner monologue strategies discussed in the previous section empower agents to reactively correct errors and refine plans through language, a fundamental challenge remains: generating physically feasible, multimodal trajectories in high-dimensional continuous spaces. Traditional autoregressive methods often struggle with the compounding errors inherent in sequential action generation, particularly when navigating complex physical constraints or long-horizon tasks. To address these limitations, the integration of diffusion models into embodied reasoning represents a paradigm shift, offering a robust alternative for Task and Motion Planning (TAMP) and world modeling. Unlike sequential approaches, diffusion models excel at modeling complex, multimodal trajectory distributions in a single pass, enabling more coherent planning and a deeper adherence to physical laws. This subsection explores the dual role of diffusion models as hierarchical planners within TAMP solvers and as generative simulators that endow agents with intuitive physics, serving as a critical bridge between the high-level cognitive reasoning of the prior section and the memory-intensive cognitive architectures of the next.

In the realm of TAMP, diffusion models have been increasingly adopted to overcome the computational bottlenecks associated with searching high-dimensional continuous spaces, where the coupling between discrete task sequencing and continuous motion feasibility often stymies traditional approaches. By serving as powerful samplers, diffusion models can propose feasible action parameters or entire trajectories conditioned on symbolic constraints, effectively decoupling the search problem. For instance, [210] proposes leveraging diffusion models to learn constraints and samplers that capture difficult-to-engineer aspects of the planning domain. By defining these sam-

plers on low-dimensional latent embeddings of object states, the approach enables long-horizon constraint-based reasoning even in partially observable environments with unseen objects. Similarly, [211] introduces a hierarchical framework where discrete skill representations learned from visual and linguistic inputs condition a diffusion model. This allows for the generation of diverse state trajectories that adhere to abstract skills, effectively bridging the gap between high-level instructions and low-level execution.

Building on these foundational planning capabilities, recent advancements have integrated diffusion models into broader hierarchical architectures to enhance both efficiency and generalization. [212] demonstrates that a “jumpy” planning strategy at a higher level, combined with a low-level diffusion planner, can significantly reduce computational costs while expanding the receptive field for long-horizon tasks. This hierarchical decomposition allows the agent to plan over compressed time steps, guiding the fine-grained diffusion policy to generate kinematically feasible motions. Furthermore, [213] factorizes the manipulation policy into a high-level agent predicting distant next-best poses and a low-level goal-conditioned diffusion policy. This structure specifically addresses kinematic constraints by distilling accurate end-effector pose predictions into joint-position trajectories, ensuring that the generated plans are physically executable.

The utility of diffusion models extends beyond static planning into reactive and adaptive scenarios, complementing the replanning strategies highlighted in Section 4.4 by providing a probabilistic foundation for decision-making under uncertainty. [214] addresses the critical challenge of when and how to replan in stochastic settings. The authors propose a principled approach based on the diffusion model’s estimated likelihood of existing plans to trigger replanning only when necessary, thereby balancing computational cost with execution robustness. In a similar vein, [215] combines an Active Inference planner with a Multi-Modal Model Predictive Path Integral controller that utilizes physics simulators for trajectory rollouts. While not purely diffusion-based in its controller, this work highlights the synergy between generative sampling and reactive control, a theme central to modern diffusion-based TAMP where diverse trajectory samples are weighed to handle dynamic obstacles.

Parallel to their role in planning, diffusion models are revolutionizing the construction of generative simulators, or “world models,” which are essential for training agents to understand physical dynamics without exhaustive real-world interaction. [216] introduces Policy-Guided Trajectory Diffusion (PolyGRAD), a novel approach that generates entire on-policy trajectories in a single pass rather than autoregressively. This non-autoregressive generation prevents the compounding of prediction errors over long horizons, enabling agents to perform reinforcement learning “in imagination” with high fidelity. Additionally, [217] explores a complementary strategy where language models are fine-tuned with embodied experiences acquired from simulators. Although this work focuses on fine-tuning LLMs, it underscores the necessity of high-quality simulated physical interactions—often generated or validated by generative models—to instill intuitive physics and object permanence in cognitive agents.

The synergy between diffusion-based planning and generative simulation is further evident in methods that utilize these models for data synthesis and policy initialization, creating a fertile ground for learning robust policies. [218] argues that learning diffusion models as priors on trajectory distributions can significantly accelerate motion planning optimization. By sampling directly from the posterior trajectory distribution conditioned on task goals, agents can bootstrap complex planning problems with high-quality initializations. Moreover, [219] demonstrates that multi-task diffusion models can generate high-quality synthetic data for unseen tasks, effectively acting as a generative simulator that expands the training distribution for generalist agents.

Finally, the integration of temporal conditions and task awareness into diffusion processes enhances their applicability to procedure planning and instructional understanding, laying the groundwork for the sophisticated cognitive architectures discussed in the following section. [220] employs a masked diffusion model to focus on relevant action types within a large decision space, utilizing joint visual-text embeddings to capture semantic relations between actions and goals. Similarly, [221] proposes a temporally-composable diffuser that extracts and utilizes historical, immediate, and prospective conditions to guide generation. These advancements illustrate how diffusion models are evolving from simple trajectory generators into sophisticated cognitive tools capable of reasoning over time, tasks, and physical constraints. By providing a reliable mechanism for simulating futures and generating feasible actions, diffusion-based approaches form the structural backbone that supports the memory systems and multi-agent coordination explored in Section 4.6.

4.6 Cognitive Architectures, Memory Systems, and Multi-Agent Coordination

4.6 Cognitive Architectures, Memory Systems, and Multi-Agent Coordination

While Section 4.5 established diffusion models as the structural backbone for generating physically feasible trajectories and simulating futures, the realization of truly robust embodied agents requires more than just high-fidelity motion planning. The next frontier involves integrating these generative capabilities into cohesive cognitive architectures that sustain long-horizon reasoning through persistent memory and navigate complex social dynamics via multi-agent coordination. Traditional modular approaches often treat perception, memory, and control as distinct pipelines; however, recent advances suggest that a unified framework is essential for dynamic environments where high-level planning cannot be decoupled from the mechanisms of memory retention and social interaction. A pivotal development in this domain is the integration of Large Language Models (LLMs) as central controllers that unify egocentric memory with real-time control loops. The [222] framework exemplifies this shift by proposing a system where an embodied LLM maintains a continuous, language-based record of past observations and actions. This approach enables closed-loop multi-round dialogues that encompass perception, planning, and control, allowing agents to answer questions based on prior explorations and execute active exploration tasks without relying on separate memory modules. By treating memory as an active, queryable component within the cognitive architecture, such systems effectively bridge the gap between the static knowledge retrieval of traditional systems and the dynamic sensorimotor engagement required for embodied reasoning.

Beyond the unification of memory and control, the structural design of these cognitive architectures is undergoing a renaissance driven by the need for biological plausibility and computational efficiency. Early models often relied on rigid, feedforward processing, but contemporary architectures are embracing nonmodular designs inspired by active inference principles to better handle the uncertainties inherent in physical interaction. As argued in [1], separating motor and perceptual functions can lead to fragility when agents encounter unknown external forces or perturbations. Instead, nonmodular architectures that implement integral control mechanisms demonstrate superior robustness, allowing agents to adapt to environmental interference seamlessly. This perspective is further supported by [223], which advocates for the integration of heterogeneous AI sub-fields—ranging from deep learning to addressable memory systems—into a unified cognitive framework. Such integration is critical for bootstrapping complex cognitive skills through ecologically valid conditions, moving beyond simple benchmark tasks toward genuine adaptive behavior that complements the reactive capabilities of diffusion-based planners.

A crucial component of these advanced architectures, extending beyond individual adaptability, is the capacity for social intelligence, specifically the emergence of Theory of Mind (ToM). For embodied agents to collaborate effectively in human-centric or multi-agent environments, they must infer the beliefs, intentions, and mental states of others, a capability that builds upon the intuitive physics provided by generative simulators. The [224] introduces the ToMnet, a neural network that utilizes meta-learning to build models of other agents solely from behavioral observations, successfully passing classic false-belief tasks like the “Sally-Anne” test. This capability is foundational for multi-agent coordination, where understanding a partner’s hidden goals is as important as executing one’s own actions. Building on this, [225] proposes embodied agent models based on distributed adaptive control theory. These models employ top-down predictions and bottom-up error feedback to simulate the mental states of others, demonstrating that probabilistic learning agents with predictive capabilities outperform pure reinforcement-based strategies in game-theoretic scenarios.

The challenge of scaling ToM to complex, multi-agent settings has led to innovative frameworks that optimize cooperative efficiency by leveraging collective representations. The [226] introduces the Theory of Collective Mind (ToCM), which predicts observations of all agents into a unified plural representation. By simulating multi-agent interactions within an imaginative space, ToCM enables agents to discern how their own actions affect the collective mental state, significantly improving cooperation efficiency in complex decision-making environments compared to standard Multi-Agent Reinforcement Learning baselines. Similarly, [227] presents the HAICA model, where “belief resonance” allows the inferred mental states of one agent to directly influence another’s predictive beliefs. This mechanism facilitates automatic coordination without explicit reasoning, proving particularly beneficial in settings with asymmetric knowledge where communication bandwidth is limited.

In scenarios involving decentralized control and costly communication, LLMs are increasingly being leveraged to foster emergent collaborative behaviors, acting as the semantic glue between individual agents. The [228] presents CoELA, a framework that integrates LLMs with perception and memory modules to enable agents to plan and communicate effectively. Experiments reveal that LLM-driven agents can surpass traditional planning methods and exhibit emergent communication strategies that earn greater trust from human partners. Furthermore, [229] addresses the challenge of partial observability by training generative models to estimate the global world state from egocentric views. By combining these compositional world models with Vision Language Models, agents can perform tree search procedures to infer the actions of others and facilitate online cooperative planning, effectively synthesizing the generative power of world models with the reasoning depth of language models.

The intersection of memory and social cognition also extends to the concept of a shared global workspace, providing a theoretical foundation for how diverse cognitive modules interact. [230] explores how functionally specialized components can share information through a bandwidth-limited channel, encouraging specialization and synchronization among independent modules. This mirrors human cognitive theories where a global workspace integrates disparate sensory and motor processes. Additionally, [231] emphasizes that memory must be an active, introspective component that mediates between semantic and sensorimotor representations, orchestrating data flows to support high-level abilities like prospection and simulation. These mechanisms ensure that the rich trajectory data generated by diffusion models (Section 4.5) can be stored, retrieved, and utilized for future social reasoning.

Ultimately, the future of embodied reasoning lies in cognitive architectures that seamlessly blend egocentric memory, adaptive control, and sophisticated social modeling. Whether through the

self-observation mechanisms proposed in [232] or the Bayesian delegation strategies in [233], the trajectory is clear: robust embodied agents must possess not only a model of the physical world but also a dynamic, reflective model of themselves and their collaborators. These integrated systems represent the culmination of the progression from reactive planning to generative simulation, and finally to socially aware, memory-endowed cognition, paving the way for agents that can navigate the complexities of open-world interactions with a depth of understanding previously reserved for biological intelligence.

5 Embodied Navigation and Goal-Oriented Movement

5.1 Cross-Modal Grounding and Instruction Alignment for Navigation

5.1 Cross-Modal Grounding and Instruction Alignment for Navigation

Before addressing the optimization strategies detailed in the subsequent section on learning paradigms, it is fundamental to establish how embodied agents achieve robust correspondence between the sequential nature of natural language instructions and the continuous, high-dimensional stream of visual observations. The core challenge in Vision-and-Language Navigation (VLN) lies not merely in feature extraction, but in dynamically grounding linguistic entities and spatial relationships within a changing 3D environment while maintaining a coherent trajectory toward a distant goal. Early approaches, often relying on simple concatenation or basic attention mechanisms, struggled to capture the nuanced structural dependencies required for this task. Consequently, research has evolved toward sophisticated architectures—ranging from dual-encoder frameworks to complex graph-based models—that explicitly model both intra-modal and inter-modal relationships to create a solid foundation for policy learning.

A primary strategy for enhancing this cross-modal alignment involves decomposing navigation instructions into finer granularities. Traditional methods that treat instructions as monolithic sequences often obscure the specific visual landmarks associated with individual sub-steps. The work titled [234] highlights the critical importance of granularity, proposing modules that enable agents to attend to single sub-instructions at each time-step. By leveraging fine-grained annotations during training, this approach allows agents to better trace their progress through an instruction, significantly improving the probability of reaching the target. This concept is further extended in [235], which introduces a pre-training paradigm specifically designed to align entity phrases in the text with visual landmarks in the environment. Through objectives such as entity phrase prediction and landmark bounding box prediction, this method bridges the semantic gap at the object level, ensuring the agent understands specific physical referents rather than merely matching global semantics.

Beyond granularity, accurately modeling the structural relationships between entities is essential for interpreting complex spatial commands. The [236] paper argues that relationships among scenes, objects, and directional clues are paramount, proposing a novel graph structure that models both inter-modal relationships between text and vision and intra-modal relationships among visual entities. Via a message-passing algorithm, information propagates between language elements and visual entities, allowing the agent to reason about environmental connectivity more effectively than methods relying solely on feature fusion. Similarly, [237] addresses the vulnerability of standard models to noise by designing a text-conditioned relation network. This architecture dynamically computes parameters via cross-modal attention to capture fine-grained spatial relations, offering

both superior performance and interpretability in grounding spatial references.

The distinction between different types of visual cues—specifically scene-level context versus object-level details—represents another critical dimension of alignment. The [SOAT A Scene- and Object-Aware Transformer for Vision-and-Language Navigation] framework utilizes two distinct visual encoders: a scene classification network and an object detector. This dual-encoder design allows the model to separately process high-level contextual information (e.g., “bedroom”) and specific object references (e.g., “green chairs”), facilitating precise alignment with instructions that rely on both cue types. This separation of concerns is echoed in [238], which processes object descriptions and action specifications separately before fusing them. By distinguishing between “what to look for” and “how to move,” this model avoids the confusion that arises when textual object and action encodings are mixed indiscriminately with visual features.

To further refine the alignment process and prepare the agent for the optimization challenges discussed in Section 5.2, recent works have explored the integration of auxiliary tasks and reinforcement learning signals directly into the grounding mechanism. The [Reinforced Cross-Modal Matching and Self-Supervised Imitation Learning for Vision-Language Navigation] study introduces a reinforced cross-modal matching approach that enforces alignment both locally and globally. By utilizing a matching critic to provide intrinsic rewards for global instruction-trajectory consistency, the agent learns to maintain a coherent understanding of the entire path rather than focusing solely on the immediate next step. Complementing this, [Contrastive Instruction-Trajectory Learning for Vision-Language Navigation] employs contrastive learning objectives to explore invariance across similar samples and variance across different ones. This framework enhances the distinctiveness of vision-language representations by contrasting the semantics of full trajectories against instructions, thereby improving robustness in unseen environments.

Memory and historical context also play a pivotal role in maintaining alignment over long horizons, a necessity for the long-term planning capabilities required in embodied reasoning. The [239] proposes encoding the full episode history of visual observations and actions alongside language inputs. This allows the agent to resolve ambiguities in complex instructions by referencing past interactions, a capability crucial for compositional tasks. Furthermore, [240] addresses the limitations of recurrent states by proposing a compartmentalized memory architecture that captures environment layouts and disentangles visual and geometric cues. This structured representation supports iterative reasoning and global planning, ensuring that the agent’s alignment with the instruction remains consistent even when navigating through previously visited or ambiguous areas.

Finally, the integration of external knowledge and semantic graphs has emerged as a powerful tool for grounding, extending the agent’s understanding beyond immediate pixel data. [241] leverages multi-step Bird’s-Eye-View representations to encode scene layouts and geometric cues, organizing them into a global scene map that informs action prediction. Additionally, [242] retrieves facts from a constructed knowledge base to complement visible content, allowing the agent to infer properties and relationships that are not immediately obvious. Collectively, these advancements illustrate a clear evolution from simple feature matching to sophisticated, structured, and knowledge-enhanced mechanisms. These architectural innovations provide the necessary representational backbone upon which the diverse learning paradigms—imitation, reinforcement, and self-supervision—discussed in the following section can effectively operate to train robust embodied agents.

5.2 Learning Paradigms: Imitation, Reinforcement, and Self-Supervision

5.2 Learning Paradigms: Imitation, Reinforcement, and Self-Supervision

Having established the architectural foundations for cross-modal grounding and instruction alignment in Section 5.1, the focus now shifts to the optimization strategies that enable embodied agents to acquire robust navigation policies. While sophisticated encoders and graph structures provide the necessary representational backbone, the efficacy of these models ultimately depends on the learning paradigms used to train them. In the domain of Vision-and-Language Navigation (VLN), early approaches often treated imitation learning, reinforcement learning, and self-supervised learning as distinct silos. However, contemporary research increasingly advocates for hybrid frameworks that synergize the sample efficiency of imitation, the optimality guarantees of reinforcement learning, and the data scalability of self-supervision to address critical challenges such as distributional shift, sparse rewards, and the scarcity of high-quality human annotations.

Behavior Cloning (BC) remains a foundational strategy due to its simplicity and stability, framing policy learning as a supervised regression or classification problem over expert trajectories. Yet, standard BC suffers significantly from compounding errors, where small deviations from the expert distribution lead to catastrophic failure in novel states—a vulnerability that persists even with the advanced grounding mechanisms discussed previously. To mitigate this, recent advancements have focused on augmenting BC with mechanisms that enhance robustness against visual variations and distributional shifts. For instance, the [243] framework introduces a two-stage self-expert cloning technique that decouples representation learning from policy optimization. By training an expert policy with weak augmentations and subsequently distilling this knowledge into a student network using strong augmentations, the approach achieves superior zero-shot generalization in visual navigation tasks, effectively handling the large distributional shifts common in unseen environments. Furthermore, the quality of the cloning process can be improved by addressing the limitations of sub-optimal demonstration data. The [244] paper proposes a trajectory stitching method that generates new, plausible transitions by connecting disconnected states in offline data, thereby incrementally improving the underlying behavioral policy without altering the core BC algorithm. This is particularly relevant for navigation where perfect expert paths are rare.

Despite these improvements, pure imitation learning lacks the inherent ability to recover from errors or optimize for long-horizon goals that were not explicitly demonstrated. This limitation necessitates the integration of Reinforcement Learning (RL) to refine policies initialized via BC, bridging the gap between static alignment and dynamic environmental interaction. A prominent paradigm involves pre-training with imitation followed by RL fine-tuning, as exemplified by [245]. This two-stage scheme demonstrates that while BC provides a strong prior grounded in human demonstrations, RL fine-tuning is essential for achieving state-of-the-art success rates in Object-Nav tasks, particularly when navigating to objects in new environments. The study further reveals that the benefits of RL fine-tuning are most pronounced when the pre-training dataset is limited, suggesting a complementary relationship where BC reduces the sample complexity of subsequent RL. However, combining these objectives is non-trivial due to conflicting optimization landscapes. The [246] work addresses this by proposing the Cycle-of-Learning (CoL) framework, which utilizes a unified loss function combining BC and Q-learning losses. This approach ensures a stable transition from imitation to reinforcement learning, preventing performance degradation and effectively handling both dense and sparse reward scenarios typical in navigation.

To further enhance sample efficiency and stability within these hybrid frameworks, researchers have explored soft imitation learning and reward relabeling techniques. The [247] paper derives a

method where the behavior-cloned policy serves as both a shaped reward and a critic hypothesis space within an entropy-regularized RL setting. This coherence allows for the seamless refinement of cloned policies using additional environment interactions, bridging the gap between offline demonstration and online adaptation. Similarly, [248] introduces a reward bonus mechanism for demonstrations and successful episodes, encouraging the agent to match demonstrated behaviors even in the absence of dense extrinsic rewards. This is crucial for navigation tasks where defining a dense reward function that captures semantic intent is notoriously difficult. Moreover, the [249] approach simplifies this by assigning a constant reward to demonstrated state-action pairs, effectively regularizing the policy to stay close to the expert distribution while still exploring via RL, thus overcoming the distribution shift inherent in pure BC.

Complementing these supervised and reinforced approaches, Self-Supervised Learning (SSL) has emerged as a powerful auxiliary paradigm to reduce reliance on scarce human-annotated trajectories. By leveraging large amounts of unlabelled interaction data, SSL methods can learn robust representations of environmental dynamics and spatial structures, providing the rich data priors necessary for the memory-intensive tasks discussed in the following section. The [250] paper presents Masked Decision Prediction (MaskDP), a scalable self-supervised pretraining method where state and action tokens in trajectories are randomly masked and reconstructed. This forces the model to infer missing information and learn underlying dynamics, resulting in policies that exhibit strong zero-shot transfer capabilities to new navigation goals and tasks. Additionally, self-supervision can be applied to improve exploration efficiency. The [251] proposes using incomplete offline demonstrations as references to guide exploration, gradually expanding the agent’s scope while avoiding myopic behaviors. This constrained optimization approach significantly enhances exploration in sparse reward settings, a common bottleneck in long-horizon navigation. Furthermore, the concept of learning from “play” or unstructured interaction, as discussed in [252], suggests that autonomously generated data through behavioral cloning of human play can augment limited expert datasets, enabling agents to acquire diverse skills and generalize better to user-specified navigation tasks.

In summary, the most effective learning paradigms for embodied navigation are no longer monolithic but rather integrative, building upon the alignment capabilities detailed in Section 5.1 to prepare agents for the complex spatial reasoning required in Section 5.3. They combine the stability of behavior cloning [253], the optimality of reinforcement learning with soft constraints [254], and the scalability of self-supervised representation learning. By synthesizing these approaches, modern agents can achieve the robustness required to navigate open worlds, handle ambiguous instructions, and recover from execution failures without exhaustive human supervision. These optimized policies, in turn, rely heavily on the structured environmental representations and spatial memory systems explored in the subsequent section to maintain coherence over long horizons.

5.3 Environmental Representation and Spatial Memory for Navigation

5.3 Environmental Representation and Spatial Memory for Navigation

While the learning paradigms discussed in Section 5.2 provide the mechanisms for agents to acquire policies from data, the efficacy of these policies in long-horizon navigation fundamentally depends on how the agent internalizes and structures environmental information. Unlike reactive policies that map observations directly to actions, agents equipped with robust spatial memory can maintain a coherent representation of the environment over time, facilitating loop closure, global planning, and reasoning about unobserved regions. Consequently, the landscape of environmental representation has evolved from simple recurrent states to sophisticated hybrid architectures that

integrate topological graphs, grid-based metric maps, and neural implicit fields. This structural foundation is critical for bridging the gap between the sample-efficient learning strategies of the previous section and the hierarchical planning capabilities detailed in the subsequent section.

A dominant paradigm in spatial memory construction involves the use of topological graphs, which abstract the environment into a set of nodes representing key locations and edges denoting traversable paths. This approach mirrors cognitive maps in biological systems, where navigation relies on identifiable landmarks and coarse-grained path integration rather than precise metric coordinates. For instance, methods like Neural Topological SLAM construct representations where nodes are enriched with semantic features and interconnected via coarse geometric information, allowing agents to solve long-horizon navigation problems with significant improvements over purely reactive baselines [255]. Similarly, the Graph Attention Memory (GAM) architecture explicitly builds a topological graph based on exploration priors, utilizing graph attention mechanisms to extract guided features that inform decision-making in complex 3D environments [256]. To address the scalability issues inherent in large graphs, recent works have introduced sparse graphical memory structures that aggregate states based on consistency objectives, proving that merging redundant nodes can significantly enhance performance on sparse-reward visual navigation tasks without losing optimality [257]. Furthermore, the integration of semantic objects with visual features into a unified graph structure, as seen in SymbioLCD2, has been shown to improve loop closure detection accuracy by leveraging the symbiosis between semantic and spatial awareness [258].

Parallel to topological approaches, grid-based memory maps offer a metric perspective that preserves fine-grained spatial relationships, often projecting historical egocentric observations into a unified global or local coordinate system. The Grid Memory Map (GridMM) exemplifies this by constructing a dynamically growing top-down grid that captures spatial relations from a global perspective while aggregating fine-grained visual clues locally through instruction relevance mechanisms [259]. Such grid-based methods are particularly effective in continuous environments where precise localization is critical. Advancements in this domain also include the development of hybrid maps that combine local metric maps for short-term reasoning with global topological maps for long-term planning. The BEVBert framework, for example, utilizes a map-based pre-training paradigm that explicitly aggregates incomplete panoramic observations into a local metric map while modeling navigation dependencies in a global topological graph, thereby achieving state-of-the-art results in language-guided navigation [260]. Additionally, research into area graphs derived from Voronoi diagrams demonstrates how pruned topological structures can efficiently represent rooms and passages, offering superior performance in complex indoor environments compared to standard segmentation methods [261].

Beyond discrete graphs and grids, the field is increasingly exploring continuous and neural implicit representations to capture the geometry and semantics of navigation spaces more flexibly. Neural implicit representations allow agents to query environmental properties such as occupancy or semantic labels at arbitrary coordinates, overcoming the resolution limits of fixed grids. In the context of multi-object navigation, agents have been designed to dynamically learn neural implicit representations during each episode, utilizing a “Semantic Finder” to predict object positions and an “Occupancy and Exploration Implicit Representation” to map explored areas and obstacles [262]. This approach enables the agent to leverage memory sources that are continuous and differentiable, facilitating more nuanced interactions with the environment. Furthermore, the concept of neural potential fields offers a graph-free, end-to-end navigation pipeline where navigation is achieved by minimizing a potential function defined continuously over image embeddings, effectively creating

a smooth navigational landscape without explicit graph construction [263].

The challenge of maintaining these rich representations over extended periods has led to innovations in memory management and fragmentation strategies. Inspired by neuroscience, the Fragmentation-and-Recall (FARMap) strategy mimics grid cell behavior by truncating local maps upon high surprise events and storing them in long-term memory as part of a topological graph, thereby enabling efficient exploration of large spaces with active memory usage [264]. Similarly, brain-inspired compact cognitive mapping systems utilize neighborhood fields determined by movement information to restrict map growth, ensuring real-time performance in large-scale environments through parallel computing and scene integration [265]. Other approaches, such as Multigrid Neural Memory, propose internal, distributed memory architectures with multigrid connectivity that emerge coherent memory subsystems capable of retaining trajectories for thousands of time steps without external controllers [266].

Finally, the utility of these representations is fully realized when integrated with active inference and planning mechanisms. Hierarchical active inference models combine cognitive maps with allocentric and egocentric world models to facilitate flexible behavior across different levels of reasoning, from context to place to motion [267]. Agents equipped with such hierarchical structures can perform curiosity-driven exploration while maintaining goal-oriented behavior, effectively inferring the underlying structure of multi-room maze environments [268]. Moreover, learning-augmented model-based planning approaches utilize graph neural networks to predict the value of information in unseen spaces, allowing agents to actively gather data that improves long-horizon navigation under uncertainty [269]. These diverse methodologies collectively underscore the shift towards structured, scalable, and semantically rich environmental representations. By providing the necessary spatial grounding, these memory systems serve as the essential substrate for the hierarchical planning and topological navigation strategies explored in the following section.

5.4 Hierarchical Planning and Topological Navigation Strategies

5.4 Hierarchical Planning and Topological Navigation Strategies

Building upon the structured environmental representations and spatial memory systems detailed in Section 5.3, the challenge of navigating large-scale, unseen environments necessitates a shift from monolithic end-to-end policies toward modular, hierarchical architectures. While the previous section established how agents internalize space through topological graphs, metric grids, and neural implicit fields, this section focuses on the mechanisms that leverage these representations for high-level reasoning and decision-making. These hierarchical approaches fundamentally decouple symbolic planning, which manages long-term goal decomposition and route selection over global maps, from low-level control mechanisms responsible for executing fine-grained motor actions. This separation of concerns allows agents to harness the global reasoning capabilities of graph-based search algorithms while retaining the adaptability of learned policies for local obstacle avoidance and continuous control. By operating on the relational structures rather than raw metric details provided by the memory systems in Section 5.3, these strategies significantly reduce the computational burden of planning and facilitate generalization to novel layouts.

Transformer-based architectures have emerged as powerful engines for processing the topological representations constructed in the preceding section, enabling agents to reason over global action spaces while simultaneously grounding instructions in local visual observations. For instance, the Dual-scale Graph Transformer (DUET) exemplifies this synergy by dynamically combining fine-scale encoding of local observations with coarse-scale encoding on a globally constructed topological

map. This dual-scale approach allows the agent to balance the complexity of large action space reasoning with the precision required for fine-grained language grounding, leading to significant improvements in success rates on benchmark vision-and-language navigation tasks [270]. By building topological maps on-the-fly, such systems enable efficient exploration and robust path planning even when the agent encounters previously unseen environmental configurations, effectively bridging the gap between the static memory structures discussed earlier and dynamic navigation needs.

Beyond static graph representations, evolving graph strategies that integrate reinforcement learning with classical planning offer a robust solution for navigation in continuous and uncertain environments. The Subgoal Graphs-Reinforcement Learning (SG-RL) framework illustrates this by employing a two-level hierarchy: a geometric path-planning method generates optimal abstract paths as sequences of subgoals based on the agent’s spatial memory, while a reinforcement learning policy learns to execute the motion planning between adjacent subgoals. This architecture effectively addresses the sparse reward problem inherent in long-horizon navigation and allows the agent to handle unexpected environmental changes, such as appearing obstacles, without needing to reconstruct the entire subgoal graph [271]. Similarly, the ReVoLT framework decouples navigation into distinct modules for object detection, high-level semantic reasoning, intermediate-level geometric planning, and low-level control. It operates on a multi-layer semantic-spatial topological graph, using a high-level reasoner to infer semantic sub-goals via tree search and a lightweight planner to generate instantaneous spatial locations through an online constructed Voronoi local graph [272].

The integration of graph-augmented deep reinforcement learning further extends these capabilities to challenging 3D environments with disconnected regions. In the GameRLand3D environment, a hybrid technique combining a low-level RL policy with a graph-based high-level classical planner demonstrated a substantial increase in success rate compared to purely end-to-end recurrent agents, particularly in large-scale, unseen maps [273]. This highlights the critical role of high-level symbolic guidance in overcoming the limitations of pure deep learning approaches when scaling to vast, complex worlds. Furthermore, hierarchical planning is not limited to spatial navigation but extends to the temporal abstraction of tasks. Frameworks like Forecaster utilize temporally abstract world models to plan over high-level goals via tree-search procedures, effectively capturing long-horizon dependencies that are often missed by step-by-step rollout methods [274].

Recent advancements have also incorporated Chain-of-Thought (CoT) reasoning to facilitate hierarchical decision-making in navigation and manipulation. By marking the completion of subgoals within demonstration trajectories, CoT structures provide coherent, long-term action guidance that helps agents decompose complex routes into executable segments. The Chain-of-Thought Predictive Control (CoTPC) method leverages this by learning to predict entire CoT sequences as structured guidance, significantly outperforming standard two-stage subgoal-conditioned policies in contact-rich and long-horizon tasks [275]. This concept of decomposing tasks into logical steps is further reinforced by neuro-symbolic approaches where Large Language Models (LLMs) are employed to break down long-term goals into autoregressive sequences of sub-goals. The DELTA framework, for example, uses LLMs to decompose tasks while leveraging scene graphs for precise environmental grounding, achieving higher planning success rates and efficiency [181].

Moreover, the synthesis of hierarchical controllers based on deep reinforcement learning policies offers a formal approach to managing complexity in Markov Decision Processes (MDPs). By treating the environment as a graph of “rooms,” each governed by a low-level DRL policy, and synthesizing a high-level planner to select among these policies, agents can navigate amid moving obstacles with guaranteed performance bounds [276]. This aligns with the broader trend of using graph neural networks to process hierarchical scene representations, as seen in visually grounded planning algo-

rithms that utilize both geometric and symbolic scene graphs to predict high-level task plans and low-level motions efficiently [190].

In summary, hierarchical planning and topological navigation strategies represent a critical frontier in embodied AI, effectively operationalizing the spatial memories described in Section 5.3 to bridge the gap between abstract symbolic reasoning and continuous physical control. By leveraging transformer-based topological planners, evolving graph strategies, and chain-of-thought decomposition, modern agents can tackle the challenges of long-horizon navigation in dynamic, unseen environments. However, the efficacy of these sophisticated planning architectures ultimately depends on the robustness of the underlying representations when faced with domain shifts. Consequently, ensuring that these hierarchical systems generalize to novel textures, lighting, and layout configurations remains a paramount challenge, motivating the exploration of generalization strategies and data augmentation techniques discussed in the following section.

5.5 Generalization Strategies and Data Augmentation in Navigation

5.5 Generalization Strategies and Data Augmentation in Navigation

While hierarchical planning and topological strategies (Section 5.4) provide the architectural scaffolding for decomposing complex navigation tasks, the efficacy of these systems ultimately hinges on their ability to generalize to unseen environments. The capacity of embodied agents to transfer navigation policies from training settings to novel, real-world configurations remains one of the most persistent challenges in Vision-and-Language Navigation (VLN). Despite achieving high success rates in known domains, agents often suffer significant performance degradation when deployed in new settings due to overfitting to specific visual textures, layout configurations, and lighting conditions. To bridge this sim-to-real and seen-to-unseen gap, the research community has developed a multifaceted arsenal of strategies centered on large-scale data synthesis, sophisticated domain randomization, dynamic curriculum learning, and meta-learning frameworks. These approaches aim to cultivate environment-invariant representations that allow the high-level planners discussed previously to operate robustly regardless of the underlying sensory input.

A primary avenue for improving generalization is the expansion of training data diversity through large-scale synthetic instruction generation. The scarcity of annotated trajectory-instruction pairs in real-world or high-fidelity simulated environments often bottlenecks model robustness. Recent paradigms have shifted towards leveraging vast libraries of photo-realistic environments to synthesize millions of instruction-trajectory pairs. For instance, scaling data generation by utilizing over 1,200 environments from datasets like HM3D and Gibson has demonstrated that simple imitation learning on such massive, diverse datasets can push success rates to new heights while reducing the generalization gap between seen and unseen environments to less than 1% [277]. Furthermore, methods have been proposed to automatically generate structured instructions via large-scale cross-modal pretrained models like CLIP, enabling agents to self-explore environments and create in-domain datasets without human supervision [278]. This approach not only alleviates the reliance on labeled data but also facilitates fast cross-domain adaptation by exposing the agent to a wider variety of semantic and geometric configurations during pre-training, thereby strengthening the grounding capabilities required for the topological reasoning described in the preceding section.

Complementary to data scaling is the technique of domain randomization and augmentation, which aims to learn environment-invariant representations by introducing controlled variability during training. Traditional domain randomization often involves altering low-level visual properties such as color, texture, and lighting. However, effective generalization requires addressing both low-level

variations and high-level structural changes. Environments like House3D have been instrumental in this regard, offering tens of thousands of diverse 3D scenes that enable scene-level augmentation alongside pixel-level domain randomization [279]. Beyond static augmentation, advanced methods employ “environmental dropout” to mimic unseen environments during the fine-tuning stage, effectively forcing the agent to rely on robust semantic cues rather than memorizing specific environmental features [280]. Additionally, techniques such as Directional Domain Augmentation (DDA) simulate unseen target features by mapping source data through a domain transformer, specifically addressing non-stationary environments where data distributions shift gradually, such as changing illumination from dawn to dusk [281]. By hardening the perception backbone against such variations, these methods ensure that the symbolic planners and graph representations remain valid even when visual inputs deviate from the training distribution.

Curriculum learning has emerged as a critical strategy for guiding agents through increasingly difficult tasks, thereby preventing premature convergence to suboptimal policies and enhancing zero-shot generalization. In the context of reinforcement learning, Unsupervised Environment Design (UED) allows for the automatic generation of curricula that maximize the agent’s regret, effectively creating a dynamic difficulty scaling mechanism [282]. Algorithms like CLUTR (Curriculum Learning via Unsupervised Task Representation Learning) decouple task representation from curriculum generation, utilizing a latent task manifold to create stable and effective training curricula that outperform static task distributions [283]. Similarly, Meta Automatic Curriculum Learning (Meta-ACL) generalizes the curriculum generation process itself, learning to adapt task sampling distributions for unknown learners, which is particularly beneficial in complex procedural task spaces where manual design is infeasible [284]. These adaptive curricula ensure that agents are consistently challenged at the edge of their capabilities, fostering policies that are robust to environmental perturbations and better prepared for the uncertainty inherent in real-world deployment.

Meta-learning frameworks provide another powerful lens through which to view generalization, framing the problem as “learning to learn” how to adapt quickly to new domains with minimal interaction. Meta-learning approaches aim to extract an inductive bias from a distribution of tasks that facilitates rapid adaptation to unseen target domains. For navigation, this often involves meta-training on a diverse set of environments so that the agent can internalize generalizable spatial and semantic priors. Methods such as Self-Adaptive Visual Navigation (SAVN) employ meta-reinforcement learning to learn a self-supervised interaction loss, allowing agents to adapt to novel scenes at test time without explicit supervision [285]. Furthermore, domain-augmented meta-learning frameworks have been developed to handle open domain generalization, where source and target domains may have different label sets and distributions, by augmenting domains at both feature and label levels [286]. The integration of meta-learning with offline datasets has also shown promise in solving long-horizon navigation tasks with significantly fewer environment interactions, leveraging reusable skills extracted from prior experiences [287]. This capability for rapid adaptation serves as a crucial bridge to the active perception strategies discussed next, as it enables agents to efficiently update their internal models when encountering novel uncertainties.

Finally, the synergy between these strategies is evident in hybrid approaches that combine data augmentation with meta-learning objectives. For example, some frameworks utilize adversarial shift layers to simulate domain shifts during meta-training, generating pseudo-tasks that enhance the model’s robustness to cross-domain few-shot classification and regression [288]. Others employ multi-view regularized meta-learning to reduce overfitting by utilizing multiple optimization trajectories, theoretically proving that increasing the number of tasks in each trajectory reduces

the generalization bound [289]. Collectively, these advancements in data synthesis, augmentation, curriculum design, and meta-learning represent a paradigm shift from training agents for specific environments to cultivating generalist navigators. By ensuring that agents possess robust, adaptable representations, these strategies lay the necessary foundation for the active perception, uncertainty handling, and resilience mechanisms explored in the following section, which are essential for operating reliably in the open world.

5.6 Active Perception, Uncertainty Handling, and Robustness in Navigation

5.6 Active Perception, Uncertainty Handling, and Robustness in Navigation

While the generalization strategies and data augmentation techniques discussed in Section 5.5 provide agents with robust, environment-invariant representations, these static priors are often insufficient for navigating the dynamic, partially observable realities of the open world. To bridge the gap between training distributions and real-world deployment, embodied agents must transcend passive observation and reactive control. This subsection investigates the convergence of active perception strategies, uncertainty-aware planning, and resilience mechanisms that enable agents to actively gather information, quantify ambiguity, and recover from unforeseen perturbations. These capabilities serve as the critical operational layer that allows the generalized policies from the previous section to function reliably before scaling to the continuous and unstructured domains explored in Section 5.7.

Active perception transforms the navigation problem from a static mapping task into a dynamic information-gathering process, allowing agents to intelligently select viewing directions or sensing actions to minimize environmental uncertainty. Rather than adhering to a pre-determined path, agents leverage internal model states to guide exploration. A promising approach involves utilizing the inherent uncertainty within transformer-based attention mechanisms; for instance, the concept of Attention-Map Entropy (AME) identifies the most informative observations, enabling efficient scene reconstruction or object classification without requiring additional loss components [290]. Similarly, self-attention mechanisms can guide visual exploration in scenarios with limited fields of view, empowering agents to dynamically attend to critical regions rather than relying on static, task-specific uncertainty maps [291]. These methods facilitate “look-around” behaviors where agents are rewarded for reducing uncertainty about unobserved portions of their environment, fostering policies that generalize effectively to new tasks and unseen settings [292; 293]. Furthermore, active perception can be framed as an informative path planning problem, where agents collect data specifically to adapt semantic segmentation networks to new domains, leveraging extracted uncertainty to safely gather relevant training data [294].

Complementing active exploration is the paramount need to handle uncertainty explicitly, distinguishing between aleatoric noise (inherent environmental stochasticity) and epistemic uncertainty (model ignorance) to make calibrated decisions. Effective navigation systems utilize these distinctions to formulate safer path selection policies. Recent frameworks propose learning occupancy priors over indoor maps and leveraging model uncertainty over generated areas to guide navigation; for point-goal tasks, agents may adopt an upper confidence bound policy to select efficient paths, while for pure exploration, they maximize model uncertainty to discover unknown regions [295]. In stochastic driving environments where standard sequence modeling often fails due to overly optimistic assumptions, uncertainty-aware decision transformers estimate state uncertainties via conditional mutual information. By segmenting sequences to learn from true outcomes rather than noisy transitions, these models enable cautious planning during inference [296]. Ad-

ditionally, incorporating class uncertainty into trajectory forecasting allows agents to exploit full class probability distributions from perception models, improving prediction accuracy in crowded scenes where object identities are ambiguous [297]. For safety-critical applications like collision avoidance, computationally tractable methods such as MC-Dropout and bootstrapping quantify model uncertainty, enabling agents to recognize unseen pedestrian behaviors and exercise caution accordingly [298].

Beyond quantifying uncertainty, robustness in navigation demands the ability to recover from deviations and adapt to non-stationary environments. Real-world navigation is rarely disturbance-free; agents frequently encounter sudden obstacles, human interruptions, or sensor noise that cause route deviations. To enhance deviation robustness, progressive perturbation-aware contrastive learning paradigms have been introduced, training agents to navigate successfully even when trajectories are perturbed by contrasting perturbation-free encodings with perturbed counterparts to instill resilience [299]. In scenarios where environment dynamics change over time, hybrid approaches combining reinforcement learning policies with online search methods, such as Policy-Augmented Monte Carlo Tree Search (PA-MCTS), allow agents to integrate up-to-date environmental models with prior policy knowledge, ensuring optimal decision-making despite non-stationarity [300; 301].

Moreover, the capacity for self-monitoring and active inquiry constitutes a critical component of robust navigation. Agents equipped with progress estimation modules can evaluate their likelihood of success and decide whether to backtrack or continue, significantly outperforming greedy strategies that lack such introspective capabilities [302]. When uncertainty regarding the goal location becomes prohibitive, active inquiry strategies enable agents to ask for help. By formulating precise uncertainty measures and employing curriculum learning where feedback is not always available, agents learn to request assistance only when necessary, balancing the cost of interaction with the benefit of reduced ambiguity [303; 304]. Finally, lookahead exploration techniques, often grounded in model-based reinforcement learning, allow agents to simulate future states to refine current state estimates. By applying iterative inference at decision-time to maximize the coherence of future state representations, agents can improve reconstruction accuracy and task performance, particularly in partially observable environments where immediate sensory data is insufficient [207]. Collectively, these advances in active perception, uncertainty quantification, and adaptive recovery mechanisms form the bedrock of next-generation embodied navigation systems. They provide the necessary resilience and situational awareness to transition from controlled indoor settings to the continuous, outdoor, and web-based frontiers discussed in the following section.

5.7 Emerging Frontiers: Continuous Control, Outdoor, and Web Navigation

5.7 Emerging Frontiers: Continuous Control, Outdoor, and Web Navigation

Building upon the robust mechanisms for active perception and uncertainty management established in complex indoor settings, embodied navigation research is now rapidly expanding beyond the confines of discrete, graph-based environments toward more intricate, continuous, and open-world domains. This evolution necessitates a fundamental shift in how agents perceive space, plan trajectories, and ground linguistic instructions, moving from high-level topological decisions to low-level atomic control in unstructured settings. A primary frontier in this domain is the transition to continuous space navigation, where the assumption of pre-defined navigability graphs is removed. In these settings, agents must execute fine-grained motor commands to traverse free space, a challenge that significantly lowers performance compared to discrete counterparts due to the loss of oracle navigation and perfect localization assumptions [305]. To address the inefficiencies inherent

in low-level action spaces within continuous environments, researchers have developed language-conditioned waypoint prediction networks. These models effectively bridge the gap between high-level semantic intent and low-level control, demonstrating that more expressive architectures can generate simpler, faster-to-execute trajectories while maintaining robust navigation metrics [306]. Furthermore, advanced architectures like the Multi-Level Attention Network (MLANet) have been introduced to meet the real-time execution demands of continuous VLN by dynamically generating sub-instructions and fusing multi-level semantic features to mitigate noise in long action sequences [307].

Parallel to the shift in action spaces is the critical expansion into outdoor and large-scale natural environments, which present unique challenges distinct from structured indoor scenes. Unlike the controlled interiors discussed previously, outdoor environments are characterized by a lack of clear spatial delineations, vast scales, and inherent semantic ambiguities that render indoor-trained policies ineffective. To tackle these issues, new benchmark tasks and mechanisms have been proposed to enable robots to reason about the unseen in outdoor terrains, leveraging Large Language Models to hallucinate plausible futures and navigate without prior mapping [308]. The complexity of outdoor navigation is further compounded in aerial contexts, where agents such as Unmanned Aerial Vehicles (UAVs) must reason about three-dimensional flight paths, altitude constraints, and complex spatial relationships. The introduction of tasks like AerialVLN highlights the significant performance gap between current baseline models and human capabilities in sky-based navigation, underscoring the need for specialized simulators and cross-modal alignment methods tailored for 3D flight [309]. Additionally, visual place recognition in these expansive outdoor settings faces distinctive hurdles, such as extreme perspective changes in bird’s-eye views, necessitating datasets like VPAIR that capture high-altitude imagery over diverse landscapes to train robust localization systems [310].

To support navigation in these unstructured outdoor domains, the representation of the environment has also evolved to accommodate scale and sparsity. Traditional volumetric representations often prove memory-inefficient for large-scale outdoor spaces, prompting a shift toward point cloud-based approaches. Innovations such as OpenGraph propose open-vocabulary hierarchical graph representations specifically designed for large-scale outdoor environments, utilizing LiDAR point clouds and visual-language models to achieve superior segmentation and query accuracy in complex urban and natural settings [311]. Similarly, benchmarks like WildScenes provide essential bi-modal data for natural environments, facilitating research into semantic perception for applications ranging from conservation to search and rescue, where domain adaptation over varying natural conditions is crucial [312]. The integration of cooperative vehicle-infrastructure data, as seen in PointSSC, further enhances semantic scene completion for long-range perception, addressing the sparsity of LiDAR data in outdoor scenarios [313].

Beyond physical navigation, the concept of embodied agency is extending into virtual digital spaces, specifically web navigation, which serves as a testbed for high-level reasoning and interaction in open worlds. In this domain, agents must follow natural language instructions to execute complex tasks on real-world websites. A key challenge here is the disconnect between the structural HTML representation and the visual context required for task completion. Recent advancements propose dual-view visual contextualization, where HTML elements are enriched with their corresponding visual content from webpage screenshots. This approach leverages the spatial arrangement of task-related elements to provide clearer context, significantly improving agent performance across diverse tasks and domains compared to methods relying solely on HTML documents [314]. This extension into cyber-physical and purely digital realms demonstrates that the principles of embod-

ied reasoning—perception, cognition, and action—are universally applicable, whether the agent is navigating a cluttered forest, flying over a city, or browsing the internet. Collectively, these emerging frontiers illustrate a maturing field that is increasingly focused on generalization, robustness, and the seamless integration of semantic understanding with precise control in any environment an agent may encounter.

6 Language-Guided Manipulation and Physical Interaction

6.1 Bridging the Semantic-Physical Gap in Affordance Reasoning

6.1 Bridging the Semantic-Physical Gap in Affordance Reasoning

The integration of Vision-Language Models (VLMs) into robotic manipulation marks a paradigm shift from static perception to dynamic physical interaction. However, a critical bottleneck persists: the semantic-physical gap. While contemporary VLMs excel at recognizing objects and describing scenes based on internet-scale data, they often lack the nuanced understanding of physical properties required for safe and effective manipulation. Generic pre-training datasets predominantly consist of image-text pairs that capture visual appearance and semantic labels but frequently omit essential robotic knowledge such as material composition, fragility, weight distribution, and functional affordances. This deficiency hampers the efficacy of robots when tasked with instructions requiring reasoning about how an object behaves under physical stress or how it should be manipulated to achieve a specific goal. To bridge this gap, recent research has focused on injecting physically grounded knowledge directly into multimodal architectures through specialized datasets and novel training frameworks, laying the foundational reasoning capabilities necessary for the precise grasp generation discussed in the subsequent section.

A primary strategy for addressing this limitation involves the curation of datasets explicitly designed to annotate physical concepts. Traditional datasets often label an object simply as a “cup,” failing to distinguish between a fragile ceramic mug and a durable plastic tumbler—a distinction critical for determining grasp force and placement strategy. To remedy this, the [315] paper introduces PhysObjects, a comprehensive object-centric dataset containing hundreds of thousands of annotations regarding physical concepts such as material, fragility, and deformability. By fine-tuning VLMs on such data, models learn to infer physical priors from visual appearance, effectively capturing human-like intuitions about object properties. The authors demonstrate that incorporating this physically grounded understanding into a planning framework significantly improves task success rates, as the robot can reason about the consequences of its actions—such as avoiding excessive force on a brittle object—before execution. This approach marks a departure from purely correlational learning toward causal physical reasoning, enabling agents to generalize physical concepts to held-out object categories.

Complementing dataset-driven approaches are framework-level innovations that restructure the learning objective to prioritize manipulation-centric knowledge. The [316] study proposes ManipVQA, a novel framework that endows Multimodal Large Language Models (MLLMs) with robotic-specific insights through a Visual Question-Answering (VQA) format. Unlike standard VQA tasks that focus on identification or counting, ManipVQA prompts the model to reason about tool detection, affordance recognition, and physical concept prediction. This method allows the model to preserve its inherent vision-reasoning capabilities while seamlessly integrating new knowledge about how objects interact with the physical world. Empirical evaluations in robotic simulators confirm that models trained with this framework exhibit robust performance in manipulation tasks, successfully translating high-level semantic instructions into physically viable action

plans. The success of ManipVQA underscores the importance of aligning the pre-training objectives of foundation models with the specific demands of embodied agency.

Beyond explicit fine-tuning, researchers are also exploring how to extract and leverage implicit affordance knowledge already latent within large pre-trained models. The [317] paper investigates the potential of off-the-shelf models like CLIP to perform affordance segmentation without task-specific supervision. The authors propose AffordanceCLIP, which leverages the rich semantic space of pre-trained VLMs to segment object parts based on action prompts. This approach demonstrates that while these models were not explicitly trained for affordance detection, they retain valuable information linking object parts to potential actions. By utilizing simple parameter-efficient tuning, AffordanceCLIP achieves competitive zero-shot performance, offering a scalable solution that eliminates the need for costly pixel-level annotations. Similarly, the [318] work further advances this direction by grounding affordances for unseen objects and actions in the wild, utilizing the abstract and human-object-interaction knowledge embedded in large-scale VLMs to achieve significant performance gains over competing methods.

However, relying solely on 2D visual cues is often insufficient for complex manipulation scenarios where 3D geometry and environmental context play pivotal roles. The [319] framework addresses this by introducing a 3D pre-training approach that captures semantic, geometric, and affordance properties directly from point clouds. By focusing on cluttered scenes and employing tasks such as grasping pose synthesis and referring expression grounding, SUGAR learns representations that are inherently more suitable for robotics than their 2D counterparts. Furthermore, the importance of environmental context in affordance reasoning is highlighted in [320], which argues that grasping regions cannot be deduced from object attributes alone but must consider the surrounding environment. This work utilizes knowledge graphs to rank visual semantic attributes, enabling zero-shot affordance prediction that accounts for contextual constraints.

The challenge of generalizing affordance knowledge to novel objects and tasks is also being tackled through generative and imaginative approaches. The [321] paper introduces a paradigm where large language models are coupled with physics simulators to “imagine” novel affordances for unseen object classes. By analyzing interaction definitions and simulating virtual scenarios, the system can predict optimal poses and interaction strategies for objects it has never encountered, outperforming traditional learning-based baselines. Additionally, the [322] study explores data-efficient learning, demonstrating that with effective alignment between visual features and affordance text embeddings, models can identify novel affordances with merely one example per category.

Despite these advancements, gaps remain in the seamless integration of these physical reasoning capabilities into real-time control loops. The [323] study highlights that even state-of-the-art LVLMS suffer from misgrounding regarding fine-grained attributes like material and shape, which are crucial for physical interaction. Addressing these hallucinations and improving the reliability of physical commonsense remains a critical frontier. As the field evolves, the synthesis of large-scale linguistic knowledge with rigorous physical grounding, as exemplified by frameworks like [316] and datasets like [315], establishes the cognitive foundation required to transition from understanding *what* an object affords to executing *how* to grasp it. This progression naturally leads to the next critical step: translating these implicit affordance insights and human intentions into explicit, precise 6-DoF grasp poses.

6.2 From Implicit Intent to Explicit Grasp Generation

6.2 From Implicit Intent to Explicit Grasp Generation

Building upon the cognitive foundation of physical affordance reasoning established in the previous section, the next critical challenge lies in translating these implicit understandings and high-level natural language instructions into precise, executable 6-DoF grasp poses. While Section 6.1 focused on bridging the semantic-physical gap by endowing models with knowledge of material properties and functional affordances, this subsection addresses the operationalization of that knowledge: converting vague human intent and task-level descriptions into specific geometric configurations. This transition marks a shift from understanding *what* an object affords to determining exactly *how* and *where* to grasp it to fulfill a specific goal, ranging from ambiguous commands to strict part-level constraints.

A primary hurdle in this translation process is resolving the inherent ambiguity of human communication, which often lacks the precision required for direct control. Existing methods frequently falter when limited to broad task-level language, necessitating frameworks that can guide grasp synthesis through fine-grained text prompts specifying object parts. To address this, the Text2Grasp framework employs a two-stage approach where a text-guided diffusion model first generates a coarse grasp pose, which is subsequently refined through a hand-object contact optimization process [324]. By leveraging Large Language Models to interpret both task-level and personalized text descriptions without requiring additional manual annotations, this method achieves accurate part-level control [324]. Extending this capability, the concept of “reasoning grasping” has been formalized to handle scenarios where robots must infer grasp poses from indirect verbal cues rather than explicit commands. By integrating a multi-modal Large Language Model with a vision-based robotic grasping framework, these models generate grasps based on implicit intentions, significantly outperforming direct integrations of standard vision-language models like CLIP or LLaVA in challenging benchmark datasets [325].

Beyond resolving linguistic ambiguity, effective manipulation requires that generated grasps be semantically aligned with the intended downstream task, moving beyond mere geometric stability. Sole reliance on object geometry often limits applicability, prompting approaches that embed semantic information directly into the grasp representation. The SemGrasp method addresses this by introducing a discrete representation that aligns the grasp space with the semantic space, enabling the generation of postures that adhere strictly to language instructions [326]. This approach utilizes a Multimodal Large Language Model fine-tuned to integrate object, grasp, and language within a unified semantic space, supported by large-scale datasets featuring detailed captions and diverse grasps [326]. Furthermore, incorporating the physical commonsense discussed in Section 6.1 is vital for handling long-tailed scenarios involving uncommon materials or shapes. The PhyGrasp model exemplifies this synergy by leveraging inputs from natural language and 3D point clouds to assess physical properties, determining optimal grasping poses that align with human preferences and intuitive physics [327].

The fidelity of this intent-to-action translation also depends on robustly grounding language within complex, cluttered environments. Traditional pipelines that decouple visual grounding from grasp proposal generation often suffer from error propagation. In contrast, joint modeling approaches propose unifying vision, language, and action through object-centric representations, allowing for more flexible language instructions and improved sample efficiency [328]. Methods like CROG leverage the visual grounding capabilities of CLIP to learn grasp synthesis directly from image-text pairs, demonstrating significant improvements in both grounding and grasping accuracy within

cluttered indoor scenes [329]. Additionally, the VL-Grasp policy combines visual grounding with 6-DoF grasp pose detection to extend the universality of interactive grasping, incorporating a filter module to enhance performance across different views and postures [330].

Precision in grasp generation is further enhanced by focusing on specific contact points and spatial constraints, effectively bridging the gap between semantic selection and geometric execution. The CoPa framework decomposes manipulation into task-oriented grasping and task-aware motion planning, utilizing foundation vision-language models to select grasping parts through a coarse-to-fine grounding mechanism and identifying spatial geometry constraints for post-grasp poses [331]. For even finer control, keypoint-conditioned policies such as KITE attend to both scene and object semantics, grounding instructions via 2D image keypoints to provide an accurate object-centric bias for downstream action inference [332]. This allows robots to interpret instructions at varying levels of specificity, from high-level commands to detailed requests like grabbing a specific part of an object [332]. Moreover, zero-shot precise manipulation has been achieved by grounding fine-grained part descriptors using visual diffusion descriptors, where a user-defined click on a source image guides the gripper pose for a visually different instance of the same object [333].

Finally, the integration of language with traditional grasp planners enables deeper semantic understanding without extensive retraining, creating a modular yet cohesive system. The LAN-grasp approach prompts a Large Language Model to identify appropriate object parts for grasping, which are then localized by a Vision Language Model to generate region-specific grasp proposals [334]. This combination allows robots to grasp and utilize objects in a safer and more meaningful manner, consistently ranking higher in human evaluations compared to conventional planners [334]. Similarly, QwenGrasp combines a large vision-language model with a 6-DoF grasp neural network to handle vague instructions and provide feedback when tasks are infeasible, thereby improving safety in open language environments [335]. Collectively, these advancements illustrate a paradigm shift where implicit human intent is systematically translated into explicit, physically plausible, and semantically grounded 6-DoF grasp actions. Having established the machinery for precise grasp generation, the field is now poised to scale these capabilities to open-vocabulary settings, where robots must generalize these skills to novel objects and tools beyond their training distribution, as explored in the following subsection.

6.3 Open-Vocabulary Manipulation and Tool Use

6.3 Open-Vocabulary Manipulation and Tool Use

Building upon the capacity to translate implicit intent into explicit grasp poses, the field now faces the challenge of scaling these capabilities to open-vocabulary settings where robots must interact with novel objects and utilize tools beyond their training distribution. While Section 6.2 focused on refining grasp generation for specific instructions, this subsection addresses the paradigm shift from closed-set, task-specific manipulation to generalist interaction. Traditional robotic systems often fail when confronted with unforeseen environmental configurations or unknown object categories due to policies tightly coupled to finite training sets. In contrast, open-vocabulary manipulation leverages the extensive world knowledge encoded within Large Language Models (LLMs) and Vision-Language Models (VLMs) to enable agents to reason about, grasp, and utilize objects they have never explicitly encountered. This transition relies heavily on compositional reasoning, allowing agents to decompose complex tasks into manageable sub-goals and infer the functional affordances of unseen items based on semantic similarity and physical context.

A critical component of this evolution is the generalization of grasping strategies to arbitrary

objects without task-specific fine-tuning. Recent approaches demonstrate that by leveraging the geometric decomposition capabilities of LLMs, robots can achieve zero-shot task-oriented grasping. For instance, the ShapeGrasp framework illustrates how a robot can decompose an unfamiliar object into simple convex shapes represented as a graph, utilizing the commonsense reasoning of an LLM to assign semantic meaning to each part and determine the optimal grasp pose for a specific task [336]. This method bypasses the need for exhaustive labeled datasets by relying on the model’s inherent understanding of shape utility and spatial relationships. Furthermore, the scalability of such approaches has been significantly enhanced by synthesizing large-scale grasp datasets directly from foundation models. The Grasp-Anything initiative highlights how the universal knowledge embedded in models like ChatGPT can be harnessed to generate millions of diverse grasp samples with text descriptions, thereby facilitating zero-shot grasp detection on vision-based tasks and real-world robotic experiments [337]. By moving beyond limited object diversity, these data-driven strategies empower robots to handle the vast variability of real-world household and industrial environments.

Beyond mere grasping, true open-vocabulary competence requires the improvisational use of tools. Robots must not only recognize standard tools but also identify novel objects that can serve as functional substitutes when conventional tools are unavailable. This concept of “improvisational tool use” demands a deep physical understanding of multi-object interactions. Research has shown that combining diverse demonstration data with self-supervised interaction data allows models to learn visual foresight, enabling them to perceive and use novel objects as tools dynamically [338]. Such systems can evaluate whether an object is required for a specific goal, such as sweeping clutter, and execute the necessary motions even if the object is not a canonical tool. Complementing this, neural models like ToolNet have been developed to predict the best tool from available objects for a declarative goal by incorporating metric and semantic properties alongside pre-trained object embeddings from commonsense knowledge repositories [339]. These models demonstrate significant improvements in generalizing to unseen tools, proving that augmenting environmental representations with external knowledge bases is a viable strategy for robust tool selection.

The complexity of tool use often extends to sequential interactions, where multiple tools must be composed to achieve a high-level task. Addressing this, frameworks like ToolTango and TANGO introduce neural models capable of predicting sequential tool interactions by encoding the world state and symbolic relationships via graph neural networks [340][341]. These approaches attend to the scene using knowledge of the goal and action history, allowing the robot to synthesize robust plans in novel environments where known tools may be missing. The ability to generalize in these settings is crucial for service robots operating in dynamic human spaces, where the availability of specific objects cannot be guaranteed.

Furthermore, the integration of physically grounded VLMs has proven essential for bridging the gap between semantic understanding and physical execution, ensuring that open-vocabulary reasoning translates into safe actions. Standard VLMs often lack nuanced knowledge of physical concepts such as material properties and fragility, which are critical for safe manipulation. The PhysObjects dataset and associated frameworks address this limitation by fine-tuning VLMs on physical concept annotations, thereby improving planning performance on tasks that require reasoning about object physics [315]. When combined with LLM-based planners, these physically grounded models enable robots to make informed decisions about how to interact with objects, reducing the risk of damage and increasing task success rates. Similarly, the CoPa framework leverages foundation models to generate 6-DoF end-effector poses by identifying spatial geometry constraints of task-relevant object parts, demonstrating fine-grained physical understanding capable of handling open-set instructions

with minimal prompt engineering [331].

Finally, the deployment of these capabilities in real-world scenarios often necessitates closed-loop feedback and adaptive planning to handle the inherent uncertainty of open-world settings. Systems like COME-robot utilize GPT-4V as a central brain to perform situated reasoning and recover from failures during open-vocabulary mobile manipulation [205]. By actively perceiving the environment and verifying task progress, these agents can adapt their strategies in real-time, ensuring robustness against unexpected outcomes. Additionally, approaches that ground language plans in demonstrations through counterfactual perturbations further enhance the reactivity and interpretability of imitation learning, allowing robots to translate abstract language plans into reactive policies that generalize across 2D navigation and manipulation tasks [342]. While these foundation model-driven approaches offer remarkable flexibility, they sometimes lack the rigorous logical consistency and safety guarantees required for complex, long-horizon tasks. This limitation motivates the exploration of neurosymbolic and prompt-based control paradigms, which synergize the semantic breadth of LLMs with the precision of symbolic reasoning, as discussed in the following subsection.

6.4 Neurosymbolic and Prompt-Based Control Paradigms for Manipulation

6.4 Neurosymbolic and Prompt-Based Control Paradigms for Manipulation

While open-vocabulary manipulation and improvisational tool use (discussed in Section 6.3) empower robots with the semantic breadth to handle novel objects, the probabilistic nature of foundation models often results in actions that lack physical precision, logical consistency, or safety guarantees. Relying solely on neural generation can lead to hallucinations or physically infeasible plans, particularly in complex, long-horizon tasks. To address these limitations and bridge the gap between high-level semantic intent and low-level executable control, recent research has increasingly turned to neurosymbolic and prompt-based paradigms. These architectural strategies synergize the flexibility and commonsense reasoning of LLMs with the rigor, interpretability, and precision of symbolic reasoning engines and visual prompting mechanisms. By decoupling the planning phase from execution verification, these systems achieve robust performance in open-world scenarios where end-to-end neural policies frequently fail, setting the stage for the data-efficient affordance learning strategies discussed in the subsequent section.

A primary avenue in this domain is the use of LLMs as semantic parsers that translate natural language instructions into formal, executable programs. This approach, often termed “LLMs as programmers,” leverages the code generation capabilities of foundation models to orchestrate a library of pre-defined primitive skills or external tools. For instance, the [343] framework demonstrates how LLMs can generate Python-like modular programs that invoke off-the-shelf computer vision models and image processing routines to solve compositional visual tasks without any task-specific training. In the context of manipulation, this translates to an LLM decomposing a command like “put the red cup on the coaster” into a sequence of symbolic operations: `find_object("red cup")`, `find_object("coaster")`, `grasp()`, and `place()`. Similarly, [344] highlights how such frameworks can mitigate the need for extensive human engineering of in-context examples by automatically generating routines based on spatial and temporal abstractions, thereby enhancing robustness in novel environments. This program-aided approach ensures that the reasoning process is transparent and that errors can be traced to specific lines of code or failed module executions.

Beyond simple program synthesis, advanced neurosymbolic architectures incorporate iterative feedback loops and reflection mechanisms to correct planning errors dynamically, addressing the un-

certainty inherent in open-world settings. The [345] framework exemplifies this through a “Think-Program-reCtify” loop designed for 3D situated reasoning. In this paradigm, the LLM first decomposes a query into a plan (Think), grounds each step into code calling 3D perception modules (Program), and crucially, adjusts the plan and code if execution fails (Rectify). This self-correction capability is vital for manipulation tasks where initial perception may be occluded or ambiguous. Complementing this, [346] proposes interleaving reasoning traces with task-specific actions, allowing the agent to gather environmental feedback and update its action plan in real-time. This synergy between reasoning and acting enables the system to handle exceptions and recover from failures more effectively than methods that rely on static, pre-computed plans.

To further enhance interpretability and sample efficiency, researchers have developed hybrid frameworks that blend neural perception with symbolic goal learning. The [347] paper presents an architecture where a language parser maps unconstrained natural language queries to executable programs composed of a shared library of primitive skills. These primitives range from purely symbolic operations, such as counting or logic checks, to trainable neural functions for visual grounding. This marriage of discrete symbolic approaches with deep networks yields significant benefits in accuracy and robustness to vocabulary variations, while remaining transferable to real-world scenes with minimal fine-tuning. Furthermore, [185] investigates a modular framework that formulates instruction following as symbolic goal learning via deep neural networks, followed by task planning via symbolic planners. By bridging connectionist and symbolic modules with Planning Domain Definition Language (PDDL), this approach compensates for ambiguous natural language requests by leveraging visual observations to refine symbolic goals.

Visual prompting and region-specific reasoning also play a critical role in refining manipulation commands, ensuring that symbolic plans are grounded in precise spatial contexts. Precise manipulation often requires attending to specific regions of interest (RoI) rather than global image features. [348] introduces spatial instruction tuning, enabling models to interact via both language and bounding boxes. This allows for fine-grained multimodal understanding where the model can reason about attributes like color, shape, and material within specific RoIs, which is essential for distinguishing between similar objects during grasping tasks. Additionally, [349] incorporates visual Chain-of-Thought reasoning, where the model dynamically focuses on visual inputs and provides interpretable thoughts annotated with bounding boxes. This method enhances the model’s ability to identify key regions necessary for answering complex queries or executing precise actions, thereby reducing hallucinations and improving grounding.

Finally, the efficiency and scalability of these neurosymbolic systems are being addressed through innovative learning paradigms, making them viable for resource-constrained robotic platforms. [350] suggests that even smaller language models can achieve reasonable reasoning performance if tasked with formalizing problems for a symbolic solver, rather than solving them directly. This “formalize-then-solve” strategy reduces the computational burden and makes these sophisticated architectures accessible for deployment. Moreover, [351] explores the automatic synthesis and compression of code libraries, enabling agents to learn reusable, interpretable abstractions that can be deployed across diverse manipulation tasks. By compressing complex behaviors into documented, reusable modules, such systems facilitate lifelong learning and adaptation in dynamic physical environments.

In summary, neurosymbolic and prompt-based control paradigms represent a pivotal shift in embodied AI, moving away from opaque, end-to-end black boxes toward transparent, modular, and verifiable systems. By leveraging LLMs for high-level abstraction and symbolic solvers or visual prompts for low-level precision, these frameworks offer a promising pathway toward generalist agents capable of safe, interpretable, and robust physical interaction. However, the efficacy of

these symbolic and prompt-driven approaches fundamentally relies on the agent’s ability to accurately perceive *where* and *how* to interact with the environment. While Section 6.4 provides the logical framework for planning, the acquisition of the necessary perceptual grounding—specifically, the understanding of object affordances—often requires moving beyond static libraries to dynamic, self-supervised learning. This necessity motivates the exploration of learning affordances directly from observation and interaction, as detailed in the following subsection.

6.5 Learning Affordances from Observation and Interaction

6.5 Learning Affordances from Observation and Interaction

While neurosymbolic and prompt-based paradigms (discussed in Section 6.4) provide robust frameworks for decomposing tasks and ensuring logical consistency, their efficacy fundamentally depends on the agent’s ability to accurately perceive *where* and *how* to interact with the environment. Traditional approaches to acquiring these manipulation skills have relied heavily on extensive, manually annotated datasets that explicitly label object parts and their corresponding affordances. However, the scalability of such supervised methods is severely constrained by the high cost of data collection and the inherent difficulty in generalizing to novel objects and unseen environments. Consequently, recent research has pivoted toward self-supervised and active learning frameworks where agents autonomously construct rich affordance maps by observing human demonstrations, engaging in unstructured play, or leveraging uncertainty-driven exploration. These paradigms bridge the gap between the high-level semantic planning enabled by LLMs and the low-level physical execution required for dexterous manipulation, allowing robots to infer “what can be done” with minimal human intervention.

A primary avenue for reducing reliance on labeled data is learning from observation, specifically by leveraging large-scale human video data. Humans possess an innate ability to learn new tool uses and manipulation strategies simply by watching others, a capability that embodied agents are increasingly mimicking through video-based learning paradigms. By passively consuming abundant videos sourced from the internet, robots can acquire scalable supervision that mitigates the bias inherent in curated robotic datasets [352]. This approach allows agents to build an understanding of useful actions and their effects on the world without direct physical interaction. For instance, frameworks have been developed to learn human-object interaction “hotspots” directly from video, inferring spatial maps that indicate how an object would be manipulated even when it is at rest [353]. These weakly supervised methods have proven competitive with strongly supervised affordance techniques and demonstrate the ability to anticipate interactions for novel object categories. Furthermore, by augmenting instructional video resources with causal modeling, agents can acquire affordance properties such as behavior and entity equivalence through pretraining tasks like Masked Action Modeling, effectively learning physical commonsense from the visual domain [354]. To address the morphological mismatch between human demonstrators and robotic actuators—a critical challenge when translating symbolic plans into motor commands—advanced algorithms perform unsupervised human-to-robot translation and keypoint detection, enabling the extraction of structured state representations from single videos for policy learning [355].

Complementing observation-based learning is the paradigm of learning from interaction, often conceptualized as autonomous “play.” Just as infants develop motor skills and causal models of the world through playful experimentation, embodied agents can generate their own training data by interacting with their environment. This self-supervised exploration allows agents to discover objects, learn their geometric extents, and infer physical properties like relative mass without external

guidance [356]. Approaches such as “Learning from Play” utilize unstructured human teleoperated data to construct latent spaces of object affordances, linking human behaviors to environmental changes to improve multi-task policy learning [357]. By organizing play behaviors in a latent space, agents can reuse these skills at test time to achieve specific goals, demonstrating robustness to perturbations and exhibiting retrying behaviors that expert-trained policies often lack [358]. Moreover, extracting self-supervised visual affordance models from play data enables efficient policy learning, where agents favor object regions identified as manipulable by humans, significantly accelerating training compared to baseline reinforcement learning methods [359]. In some frameworks, agents actively discover the affordance landscape of unmapped 3D environments by maximizing successful interactions while simultaneously training image-based segmentation models, effectively densifying rewards for exploration [360].

To further enhance data efficiency and robustness, uncertainty-driven active learning has emerged as a critical strategy for affordance discovery. Traditional deterministic models often fail to capture the nuances of complex scenes or the confidence required for safe interaction, a limitation that can be catastrophic when executing symbolic plans generated by LLMs. Bayesian deep learning methods address this by predicting affordances directly from RGB images while quantifying both aleatoric and epistemic uncertainty, ensuring that agents can distinguish between known and unknown interaction possibilities [361]. By leveraging this uncertainty, agents can drive an interactive discovery process, selectively exploring regions or actions where the model is most uncertain. This approach has been shown to enable the efficient discovery of visual affordances for primitives such as grasping and stacking, allowing real-world robots to learn effective policies in a small number of trials [362]. Similarly, frameworks inspired by active learning train ensembles of dynamics models and incentivize the agent to explore states that maximize disagreement among these models, facilitating sample-efficient skill acquisition without external rewards [363]. This principle extends to navigating and segmenting unknown environments, where agents plan paths to frontiers with high model uncertainty to collect informative data for refining their perception systems [364].

The integration of these observational and interactive strategies often involves sophisticated memory and representation mechanisms that serve as the foundation for higher-level reasoning. Agents can learn instance segmentation purely through interaction, refining their models by inferring object pixels and testing hypotheses against the physical world, thereby generalizing to novel objects without manual labels [365]. Furthermore, curiosity-driven intrinsic motivation allows agents to construct world models that predict dynamic consequences, using self-models to adversarially challenge their own predictions and generate novel, structured behaviors in unstructured environments [366]. By combining model-based planning with model-free deep reinforcement learning guided by visual affordances, agents can achieve sample-efficient policy optimization that generalizes well to unseen objects and tasks [367]. Ultimately, the convergence of learning from observation, autonomous play, and uncertainty-driven exploration represents a significant shift toward generalist embodied agents. These agents possess the perceptual grounding necessary to execute the complex, contact-rich interactions and dexterous manipulations discussed in the following section, adapting to open-world challenges with minimal human supervision.

6.6 Dexterous Manipulation and Complex Physical Interactions

6.6 Dexterous Manipulation and Complex Physical Interactions

Building upon the perceptual grounding and affordance discovery mechanisms established in the preceding section, the field now faces the critical challenge of translating these high-level interaction

possibilities into precise, contact-rich physical execution. The transition from identifying *where* to interact to successfully performing dexterous manipulation represents a significant leap in embodied artificial intelligence. This shift requires agents to navigate high-dimensional action spaces, manage complex contact dynamics, and adapt to diverse physical properties that static affordance maps alone cannot fully capture. Unlike parallel-jaw grippers, multi-fingered anthropomorphic hands offer human-like versatility, enabling sophisticated tasks such as in-hand reorientation, tool use, and the manipulation of articulated or deformable objects. However, this morphological advantage introduces substantial control challenges, including the coordination of numerous degrees of freedom (DoF), the handling of intermittent contact forces, and the necessity for seamless sensorimotor integration. Recent advancements have leveraged deep reinforcement learning (RL), imitation learning, and novel simulation-to-real (Sim2Real) transfer techniques to address these complexities, pushing the boundaries of what autonomous agents can achieve in unstructured environments.

A primary hurdle in dexterous manipulation is mastering contact-rich interactions where forces must be finely balanced to prevent slippage or object damage while achieving task goals. Deep RL has emerged as a scalable solution for learning these complex behaviors directly from sensor data, effectively bypassing the need for explicit analytical models of contact dynamics. For instance, policies trained via model-free deep RL have successfully demonstrated the ability to perform intricate tasks like twisting lids off bottles using two hands, showcasing generalization across unseen objects through careful reward design and physical modeling [368]. Similarly, systems have been developed to learn high-DOF reaching-and-grasping by utilizing dynamic representations of gripper-object interactions, such as the Interaction Bisector Surface (IBS), which effectively informs fine-grained finger control and improves sample efficiency [369]. The integration of tactile feedback further enhances robustness; frameworks like TAVI (Tactile Adaptation from Visual Incentives) optimize tactile-based policies using vision-derived rewards, significantly outperforming methods that rely solely on tactile or visual inputs for tasks such as peg insertion and unstacking [370]. Moreover, leveraging contact forces explicitly has been shown to improve grasping robustness under uncertainty, allowing robots to react to unforeseen effects during manipulation [371].

Beyond rigid object manipulation, dexterous agents must also handle articulated objects and deformable bodies, which introduce additional layers of complexity due to changing kinematic structures and continuous shape variations. These scenarios demand not only an understanding of static affordances but also a dynamic prediction of how object states evolve under manipulation. Benchmarking efforts like DexArt have been established to evaluate generalizable dexterous manipulation on articulated objects, highlighting the limitations of traditional grippers and the potential of multi-fingered hands to operate daily objects like faucets and doors [372]. For deformable objects, where physics interactions are notoriously difficult to model, frameworks such as DexDeform combine human demonstrations with differentiable physics to refine skills and explore goal spaces, enabling manipulation of soft materials that would stump standard RL approaches [373]. Complementary approaches like VIRDO++ utilize multimodal neural implicit representations to achieve high-fidelity state estimation and dynamics prediction for deformable objects using partial visuo-tactile feedback, facilitating real-world application without privileged contact information [374].

Long-horizon dexterous tasks often require chaining multiple sub-policies or adapting strategies dynamically, moving beyond single-step interactions to coordinated sequences. The concept of “Sequential Dexterity” addresses this by introducing a transition feasibility function that progressively fine-tunes sub-policies, enabling autonomous switching and recovery from failures in long-horizon tasks [375]. In scenarios involving soft robotic hands, which offer inherent compliance but pose control difficulties, continual policy distillation frameworks have been proposed to con-

solidate knowledge from multiple expert policies, mitigating catastrophic forgetting and enabling versatile in-hand manipulation of objects with varying shapes [376]. Furthermore, finger gaiting—breaking and remaking contacts to expand the manipulation workspace—has been achieved through compliance-enabled planning, allowing for full $SO(3)$ control of objects within the hand without external support [377].

The bridge between simulation and reality remains a pivotal hurdle, yet recent works demonstrate remarkable progress in Sim2Real transfer for dexterous systems. Techniques such as domain randomization, combined with robust pose estimation and adaptive policies, have enabled vision-based policies trained in simulation to perform agile in-hand manipulation on real hardware, competing with systems utilizing motion capture [378]. Point cloud-based reinforcement learning frameworks like DexPoint have also achieved generalization to new objects in the real world by incorporating imagined hand point clouds and contact-based rewards during training [379]. Additionally, methods that leverage human grasp affordances from in-the-wild videos, such as DexVIP, allow policies to learn from free-form visual data, sidestepping the need for expensive laboratory demonstrations and accelerating the acquisition of dexterous skills [380].

Finally, the exploration of bimanual coordination and dynamic interactions opens new possibilities for complex physical tasks that mirror human dexterity. Systems capable of throwing and catching objects with bimanual hands utilize multi-agent RL and trajectory prediction models to handle high-speed dynamic actions, demonstrating significant improvements over baseline methods [381]. Similarly, learning visuotactile skills with two multifingered hands has been facilitated by low-cost teleoperation systems and repurposed prosthetic hardware, enabling the collection of rich multimodal datasets for long-horizon, high-precision tasks [382]. These collective advancements underscore a paradigm shift towards generalist embodied agents capable of executing dexterous, long-horizon, and physically complex interactions in real-world settings. As these capabilities mature, the focus naturally shifts to how we rigorously evaluate whether these agents truly understand the linguistic instructions guiding such sophisticated behaviors, a topic addressed in the following section on benchmarking and evaluation.

6.7 Benchmarking and Evaluation of Language-Guided Manipulation

6.7 Benchmarking and Evaluation of Language-Guided Manipulation

Building upon the advancements in dexterous manipulation and complex physical interactions described previously, the rapid integration of Vision-Language Models (VLMs) into embodied agents necessitates a rigorous re-evaluation of how we benchmark and assess language-guided manipulation systems. While early evaluations focused primarily on task success rates in controlled environments, the field is increasingly recognizing the need for multidimensional metrics that capture compositionality, generalization capabilities, and the integration of physical commonsense. A critical challenge in this domain is the discrepancy between benchmark performance and true model understanding, where models may exploit dataset biases rather than demonstrating robust reasoning. For instance, research into vision-language compositionality has revealed that many existing benchmarks are “hackable,” allowing blind models to outperform state-of-the-art systems by leveraging syntactic biases rather than visual grounding [383]. This finding underscores the urgent need for adversarial refinement mechanisms and debiased datasets to ensure that evaluation metrics truly reflect an agent’s ability to parse complex linguistic constraints and map them to the physical actions enabled by dexterous hardware.

To address the specific demands of robotic manipulation, recent efforts have introduced bench-

marks designed to test fine-grained instruction following and compositional generalization. The VLMbench represents a significant step forward, offering a comprehensive suite of tasks that require agents to execute diverse manipulation commands involving various object shapes, action types, and motion constraints [384]. Unlike simpler navigation tasks, manipulation requires precise adherence to spatial and physical modifiers, such as “keep it upright” or “grasp by the handle.” However, evaluating whether a model genuinely understands these nuances remains difficult. Studies on compositional generalization in grounded language understanding indicate that even state-of-the-art models often fail dramatically when required to apply systematic compositional rules to novel situations, such as interpreting new combinations of adjectives and verbs [385]. This limitation suggests that current benchmarks must evolve to include more rigorous tests of structural generalization, where agents must interpret syntactic structures and semantic combinations never encountered during training [386].

Furthermore, the assessment of physical commonsense in manipulation tasks is paramount for deploying agents in real-world settings, particularly when interacting with the articulated and deformable objects discussed in the preceding section. Models must not only follow linguistic instructions but also possess an intuitive understanding of physics, such as object fragility, stability, and affordances. While benchmarks like RICA have been developed to evaluate robust inference capabilities based on commonsense axioms in textual domains, extending these evaluations to the visuo-motor realm is an emerging frontier [387]. Current evaluations often overlook whether a model’s successful action execution stems from genuine physical reasoning or statistical correlation. The lack of datasets that explicitly probe for physical commonsense—such as predicting the outcome of a push or the stability of a stack under specific linguistic constraints—creates a significant gap in our ability to certify the reliability of embodied agents. Research highlights that fine-tuned models often struggle to generalize their commonsense reasoning when faced with even moderate changes in experimental setups, suggesting susceptibility to dataset bias [388].

Another critical dimension of evaluation is the robustness of models to instruction ambiguity and variation. In real-world human-robot interaction, instructions are rarely perfect; they may be ambiguous, under-specified, or phrased in novel ways. The AmbiBench framework investigates how models handle task ambiguity, revealing that while scaling and human feedback training improve performance, models still lag behind humans in deducing intended behaviors from poorly specified tasks [389]. Similarly, evaluations of instruction-following capabilities have shown that models are sensitive to the specific phrasing of instructions, with performance degrading significantly when faced with novel but semantically equivalent phrasings [390]. This sensitivity poses a severe risk for manipulation tasks where slight misinterpretations can lead to catastrophic failures, especially in high-stakes dexterous scenarios. To mitigate this, new metrics like the Decomposed Requirements Following Ratio (DRFR) have been proposed to break down complex instructions into simpler criteria, allowing for a more granular analysis of where and why a model fails to comply with specific constraints [391].

Despite these advances, significant gaps remain in current evaluation landscapes, particularly regarding fine-grained instructions and long-horizon dependencies. Many existing benchmarks rely on single-turn interactions or simplified action spaces that do not reflect the complexity of real-world manipulation involving the sophisticated control strategies mentioned earlier. The need for benchmarks that assess multi-turn dialogue and iterative error correction in manipulation scenarios is evident, as agents must often ask clarifying questions or adapt their strategies based on feedback. Frameworks like GVCCI attempt to address the domain gap in visual grounding for manipulation by leveraging lifelong learning and synthetic data generation, yet comprehensive benchmarks for

evaluating such continuous adaptation are still scarce [392]. Moreover, the issue of sample variance in visually grounded generation highlights the instability of current evaluation protocols, where human references can vary drastically, affecting the reliability of metrics like CIDEr [393].

Future directions for benchmarking must prioritize the creation of datasets that challenge models with complex, multi-constraint instructions and require deep physical reasoning. The CoDI-Eval benchmark, which evaluates controllable generation under diversified instructions, offers a template for constructing test suites that probe the limits of constraint satisfaction [394]. Additionally, as models become more capable, evaluation methods must shift from static success rates to dynamic assessments of safety, efficiency, and explainability. The development of fine-grained evaluation protocols like FLASK, which decomposes scoring into specific alignment skill sets, provides a promising pathway for holistically assessing the diverse capabilities required for successful language-guided manipulation [395]. Ultimately, bridging the gap between benchmark performance and real-world deployability will require a concerted effort to design evaluations that are robust to bias, sensitive to nuance, and deeply grounded in the physics of interaction, ensuring that the theoretical capabilities of VLMs translate effectively into the dexterous, reliable agents envisioned for future applications.

7 Interactive Agents, Memory, and Active Perception

7.1 Egocentric Memory Architectures for Context Retention

7.1 Egocentric Memory Architectures for Context Retention

The transition from static perception to dynamic embodied reasoning necessitates a fundamental shift in how artificial agents manage information over time. Unlike traditional computer vision systems that process isolated frames, embodied agents operate within a continuous perception-cognition-action loop where the current state is inextricably linked to historical interactions. To support long-horizon task execution, modern frameworks are increasingly adopting centralized, language-based memory modules that unify perception, planning, and control. These egocentric memory architectures serve as the cognitive backbone for agents, enabling them to maintain episodic records of past observations and actions, thereby grounding abstract instructions in the concrete reality of their lived experience. This foundational capacity for context retention is a prerequisite for the more interactive behaviors discussed later, as an agent must first possess a coherent history before it can effectively identify gaps in its knowledge or formulate strategic inquiries.

At the core of these architectures is the integration of Large Language Models (LLMs) as a unifying “brain” that manages egocentric memory alongside control mechanisms. The framework known as LLM-Brain exemplifies this approach by utilizing a large-scale language model to unify egocentric memory and control through closed-loop, multi-round dialogues [222]. In this paradigm, all components—including perception, planning, and memory—communicate via natural language, allowing the agent to maintain a coherent narrative of its journey. This linguistic unification is crucial because it allows the agent to reason about its own history using the same semantic space employed for high-level planning, effectively bridging the gap between low-level sensorimotor data and high-level cognitive abstraction. Similarly, the Multimodal Embodied Interactive Agent (MEIA) introduces a Multimodal Environment Memory (MEM) module specifically designed to integrate embodied control with large models through visual-language memory of scenes [396]. By storing scene information in a multimodal format, MEIA ensures that action planning remains aligned with the spatial and temporal context of the environment, overcoming the limitations of unimodal memory representations that often fail to capture the nuances of dynamic interactions.

The structural organization of these memory systems often draws inspiration from cognitive science, distinguishing between different types of memory to optimize retrieval and retention. For instance, the Self-Controlled Memory (SCM) framework enhances LLMs by introducing a memory stream and a dedicated memory controller that determines when and how to utilize stored memories [397]. This architecture allows agents to process ultra-long contexts without fine-tuning, effectively mitigating the loss of critical historical information that plagues standard context-window-limited models. Furthermore, the concept of a “Working Memory Hub” has been proposed to provide greater continuity for nuanced contextual reasoning, addressing the isolation of distinct dialog episodes found in traditional LLM designs [199]. By incorporating an episodic buffer, such systems can retain memories across episodes, facilitating the complex reasoning required for collaborative scenarios and intricate manipulation tasks.

A critical function of egocentric memory is the ability to perform “mental time travel,” recalling specific past events in detail to inform current decision-making. The Hierarchical Chunk Attention Memory (HCAM) architecture enables this by dividing the past into chunks and performing high-level attention over coarse summaries before drilling down into relevant details [398]. This mechanism allows agents to remember where objects were hidden or how specific environmental features changed over time, significantly outperforming agents with flat memory architectures in tasks requiring long-term recall. Complementing this, the Structured Scene Memory (SSM) for vision-language navigation offers a compartmentalized architecture that accurately memorizes percepts while disentangling visual and geometric cues [240]. SSM employs a collect-read controller that adaptively gathers information to support current decisions, mimicking iterative algorithms to facilitate long-range reasoning in complex 3D environments.

Beyond simple storage, advanced egocentric memory systems actively leverage past experiences to refine future planning through retrieval-augmented mechanisms. The Retrieval-Augmented Planning (RAP) framework dynamically leverages past experiences corresponding to the current situation, enhancing the planning capabilities of multimodal LLM agents [399]. By reflecting on past successes and failures, agents can avoid repeating errors and adapt their strategies to novel situations. This reflective capability is further enhanced by frameworks like “Inner Monologue,” where agents use environment feedback to form a continuous stream of consciousness that updates their internal state and guides subsequent actions [186]. Such closed-loop feedback mechanisms ensure that the memory is not merely a static log but an active component of the reasoning process, allowing for adaptive replanning when initial strategies fail. It is precisely this ability to monitor one’s own state and history that empowers an agent to recognize uncertainty, setting the stage for the active inquiry strategies explored in the following section.

Theoretical underpinnings for these architectures often revisit the distinction between episodic and semantic memory. The “Tensor Brain” theory posits that perception, episodic memory, and semantic memory are realized through different operational modes of oscillating interactions between symbolic and subsymbolic layers, emphasizing that memory’s primary purpose is to support the agent in the present and future [400]. Similarly, dual-memory architectures have been proposed to handle lifelong learning, where one network learns fine-grained spatiotemporal representations of object instances (episodic) while another develops compact category representations (semantic) [401]. This separation allows agents to prevent catastrophic forgetting while continuously acquiring new knowledge from streaming sensory data.

Finally, the scalability of these memory systems to multi-agent and open-world settings remains a pivotal area of development. Frameworks like CoELA demonstrate how LLMs can be incorporated into modular cognitive architectures to facilitate cooperation among multiple embodied agents, uti-

lizing shared memory and communication protocols to achieve long-horizon goals [228]. Meanwhile, systems like LARP address the need for consistent long-term memory in open-world simulations, ensuring that agents maintain coherent personalities and action histories over extended periods [402]. As these architectures evolve, the integration of robust, egocentric memory modules will remain central to developing embodied agents capable of genuine autonomy, adaptability, and deep contextual understanding in unstructured real-world environments, providing the essential substrate upon which active clarification and inquiry strategies are built.

7.2 Active Inquiry and Clarification Strategies

7.2 Active Inquiry and Clarification Strategies

Building upon the robust egocentric memory architectures discussed previously, which provide agents with a coherent history of past interactions, the next critical evolutionary step is the ability to proactively seek information to resolve current uncertainties. While memory ensures an agent retains context, active inquiry and clarification strategies empower it to identify gaps in that context before executing physical actions. In dynamic, partially observable environments, semantic ambiguities and missing sensory data can lead to catastrophic failures or inefficient exploration if left unaddressed. By engaging in dialog with humans or querying external knowledge sources, embodied agents can close the gap between high-level intent and low-level execution. This paradigm shifts the agent from a passive executor reliant on static instructions to an active participant in the decision-making loop, optimizing performance by gathering necessary context prior to action commitment. This proactive stance serves as a crucial bridge: it leverages the historical context stored in memory to detect knowledge gaps, setting the stage for the reactive error correction mechanisms required when actions inevitably fail in the real world, as explored in the subsequent section.

A primary challenge in active inquiry is determining the precise moment when clarification is necessary. Asking questions for every ambiguity can severely hamper user experience and interrupt the flow of interaction, while failing to ask when needed can result in incorrect actions. Research has demonstrated that agents must leverage contextual signals and hypothesis ambiguity to trigger clarifying dialogs only when essential for user satisfaction. For instance, neural self-attentive models have been proposed to detect various types of ambiguities, such as intent or automatic speech recognition errors, ensuring that questions are posed strategically rather than indiscriminately [403]. Furthermore, uncertainty estimation plays a pivotal role in this decision process. Approaches like “intent-sim” estimate the entropy over user intents to quantify the utility of querying for clarification, showing that targeting only the most uncertain predictions can double performance gains compared to random selection [404]. This selective questioning mechanism is crucial for maintaining efficiency in large-scale conversational and embodied systems, ensuring that the agent’s cognitive resources are allocated to resolving genuine ambiguities rather than redundant confirmations.

Once the need for clarification is established, the agent must generate questions that effectively resolve the ambiguity. In situated environments, referential ambiguity often arises when natural language instructions do not uniquely identify an object or location. To address this, agents can employ decision-theoretic models that utilize entropy-based utility assignments to formulate questions across multiple modalities. Empirical studies involving virtual robots organizing tools have revealed that such computational models outperform simple slot-filling baselines by adapting question types to the specific nature of the ambiguity and dialogue context [405]. Moreover, the ability to generate polar (yes-no) questions without supervised data has been achieved by leveraging expected information gain objectives, allowing agents to improve communicative success in goal-

oriented tasks even with off-the-shelf image captioners [406]. These generative capabilities ensure that the inquiry process is not only triggered appropriately but also yields high-value information that directly facilitates task completion.

In embodied decision-making, the concept of “Asking Before Acting” (ABA) has emerged as a powerful framework to enhance efficiency in unfamiliar environments. Large Language Model (LLM) based agents often struggle with vague instructions or unknown environmental layouts, leading to suboptimal trial-and-error behaviors. By empowering agents to proactively inquire with external sources or humans for pertinent information using natural language, ABA enables them to circumvent laborious exploration steps. Experiments across text-based household tasks and real-world robot arm manipulation have shown that even modest prompt modifications allowing for inquiry can yield substantial advantages in both performance and efficiency over baseline agents that act immediately [407]. This approach is particularly effective when combined with fine-tuning on reformulated metadata, which helps the agent learn the rationale behind asking specific questions [407]. By integrating this strategic pauses for inquiry, agents can significantly reduce the likelihood of execution errors, thereby minimizing the need for the extensive recovery loops discussed later.

For mobile robots, the challenge of disambiguating referred objects is compounded by the lack of a fixed viewpoint. Systems like DoRO (Disambiguation of Referred Object) address this by aggregating observations from multiple views while exploring an area, then raising verbose queries based on grounded object instances. This method ensures that the robot conveys the specific type of ambiguity it faces, facilitating more accurate human intervention and task grounding [408]. Similarly, in navigation tasks, agents can leverage grounded question answering to drive curiosity-based exploration. By formulating questions about physical properties or spatial relationships and treating the changes in answers as intrinsic rewards, agents can uncover structured knowledge about their environment, solving sparse-reward tasks more efficiently than those relying on holistic state transitions [409]. These approaches highlight how active inquiry can be intrinsically linked to perception and movement, transforming questioning into a tool for spatial understanding and environmental mapping.

The integration of human assistance through active sensing further exemplifies the power of clarification strategies. Frameworks such as HARPS (Human Assisted Robotic Planning and Sensing) combine online sampling-based Partially Observable Markov Decision Process (POMDP) policies with multimodal semantic interaction. This allows robotic agents to actively query humans for novel semantic data, such as labeling arbitrary landmarks, thereby improving belief states in uncertain environments and significantly increasing target capture rates in search and rescue scenarios [410]. Additionally, research into interactive visual reasoning under uncertainty highlights the human-like capability to generate hypotheses and test them via active trials. Environments like IVRE challenge agents to propose effective experiments to validate hypotheses about action-effect pairs, revealing that current learning methods often lag behind human efficacy in resolving ambiguities through interaction [411]. Such frameworks demonstrate that effective embodiment requires a tight coupling between planning, sensing, and communicative action.

Finally, the quality of generated questions is paramount. Effective clarification requires identifying what information is missing and useful by comparing local context against global knowledge. Models that adopt this “global view” approach have been shown to outperform baselines by generating questions that pinpoint specific gaps in understanding [412]. Moreover, the alignment between model uncertainty and human clarification-seeking behavior is not always direct; thus, generating questions based on rigorous model uncertainty estimation rather than mimicking human patterns alone leads to significant improvements in task success [413]. As embodied agents continue to

evolve, the synthesis of uncertainty estimation, strategic questioning, and multimodal grounding will remain central to developing systems capable of seamless and safe interaction in complex, open-world environments. While these strategies aim to prevent errors proactively, they operate in tandem with the reactive feedback loops and incremental error correction mechanisms necessary for handling unforeseen execution failures, forming a comprehensive cycle of autonomous reasoning.

7.3 Feedback Loops and Incremental Error Correction

7.3 Feedback Loops and Incremental Error Correction

While active inquiry strategies discussed in the previous section enable agents to resolve ambiguities *before* acting, the dynamic nature of real-world environments inevitably leads to execution failures that require robust mechanisms for real-time detection and correction. Embodied reasoning operates within a continuous perception-cognition-action loop where errors arise from model inaccuracies, environmental disturbances, or unforeseen physical constraints. Consequently, the ability to leverage feedback loops—ranging from internal self-monitoring to external human corrections—is paramount for achieving long-horizon autonomy. A critical advancement in this domain is the concept of “inner monologue,” where agents utilize language models to process environmental feedback, such as success detection and scene descriptions, to refine their planning strategies without requiring additional training. By maintaining a closed-loop dialogue with the environment, agents form a continuous reasoning stream that significantly improves instruction completion rates across complex manipulation and mobile manipulation tasks [186]. This approach transforms the agent from a passive executor into an introspective entity capable of adapting its trajectory based on immediate sensory evidence, effectively bridging the gap between high-level intent and low-level reality.

Beyond internal reasoning, incorporating human feedback into the learning loop offers a powerful pathway for incremental policy adaptation. Traditional reinforcement learning often struggles with the sample inefficiency of learning from scratch in new environments; however, systems that can distill generalizable knowledge from online human corrections demonstrate rapid adaptation. For instance, frameworks that employ large language models to interpret arbitrary forms of language feedback—spanning high-level task plan corrections to low-level skill parameter adjustments—can effectively retrieve relevant past experiences to handle novel settings. Such systems have been shown to reduce the intervention rate drastically, often requiring minimal corrections after just a few iterations of interaction [414]. Furthermore, the integration of human feedback need not be limited to explicit corrections; implicit feedback derived from user interactions can also optimize system policies. Methodologies that leverage implicit exploration and unbiased counterfactual estimators allow agents to learn from the outcomes of decisions they did not explicitly take, thereby improving policy performance in production scenarios without deploying untested strategies [415].

A sophisticated layer of error correction involves the development of failure-aware policies that move beyond simple process-of-elimination strategies. When an action fails, the dependency between the failure and remaining untried actions contains valuable information that can guide subsequent decision-making. By representing self-assessment results of previous failures as variable states and utilizing recurrent neural networks to memorize these events, agents can learn policies that achieve higher success rates with fewer trials compared to methods that merely discard failed actions [416]. This capacity for self-assessment is further enhanced by frameworks that diagnose execution failures against learned constraints. By identifying which parameters violate precondition models, robots can generate synthetic data to correct their execution models, effectively turning failed experiences

into opportunities for improving future grasp and manipulation success [417].

In scenarios involving external disturbances, robust recovery mechanisms are essential for maintaining autonomy. Agents equipped with introspection systems can classify anomalies using non-parametric statistics and deploy specific recovery policies, such as re-enactment or adaptation, to resolve accidental and persistent failures. These systems, often built upon graph-based motion generation, enable robots to execute around-the-clock introspection and even elicit self-recovery when initial misclassifications occur [208]. Similarly, for contact-rich sequential manipulation tasks where hand-engineered recovery strategies are infeasible, meta-reasoning algorithms can monitor the progress of multiple recovery policies and allocate training resources to those most likely to improve task performance, substantially boosting success rates in both simulation and real-world deployments [418].

The challenge of incremental learning is compounded by the risk of catastrophic forgetting, where acquiring new knowledge degrades previously learned skills. To address this, novel architectures inspired by biological processes, such as the thalamocortical circuit, utilize optimization at inference time to dynamically generate internal representations of current tasks. By alternating between updating model weights and latent task embeddings, these agents can parse temporal experiences into discrete events, mitigating forgetting while organizing knowledge into flexible structures [419]. Additionally, intentional forgetting mechanisms have been proposed to systematically prioritize the adaptation of key problem-solving skills by deliberately erasing minor behaviors, thereby reducing healing time and enabling agents to adapt to drifted environments more effectively [420].

Human-in-the-loop frameworks further refine this adaptive capability by allowing users to identify personalized task-irrelevant concepts through counterfactual demonstrations. This interactive approach not only helps users understand agent failures but also reduces the number of demonstrations required for fine-tuning, aligning agent behavior with individual user preferences [421]. Moreover, systems that maintain a dynamic memory of user feedback can continually improve their reasoning capabilities over time without model retraining. By retrieving corrections from memory as additional context, these agents avoid repeating previous mistakes, demonstrating significant performance gains after limited teaching interactions [422]. Finally, the integration of language corrections into hierarchical policies enables robots to adapt to real-time verbal feedback, incorporating fine-grained suggestions into an iterative training scheme that enhances both low-level execution and high-level decision-making [423]. Collectively, these diverse mechanisms illustrate a shifting paradigm where embodied agents are no longer static learners but dynamic systems capable of continuous self-improvement. As agents become more proficient at correcting their own errors and adapting to human guidance, the focus naturally shifts toward the nuances of the interaction itself, specifically how these corrections are communicated within the fluid turn-taking and social dynamics of human-robot collaboration, as explored in the following section.

7.4 Multimodal Turn-Taking and Social Interaction Dynamics

7.4 Multimodal Turn-Taking and Social Interaction Dynamics

As agents evolve from static learners capable of self-correction into dynamic systems engaging in continuous adaptation, the focus naturally shifts to the nuances of the interaction itself. While the preceding section detailed mechanisms for error correction and policy refinement through feedback loops, successful deployment in human-centric environments demands more than just functional accuracy; it requires the ability to participate in natural, fluid conversational flows. This subsection surveys the evolving landscape of techniques designed to model these social dynamics, emphasizing

the critical transition from unimodal text processing to sophisticated multimodal fusion architectures that capture the subtleties of turn-taking, non-verbal cues, and collaborative intent. These capabilities form the essential bridge between the immediate reactive corrections discussed previously and the long-term personalized memory systems explored in the following section.

Turn-taking, the fundamental mechanism governing who speaks when, presents a significant challenge for robotics due to the inherent ambiguity and rapid timing of human communication. Traditional approaches often relied on simplistic pauses or single-modality features, which frequently fail in noisy, real-world scenarios characterized by class imbalance where short pauses are often misinterpreted as turn endings. To address this, recent research has introduced advanced fusion mechanisms. For instance, [424] proposes a novel gated multimodal fusion mechanism that seamlessly utilizes information from both speech and text modalities. By employing contrastive learning and a specialized data augmentation strategy to construct negative instances, this approach effectively tackles data imbalance, demonstrating superior competitiveness over state-of-the-art baselines in predicting when a human partner intends to yield or hold the floor. Furthermore, the integration of Large Language Models (LLMs) with acoustic models has opened new avenues for continuous prediction. [425] illustrates how fusing neural acoustic models with LLMs, enhanced by multi-task instruction fine-tuning, allows agents to better understand conversational context and predict backchanneling locations, leading to more natural interactions between humans and speech-enabled AI agents.

Beyond simple turn prediction, the ability to anticipate transitions early is critical for proactive collaboration, particularly in time-sensitive domains like surgical assistance. [426] introduces the Cognitive Turn-taking Model (CTTM), which leverages spiking neural networks to process implicit and explicit multimodal cues. This framework achieves early prediction of human turn-taking intentions, outperforming human baselines when presented with partial observations, thereby enabling robots to initiate actions with minimal latency. Similarly, [427] highlights the necessity of such proactive behavior in physical human-robot collaboration, showing that algorithms capable of comprehending multimodal cues can achieve high accuracy in predicting turn transitions even before the human action is fully completed, thus smoothing the collaborative workflow.

Social interaction dynamics extend beyond turn-taking to include the interpretation of non-verbal gestures, gaze, and body posture, which are essential for grounding communication in physical reality. In collaborative tasks, disambiguating referring expressions often requires synthesizing temporal dependencies across multiple modalities. [428] formulates a hypothesis that modeling the temporal dependencies of events across head movements, hand gestures, and speech significantly increases predictive power for disambiguating fetching requests. Moreover, the complexity of group settings requires robust representations of social dynamics that go beyond dyadic interactions. [429] proposes using adapted Temporal Graph Networks to represent social interaction dynamics by incorporating gaze, voice activity, and environmental context. This method treats social dynamics as a link prediction problem, achieving substantial improvements in next-speaker prediction while utilizing a highly efficient message-passing mechanism.

The generation of appropriate non-verbal responses is equally vital for maintaining social presence. Agents must not only perceive but also produce synchronized gestures and facial expressions that align with their verbal output and the interlocutor’s state. [430] introduces a probabilistic method based on normalizing flows to synthesize facial gestures that are aware of the interlocutor’s multimodal cues, avoiding the repetitive motions typical of deterministic systems. Similarly, [431] utilizes a diffusion-based approach to generate gestures that are semantically aligned with utterances, offering controllability to emphasize specific words and adapting to both monologue and conversational contexts. The importance of grounding these gestures in semantic and affective information is fur-

ther underscored by [432], which argues for theory-driven approaches that combine semantic and affective inputs to generate robot gestures with appropriate shape, intensity, and speed.

Furthermore, the physical embodiment of the agent allows for unique forms of social signaling, such as deictic gestures and gaze, which can direct human attention and clarify intent. [433] demonstrates that an anthropomorphic robot communicating intent through gaze and pointing gestures leads to more natural fixation behavior and faster reaction times compared to verbal-only modes. In navigation tasks, interpreting human gestures is crucial for social compliance. [434] combines image-based imitation learning with model-predictive control to allow robots to interpret complex human gestures like “follow me” or directional pointing, generating socially compliant trajectories. The reciprocal nature of these interactions is also evident in [435], where non-verbal signals from the robot, such as arm gestures and light signals, effectively communicate turn transitions to human collaborators in an industrial setting.

Finally, the emergence of foundation models is reshaping how agents manage these complex social loops. [436] presents a system where LLMs regulate robot behavior by coordinating limb movements, head orientation, and speech to produce dynamic, multi-modal expressions based on high-level linguistic guidance. This shift towards using LLMs as central controllers for social behavior is complemented by efforts to resolve cross-modal conflicts. [437] adopts a neurorobotic paradigm to handle incongruent audio-visual inputs, enabling robots to express human-like social attention similar to biological agents. As the field advances, the integration of these diverse multi-modal capabilities—from early turn-taking prediction to generative gesture synthesis and conflict resolution—forms the backbone of truly socially intelligent embodied agents. However, for these agents to sustain such high-fidelity interactions over extended periods, they must move beyond immediate social cues to retain and leverage historical interaction data, a challenge addressed by the retrieval-augmented personalization and lifelong learning strategies discussed in the following section.

7.5 Retrieval-Augmented Personalization and Lifelong Learning

7.5 Retrieval-Augmented Personalization and Lifelong Learning

While the preceding section detailed how agents manage the immediate temporal dynamics of social interaction, this subsection addresses the critical challenge of maintaining consistency and adaptability over extended time horizons. The transition from static, one-size-fits-all embodied agents to adaptive, user-centric companions necessitates a fundamental shift in how agents store, retrieve, and utilize historical interaction data. Traditional large language model (LLM) agents often operate with limited context windows or implicit parameter-based memory, constraining their ability to maintain long-term consistency and adapt to individual user idiosyncrasies. To overcome these limitations, recent research has pivoted toward retrieval-augmented architectures that leverage external memory banks. These systems enable agents to dynamically retrieve relevant past dialogue-action pairs, facilitating personalized behaviors and the continuous expansion of procedural knowledge without catastrophic forgetting.

A primary challenge in personalization is the accurate representation of user profiles. Early approaches relied on explicit, hand-crafted persona descriptions, which are labor-intensive to create and often fail to capture the dynamic nature of human preferences. In response, modern frameworks have begun to construct implicit user profiles automatically from large-scale dialogue history. For instance, methods have been proposed that utilize key-value memory networks to store historical post-response pairs, allowing the agent to build a dynamic profile that reflects what and how

a specific user has responded to similar situations in the past [438]. By retrieving these personalized wording behaviors and integrating them via customized attention mechanisms, agents can significantly improve response selection accuracy and maintain consistency with the user’s unique communication style [439]. Furthermore, to address the “stability-plasticity” dilemma inherent in lifelong learning, some frameworks incorporate memory updating mechanisms inspired by the Ebbinghaus Forgetting Curve. This allows the agent to selectively reinforce significant memories while allowing less relevant information to fade, mimicking human-like memory consolidation and ensuring that the agent adapts to evolving user personalities over time [440].

In the domain of embodied reasoning, where agents must execute physical tasks based on natural language instructions, retrieval-augmented personalization becomes critical for interpreting open-domain commands and adapting to user-specific routines. The HELPER framework exemplifies this approach by equipping an embodied agent with an external memory of language-program pairs. When faced with a new instruction, HELPER retrieves relevant memories based on the current dialogue context, visual description, or user corrections, using them as in-context examples to guide the LLM in generating appropriate action programs. Crucially, this memory is expanded during deployment; as the agent successfully executes tasks or receives corrections, it stores new language-action mappings, effectively personalizing its behavior to the user’s specific procedures and vocabulary [441]. This capability is essential for handling idiosyncratic instructions that were not present during initial prompt engineering, enabling the agent to learn from interaction rather than relying solely on pre-training.

Beyond simple retrieval, advanced systems employ sophisticated learning strategies to refine their action spaces and procedural knowledge continuously. The concept of “In-Memory Learning” proposes a declarative framework where agents distill insights from past experiences directly within memory components, updating existing notes to enhance future performance without requiring re-training of the underlying model parameters [442]. Similarly, the LearnAct framework introduces an iterative learning strategy where agents revise and update their available actions in the form of code functions based on errors identified in unsuccessful tasks. This open-action learning allows embodied agents to expand their skill set experientially, transforming trial-and-error failures into permanent procedural knowledge that can be retrieved for future similar tasks [443].

The integration of retrieval mechanisms also plays a pivotal role in mitigating hallucinations and ensuring grounded decision-making in complex environments. By augmenting LLMs with explicit action knowledge bases, agents can constrain their planning trajectories to known, valid actions, thereby reducing planning hallucinations and improving task success rates in benchmarks like ALF-World [444]. Moreover, in scenarios involving multi-turn interactions and complex web or mobile navigation, self-reflective memory-augmented planning frameworks utilize memory to maintain context across long horizons, allowing agents to reflect on past failures and adjust their strategies accordingly [445]. This is particularly relevant for mobile agents that must align with personalized user needs and standard operating procedures (SOPs); by integrating SOP information into in-context learning and retrieving user-specific confirmations for sensitive operations, agents can achieve high success rates while respecting privacy and personal preferences [446].

Furthermore, the ability to retrieve and utilize heterogeneous information sources enhances the agent’s capacity for lifelong adaptation. Frameworks like Heterogeneous Memory Networks (HMNs) demonstrate the efficacy of simultaneously utilizing user utterances, dialogue history, and background knowledge tuples stored in distinct memory structures. This allows the decoder to re-currently select relevant information from context-aware and context-free memories, significantly outperforming models that rely on a single memory source [447]. In embodied settings, this trans-

lates to an agent that can recall not only what a user said but also the physical context and relevant world knowledge required to execute a task. Additionally, approaches that reformulate dialogue state tracking as example-guided question answering leverage retrieval to minimize task shifts between services, enabling continual learning where the agent adapts to new domains without forgetting previously learned services [448].

Ultimately, the synergy between retrieval-augmented generation and external memory architectures provides a robust pathway toward truly generalist embodied agents. By decoupling knowledge storage from model parameters, these systems allow for infinite scalability of procedural knowledge and deep personalization. Whether through the dynamic updating of user profiles [449], the iterative refinement of action libraries [443], or the strategic retrieval of past successes to guide future planning [441], these memory-augmented approaches are redefining the boundaries of what embodied agents can achieve in open-world, long-horizon interactions. As these technologies mature and establish the internal mechanisms for personalized, long-term competence, the field must concurrently develop rigorous methodologies to evaluate these capabilities, a topic explored in the following section regarding benchmarks and evaluation protocols.

7.6 Benchmarks and Evaluation of Interactive Embodied Dialogue

7.6 Benchmarks and Evaluation of Interactive Embodied Dialogue

Building upon the memory-augmented architectures and personalized learning mechanisms detailed in the previous section, the field now faces the critical challenge of rigorously evaluating these advanced capabilities. The assessment of interactive embodied agents represents a complex frontier where the measurement of multi-turn dialogue is inextricably linked to physical action execution and environmental grounding. Unlike static conversational agents, embodied systems must be evaluated not only on linguistic fluency but also on their ability to execute tasks, maintain spatial awareness, and recover from execution failures through dialogue. A primary hurdle in this domain is the misalignment between traditional imitation learning metrics and the actual objectives of embodied dialogue research. Recent analyses suggest that models trained solely with imitation learning often exhibit spurious actions during evaluation and fail to effectively ground query utterances, which are essential for successful task completion [450]. Consequently, the field is shifting towards evaluation frameworks that prioritize higher-level semantic goals and task success rates over low-level token matching or action replication, ensuring that the long-term adaptability discussed previously translates into reliable real-world performance.

To address the multifaceted nature of embodied interaction, researchers have developed specialized benchmarks that integrate language, vision, and action. The TEACH dataset stands out as a foundational resource, providing over 3,000 human-human interactive dialogues set within simulated household environments. This dataset enables the assessment of three distinct benchmarks: Execution from Dialog History, Trajectory from Dialog, and Two-Agent Task Completion, allowing researchers to measure an agent’s capability in dialogue understanding, language grounding, and physical task execution [451]. Building upon such datasets, there is a growing emphasis on annotating dialogues with specific “dialog acts” to model the functional intent behind utterances. By utilizing annotated datasets like TEACH-DA, which includes over 39,500 utterances tagged with dialog acts, researchers can train models to predict appropriate non-dialog behaviors based on communicative intent. Experiments indicate that incorporating these dialog acts can improve end-task success rates by up to 2 points compared to systems that lack such structured guidance [452].

Beyond task success, evaluating the coherence and robustness of the dialogue itself is paramount,

particularly for agents designed for long-horizon interactions. Traditional turn-based metrics often fail to capture the dynamics of extended engagements, leading to the development of more sophisticated evaluation protocols. For instance, the Standardised Test Suite (STS) offers a novel approach by mining behavioral scenarios from real human interaction data and replaying them to agents offline. This method allows for controlled, interpretable, and rapid ranking of agents based on their success in completing naturalistic interaction continuations, effectively bridging the gap between slow online human evaluation and unreliable proxy metrics [453]. Furthermore, the importance of interactive evaluation cannot be overstated; static post-hoc judgments often miss the nuances of real-time cooperation. Methods approximating interactive human evaluation through self-play scenarios, where agents converse with themselves to calculate sentiment and semantic coherence proxies, have shown significant correlation with human-rated quality, providing a scalable alternative for assessing open-domain dialog systems [454].

The ability of an agent to recover from misunderstandings and maintain dialogue coherence under uncertainty constitutes another crucial evaluation dimension, directly reflecting the resilience required for lifelong learning. In embodied settings, agents must detect when a user’s instruction is ambiguous or when their own actions have failed, prompting clarifying questions or corrective strategies. Research into dialogue breakdown detection highlights the utility of semi-supervised learning and data augmentation techniques to identify moments where an agent’s utterance prevents conversation continuation, a capability vital for maintaining user trust [455]. Moreover, the evaluation of coherence has evolved to include graph-based metrics like GRADE, which model topic transitions and fine-grained dialogue flows, offering a more persuasive assessment of communication logic than surface-level features [456]. Similarly, frameworks like DynaEval unify turn-level and dialogue-level evaluation by modeling the entire dialogue as a graph, ensuring that the dynamic dependencies between utterances are accounted for in the final score [457].

Ethical and social dimensions also play an increasingly important role in benchmarking embodied agents, especially as they evolve into personalized companions. As these systems operate in human spaces, their moral integrity and ability to adhere to social norms must be rigorously tested. Resources like the Moral Integrity Corpus (MIC) provide a framework for benchmarking the ethical assumptions and social attributes of conversational agents, ensuring that they do not reflect insensitive or harmful viewpoints [458]. Additionally, psychological metrics derived from established human communication theories—such as emotional entropy, empathy, and linguistic style matching—are being adopted to evaluate the “humanness” and relational quality of embodied agents, offering insights that traditional lexical metrics miss [459].

Finally, the robustness of embodied agents is often tested against their ability to handle non-deterministic user behaviors and diverse dialogue paths, a necessity for agents that must adapt to evolving user preferences. The Conversation Graph (ConvGraph) approach facilitates the evaluation of non-deterministic agents by generating novel dialogue paths for data augmentation and multi-reference training, thereby improving success rates in complex, branching conversations [460]. As the field progresses, the integration of these diverse evaluation strategies—ranging from task-specific success metrics and dialog act analysis to coherence graphs and psychological profiling—will be essential for developing truly generalist embodied agents capable of seamless, robust, and safe interaction in open-world environments. These comprehensive evaluation protocols provide the necessary feedback loop to validate the retrieval-augmented and personalized learning strategies discussed earlier, ensuring that theoretical advances translate into tangible improvements in agent competence.

8 Evaluation Landscapes, Robustness, and Safety

8.1 Multidimensional Benchmarks for Perception-Cognition-Action Chains

8.1 Multidimensional Benchmarks for Perception-Cognition-Action Chains

The evaluation of embodied artificial intelligence has undergone a paradigm shift, moving away from isolated assessments of perception or language understanding toward comprehensive benchmarks that rigorously test the entire perception-cognition-action (PCA) chain. In real-world deployments, an embodied agent must not only recognize objects and parse instructions but also reason about causal relationships, plan over long horizons, and execute precise motor controls while maintaining robustness against environmental noise. To address the complexity of these integrated capabilities, the research community has developed multidimensional benchmark suites that simultaneously challenge an agent’s sensory processing, cognitive reasoning, and physical actuation. These benchmarks are critical for identifying systemic weaknesses that remain hidden when components are evaluated in silos, thereby driving the development of more holistic and reliable embodied systems. Crucially, establishing the fidelity of these PCA chains is a prerequisite for detecting the specific failure modes—such as hallucinations—that will be dissected in the subsequent section.

A primary dimension of modern evaluation is the assessment of long-horizon planning and complex task execution in human-centered environments. Traditional benchmarks often focused on short, atomic tasks, failing to capture the sequential dependencies and memory requirements of daily activities. The [461] benchmark represents a significant leap forward in this domain by defining 1,000 everyday activities grounded in 50 diverse scenes, ranging from houses to offices. Unlike previous datasets, [461] incorporates over 9,000 objects annotated with rich physical and semantic properties, requiring agents to engage in multi-step reasoning and complex manipulation skills. The associated OMNIGIBSON simulation environment supports realistic physics for rigid bodies, deformable objects, and liquids, ensuring that successful task completion depends on a tight coupling between high-level cognitive planning and low-level physical control. Experiments within this framework have revealed that even state-of-the-art solutions struggle with the realism and diversity of these tasks, highlighting the gap between current capabilities and the demands of generalist agents. Complementing this, the [462] benchmark offers a procedurally generated Gridworld environment that preserves the symbolic complexity of household tasks while enabling rapid prototyping. This allows researchers to efficiently evaluate decision-making strategies across countless task variations, serving as a scalable entry point for developing agents capable of handling open-ended, long-horizon challenges.

Beyond task complexity, a crucial aspect of embodied reasoning is the ability to understand and manipulate causal structures within the environment. Agents must infer how their actions influence the state of the world and transfer this knowledge to novel scenarios. The [463] benchmark addresses this by providing a simulation environment where agents construct 3D shapes from blocks, inspired by child development. Its key innovation lies in the ability to intervene on all causal variables, such as object mass, color, and size, allowing for fine-grained control over task distributions. This facilitates the evaluation of an agent’s capacity for causal structure learning and transfer, testing whether policies can generalize when underlying physical factors change. Similarly, the [464] benchmark focuses on the intersection of language grounding and long-horizon reasoning in table-top manipulation. It covers diverse reasoning aspects including arithmetic, spatial relations, and reference resolution, while specifically investigating the modality bridging problem of incorporating observation feedback into Large Language Model-based planning. These benchmarks collectively emphasize that true embodied intelligence requires more than pattern matching; it demands a deep

understanding of cause-and-effect relationships that govern physical interactions, a foundation that, when unstable, often leads to the fabrication of events or relations.

Robustness constitutes the third critical pillar of multidimensional evaluation, particularly regarding the reliability of perception systems under natural corruptions. In safety-critical applications, agents must maintain performance despite sensor failures, adverse weather, or external disturbances. The [465] benchmark is the first comprehensive suite designed to probe the robustness of 3D detectors and segmentors against out-of-distribution scenarios. It introduces eight corruption types stemming from severe weather, internal sensor failures, and external disturbances, revealing that state-of-the-art 3D perception models are often vulnerable to such corruptions despite performing well on clean data. Expanding on the theme of robustness in different modalities, the [466] suite provides a rigorous test for monocular depth estimation models across 18 corruption categories, including lighting conditions and data processing anomalies. These findings underscore that without dedicated robustness evaluations, perception modules may fail catastrophically in real-world deployments, breaking the entire PCA chain. Furthermore, the [467] benchmark pushes the envelope by evaluating agents in dynamically changing environments characterized by unexpected disasters like fire and flood. This tests an agent’s ability to perform rapid situational assessment and adaptive decision-making, moving beyond static environments to simulate the unpredictability of real-world crises where perceptual errors can have severe consequences.

Finally, the integration of these dimensions is evident in benchmarks that assess the full spectrum of embodied capabilities. The [468] explicitly targets the integrated capabilities of Multimodal Large Language Models across autonomous driving, domestic robotics, and open-world games. It requires models to seamlessly integrate perception, cognition, and action in a reasoning chain, offering error localization to scrutinize inaccuracies in specific stages of the process. Additionally, the [469] provides a structured evaluation of causal understanding, which is fundamental for reliable planning and explanation generation. By combining rigorous standards for long-horizon planning, causal reasoning, and perceptual robustness, these multidimensional benchmarks provide the necessary infrastructure to guide the field toward truly generalist and safe embodied agents. They serve not merely as scoreboards but as diagnostic tools that reveal the intricate dependencies between seeing, thinking, and acting in physical worlds, setting the stage for a deeper analysis of how failures in these chains manifest as specific forms of hallucination.

8.2 Quantifying and Categorizing Visual and Linguistic Hallucinations

8.2 Quantifying and Categorizing Visual and Linguistic Hallucinations

While Section 8.1 established multidimensional benchmarks as essential diagnostic tools for the perception-cognition-action (PCA) chain, it also highlighted that instability within these chains often manifests as specific failure modes. The most pervasive and critical of these failures is hallucination, defined as the generation of text that is coherent and plausible yet factually inconsistent with visual input or real-world constraints. As Vision-Language Models (VLMs) are increasingly deployed for embodied tasks such as robotic navigation and manipulation, the ability to rigorously quantify and categorize these errors becomes paramount. Early evaluations often relied on coarse-grained metrics that failed to distinguish between different types of factual inconsistencies. However, building on the diagnostic potential identified in the previous section, recent research has shifted toward developing fine-grained taxonomies and specialized tools to dissect the multifaceted nature of hallucinations, ranging from simple object existence errors to complex narrative fabrications that jeopardize physical safety.

A primary step in addressing this challenge is the establishment of a comprehensive taxonomy that extends beyond the limitations of traditional evaluations. While early categorizations focused heavily on object hallucinations, where models posit the existence of entities not present in the scene, this view is insufficient for capturing the full spectrum of failure modes required for robust embodied reasoning. Recent works have expanded this definition to include attribute hallucinations, where object properties such as color or material are misidentified, and relation hallucinations, which involve incorrect spatial or functional relationships between objects [470]. Furthermore, more subtle and dangerous forms have been identified, such as “event hallucinations,” where models construct entirely fictional narratives or actions around non-existent entities, a critical flaw for agents required to reason about dynamic physical interactions [471]. Another significant category involves counter-common-sense hallucinations, where the model’s internal knowledge contradicts actual visual evidence, leading to physically impossible interpretations that directly undermine the causal reasoning capabilities discussed in Section 8.1 [472]. To further refine these categories, researchers have proposed distinct orientations such as “factual mirage” and “silver lining,” distinguishing between intrinsic errors arising from model limitations and extrinsic errors caused by ambiguous inputs, while also grading them by severity levels from mild to alarming [473].

To operationalize these taxonomies, several advanced benchmarking suites and diagnostic tools have been developed to isolate specific failure mechanisms. One of the most prominent is HallusionBench, which uniquely addresses the entanglement of language hallucination and visual illusion. By utilizing control groups of images and questions, HallusionBench effectively isolates whether a model’s error stems from a lack of visual grounding or a linguistic bias, revealing that even state-of-the-art models struggle significantly with nuanced image-context reasoning [474]. Complementing this is AMBER, an LLM-free multi-dimensional benchmark designed to evaluate both generative and discriminative tasks across existence, attribute, and relation dimensions. AMBER is particularly notable for its low-cost evaluation pipeline that avoids reliance on expensive proprietary models or human annotators for every query, thereby facilitating broader and more frequent assessment of model robustness [475]. These tools serve as the necessary bridge between the high-level PCA benchmarks of the previous section and the detailed failure analysis required to improve agent reliability.

Beyond static benchmarks, dynamic evaluation frameworks have emerged to capture the diversity of hallucination instances under varying conditions, mirroring the robustness challenges outlined in Section 8.1. For instance, VHTest employs text-to-image generative models to synthesize diverse images specifically designed to trigger visual hallucinations in various modes, providing a more comprehensive stress test than static datasets allow [476]. Similarly, the CorrelationQA benchmark quantifies the “instinctive bias” of models by presenting spurious images that are highly relevant but inconsistent with the correct answer, exposing how models often prioritize semantic correlation over factual verification [477]. For open-vocabulary scenarios where fixed class sets are limiting, ALOHa offers a modernized metric that leverages large language models to extract and match groundable objects dynamically, significantly outperforming traditional metrics like CHAIR in identifying hallucinated objects in diverse environments [478].

The quantification of hallucinations has also extended to identifying specific mechanistic causes and bias patterns to inform mitigation strategies. Research has highlighted the role of spurious co-occurrences in training data, leading to frameworks like HalluciDoctor that utilize cross-checking paradigms to detect and eliminate hallucinatory toxicities in instruction datasets [479]. Additionally, studies have uncovered subtle biases, such as the “semantic shift bias” associated with paragraph breaks, where models are prone to hallucinate content following specific formatting to-

kens, suggesting that quantification must account for structural artifacts in generation [480]. To address the difficulty of detecting these errors in real-time, methods like LogicCheckGPT propose using logical closed loops to probe consistency, while others leverage internal model states and eigenvalue analysis to detect uncertainty signals indicative of hallucination without external tools [481; 482].

Ultimately, the landscape of hallucination evaluation is moving from binary accuracy checks to a multidimensional analysis of faithfulness, coverage, and logical consistency, setting the stage for the deeper cognitive assessments in the following section. Frameworks like VALOR-EVAL emphasize the trade-off between being informative and being faithful, proposing metrics that penalize hallucinations while rewarding detailed, accurate descriptions [483]. By integrating these fine-grained taxonomies with robust diagnostic tools, the community is better equipped to not only measure the current limitations of VLMs but also to guide the development of architectures that prioritize factual alignment. This rigorous quantification of hallucinations provides the foundational understanding necessary to evaluate whether an agent possesses genuine physical commonsense, a theme that will be explored in depth in Section 8.3, where the focus shifts from avoiding errors to actively mastering intuitive physics.

8.3 Evaluating Physical Commonsense and Intuitive Physics

8.3 Evaluating Physical Commonsense and Intuitive Physics

While the previous section detailed methods for quantifying factual inconsistencies and hallucinations, ensuring that an embodied agent does not merely avoid errors but actively possesses a robust understanding of intuitive physics and functional commonsense is equally critical. Unlike static recognition tasks, embodied reasoning demands that agents predict the consequences of actions, comprehend object permanence, and infer latent physical properties such as mass, density, and fragility. Consequently, the evaluation landscape has evolved from simple question-answering benchmarks to sophisticated protocols assessing an agent’s ability to reason about causal mechanisms and physical dynamics. A cornerstone of this evaluation is the adaptation of the Violation of Expectation (VoE) paradigm from developmental psychology. The [484] benchmark represents a significant advancement by raising the bar for explanatory capacities; rather than merely detecting impossible events, it challenges models to comprehend underlying physical explanations and reconstruct concealed scenes to visually expound on violations. This approach transcends surface-level pattern matching, requiring agents to maintain a generative model of physical dynamics capable of inferring occluded object states from visual sequences, thereby catalyzing the development of AI endowed with human-like intuitive physics.

Complementing the focus on rigid body dynamics, recent evaluations have expanded to encompass continuum physics and soft body behaviors, which present unique challenges for current machine learning models. The [485] dataset addresses this gap by requiring the inference of diverse physical properties across scenarios involving deforming and flowing objects. Evaluations using [485] reveal that while AI models show promise in standard physical reasoning, they struggle significantly with the nuances of continuum mechanics, highlighting a distinct divide between human and machine intelligence. Furthermore, recognizing that agents must interact with physical systems in real-time rather than merely observing stationary scenes, researchers have developed interactive benchmarks. The [486] framework specifically challenges agents to exhibit multi-step planning and in-situ intervention, where minor timing deviations can lead to task failure. By demanding simultaneous intuitive physical reasoning and dynamic manipulation, [486] exposes a notable performance gap be-

tween existing learning algorithms and human proficiency, emphasizing the imperative for research into agents that learn through active interaction with representative physical scenarios.

Beyond pure physics, effective operation within human environments requires mastery of functional commonsense—understanding not just how objects move, but how they are used, their affordances, and the societal norms governing their manipulation. The [487] dataset provides a rigorous testbed for this capability by leveraging human gameplay walkthroughs from interactive fiction games. Unlike benchmarks relying on factual knowledge retrieval, [487] focuses on assessing functional commonsense rules, requiring models to infer action outcomes through multi-hop reasoning. The significant performance gap observed between large language models and human experts on [487] underscores the difficulty machines face in utilizing functional knowledge to navigate complex, unseen situations without relying on memorized facts. Similarly, the [488] study introduces the CommonSense Object Affordance Task (COAT), which analyzes an agent’s ability to select substitute objects when executing household tasks. By probing contextual dependencies such as safety, appropriateness, and efficiency, this work highlights the intricate decision-making processes required for robust household agency, shifting evaluation metrics toward a more holistic understanding of object utility and physical state.

The evaluation of physical commonsense is increasingly intertwined with human-centric social reasoning, as physical actions often occur within social contexts involving other agents. Understanding the beliefs, intentions, and mental states of others is crucial for safe and effective collaboration. The [489] benchmark tests models’ ability to ground individuals based on context descriptions regarding their mental and physical states, demonstrating that rich visual commonsense and powerful multi-modal integration are essential for such tasks. Moreover, the [490] dataset investigates the capability of AI systems to predict human belief states in object-context scenarios where verbal communication is prohibited, utilizing nonverbal cues such as gaze and gesture. These benchmarks illustrate that true embodied intelligence requires a synthesis of physical reasoning and Theory of Mind, enabling agents to anticipate human actions and adjust their physical interactions accordingly. The convergence of these evaluation strands—intuitive physics, functional affordances, and social cognition—suggests that future progress in embodied AI will depend on developing unified models that seamlessly integrate causal physical understanding with nuanced social awareness. As evidenced by the challenges posed by datasets like [491] and [492], which demand reasoning about complex social interactions grounded in physical environments, bridging these cognitive domains is a prerequisite for the reliable sim-to-real transfer strategies discussed in the subsequent section.

8.4 Bridging the Sim-to-Real Gap and Calibration Strategies

8.4 Bridging the Sim-to-Real Gap and Calibration Strategies

While the preceding section highlighted the necessity of evaluating agents on physical commonsense and intuitive physics, translating these cognitive capabilities from controlled benchmarks to real-world deployment remains one of the most critical hurdles in embodied AI. This “sim-to-real” gap arises from inevitable discrepancies between the synthetic source domain and the physical target domain, encompassing variations in visual appearance, sensor noise, and, most critically, dynamic properties such as friction, mass distribution, and actuator latency. Although simulation offers a safe, scalable, and cost-effective platform for generating vast amounts of training data, policies trained exclusively in simulation often suffer from severe performance degradation when deployed on physical hardware due to these domain shifts. Consequently, bridging this gap requires robust transfer strategies, precise calibration frameworks, and high-fidelity digital twin methodologies to

ensure that the intuitive physics and functional reasoning assessed in Section 8.3 can be reliably executed in open-world scenarios.

A foundational approach to addressing the reality gap is Domain Randomization (DR), which involves training policies across a wide distribution of randomized simulation parameters to induce robustness against unseen physical variations. Theoretical analyses suggest that DR can succeed under mild conditions without requiring real-world training samples, provided the policy leverages memory or history-dependent strategies to infer latent domain parameters [40]. Empirical successes in robotic manipulation and locomotion have demonstrated that policies trained with dynamics randomization can generalize to real robots despite significant calibration errors and modeling inaccuracies [493]. However, standard DR often relies on fixed, heuristic distributions that may not optimally cover the target domain, leading to overly conservative policies or inefficient use of the training budget. To address this, advanced methods like Bayesian Domain Randomization (BayRn) utilize Bayesian optimization to adaptively search the space of source domain distribution parameters based on sparse real-world feedback, thereby maximizing real-world objective performance while minimizing prior knowledge requirements [494]. Furthermore, active sampling strategies such as Active Dynamics Preference (ADP) have been proposed to quantify the informativeness of sampled system parameters, selectively training on environments that prevent the policy from becoming either over-conservative or over-optimistic [495].

Beyond randomization, rigorous system identification and calibration play pivotal roles in aligning simulation fidelity with reality, ensuring that the physical dynamics assumed during training match the operational environment. Traditional system identification often struggles with complex, non-linear dynamics and requires extensive manual tuning. Recent advances propose differentiable causal discovery frameworks, such as COMPASS, which automatically learn the causal relationships between environment parameters and the sim-to-real gap, enabling automated parameter tuning and improved trajectory alignment without exhaustive domain knowledge [496]. Similarly, task-driven adaptation frameworks like AdaptSim meta-learn adaptation policies to optimize simulation parameters specifically for target task performance rather than attempting to match global dynamics, demonstrating superior data efficiency in real-world adaptation [497]. In scenarios where dynamics mismatches are severe and the source domain lacks full support for target transitions, methods focusing on skewing and extending source support have shown efficacy in crafting effective target policies [498].

Complementing these algorithmic adaptations, the concept of the Digital Twin (DT) has emerged as a comprehensive strategy for bridging the gap by creating high-fidelity, executable virtual replicas of physical systems that evolve alongside their real-world counterparts. Unlike static simulators, advanced digital twins facilitate bi-directional communication, allowing for real-time parameter updates and continuous learning. For instance, executable digital twins (xDT) have been successfully employed to transfer and adapt nonlinear model predictive controllers for autonomous vehicles by online adaptation of control parameters, achieving high performance without tedious manual tuning [499]. In industrial settings, intelligent digital twins integrated with transfer learning enable cross-phase knowledge transfer, allowing algorithms to be tested in the design phase and fine-tuned with minimal real data once the physical system is available [500]. Furthermore, frameworks supporting run-time model swapping, such as fmiSwap, address the issue of model obsolescence due to wear and tear, ensuring that the digital twin remains a faithful representation throughout the system’s lifecycle [501].

Crucially, the reliability of these transfer strategies hinges on the calibration of predictive uncertainty, a prerequisite for the safety protocols discussed in the subsequent section. The dilemma

of domain adaptation often involves a trade-off where models achieve higher accuracy at the expense of well-calibrated probabilities. To resolve this, frameworks like Transferable Calibration (TransCal) have been developed to achieve accurate calibration with lower bias and variance in a unified, hyperparameter-free optimization, ensuring that agents can reliably estimate their uncertainty in unlabeled target domains [502]. Additionally, confidence-based switching criteria have been introduced to automatically determine when a simulation-trained policy is sufficiently robust for real-world deployment, preventing the execution of unreliable behaviors during the transfer process [503].

Finally, the paradigm of “sober deployment” emphasizes rigorous validation and model selection before physical execution to minimize risks, acting as a critical gatekeeper between simulation and the complex social environments detailed next. Techniques such as Validate on Sim, Detect on Real (VSDR) combine out-of-distribution detection with simulation evaluation to rank policies without costly real-world rollouts, effectively acting as a “unit test” for sim-to-real transfer [504]. Moreover, probabilistic dynamics models can predict transfer performance metrics based on limited real-world trajectories, simplifying the selection of training setups and hyperparameters [505]. As the field advances, the integration of these calibration strategies with high-fidelity digital twins and adaptive randomization techniques promises to significantly narrow the sim-to-real gap. By ensuring that agents possess both accurate physical models and calibrated uncertainty estimates, these methods lay the necessary groundwork for the comprehensive safety protocols, social navigation standards, and human-robot interaction frameworks explored in the following section.

8.5 Safety Protocols, Social Navigation, and Human-Robot Interaction

8.5 Safety Protocols, Social Navigation, and Human-Robot Interaction

While the strategies discussed in the previous section effectively bridge the sim-to-real gap by aligning physical dynamics and calibrating uncertainty, the successful deployment of embodied agents in open-world environments demands a further evolution: a transition from purely geometric collision avoidance to comprehensive safety protocols that encompass social norms, human perception, and trustworthiness. Traditional robotics often prioritized static obstacle avoidance, but modern evaluation landscapes must account for the complex, dynamic interplay between autonomous systems and human agents. A primary challenge in this domain is defining and measuring “socially compliant” behavior, which extends beyond physical safety to include comfort, legibility, and politeness. To address the lack of standardized evaluation, researchers have proposed comprehensive frameworks that define socially navigating robots as entities respecting principles such as safety, comfort, and responsiveness to context [506]. These guidelines emphasize that fair evaluation requires not only robotic agents moving in static environments but also dynamic human agents and their subjective perceptions of the appropriateness of robot behavior.

To quantify these abstract social concepts, several benchmarking protocols and simulation environments have been developed, serving as critical testbeds before real-world exposure. For instance, the SocNavBench framework provides a grounded simulation testing environment with curated scenarios derived from real-world pedestrian data, enabling consistent and interpretable comparisons of disparate social navigation methods [507]. Similarly, the Social Environment for Autonomous Navigation (SEAN) offers a high-fidelity, open-source platform designed to facilitate reproducible evaluation across different robotic platforms and dynamic pedestrian scenarios [508]. These simulators are critical because learning and benchmarking social behaviors entirely in the real world is often data-intensive and poses significant safety risks during the training phase [509]. Furthermore,

specific protocols have been introduced to validate social navigation policies using in-situ metrics and questionnaires, ensuring that benchmarking is realistic, scalable, and repeatable across various physical spaces [510].

A crucial aspect of safety evaluation involves understanding how human perception influences the assessment of risk and realism, particularly when bridging the gap between simulated metrics and human intuition. Studies utilizing Virtual Reality (VR) have revealed that human assessments of safety and realism in autonomous driving test cases can vary significantly based on factors such as scenario complexity and the potential for interaction, highlighting a gap between software metrics and human perception [511]. VR systems have also proven effective in collecting data for safety-critical scenarios, such as jaywalking or interactions with vulnerable road users, which are ethically and logistically challenging to capture in the real world [512]. Moreover, immersive VR environments allow researchers to “hack” non-verbal communication cues, demonstrating that pedestrians infer social intentions from vehicle kinematics rather than merely observing object states [513]. This insight underscores the need for evaluation metrics that capture the nuances of non-verbal signaling and social expectation, ensuring that agents calibrated for physical reality are also aligned with human social expectations.

Beyond social compliance, rigorous situational awareness and the handling of uncertainty are paramount for safety-critical deployments, complementing the uncertainty calibration techniques mentioned earlier. Evaluation metrics must evolve to reflect the dynamic interactions between agents rather than relying on static assumptions. For example, interaction-dynamics-aware metrics utilize Hamilton-Jacobi reachability theory to construct “safety zones” that define where safety-critical obstacles lie, offering a more accurate capture of risk than traditional perception metrics [514]. These concepts can be further refined through maneuver-based decomposition, which reduces the volume of safety zones by leveraging information about the ego vehicle’s specific maneuvers, thereby improving efficiency without compromising completeness [515]. In scenarios involving occlusions, set-based safety verification methods provide provable safety guarantees by over-approximating the states of hidden obstacles, addressing the inability of probabilistic methods to guarantee safety under high uncertainty [516]. Additionally, online monitoring frameworks have been developed to provide formal guarantees on the safety of human-robot interactions by integrating pedestrian intent estimation with reachability-based schemes [517].

Trustworthiness in human-robot interaction also hinges on the system’s ability to resolve conflicts and adapt to human preferences, forming a behavioral layer of safety that operates alongside technical calibration. Bidirectional frameworks have been proposed where robots dynamically modify their routes based on detected conflicts with human decisions while maintaining verbal communication, fostering a transparent and trustworthy interaction loop [518]. The assessment of such systems often involves analyzing conflict resolution datasets to understand how autonomous vehicles impact traffic efficiency and safety compared to human-only interactions [519]. Furthermore, the concept of responsibility allocation in multi-agent settings has led to the development of social preference-aware control frameworks that quantify individual selfishness and responsibility shares, ensuring formally provable safety guarantees in semi-cooperative scenarios [520].

Finally, the evaluation of safety must consider the broader network-level implications and the mitigation of biases, ensuring that agents do not perpetuate harmful patterns while navigating complex social fabrics. Network-level safety metrics (NSM) have been introduced to assess the overall safety profile of traffic flow by processing imagery from roadside units, moving beyond individual crash analysis to consider the traffic flow as a complex dynamic system [521]. Concurrently, there is a growing recognition that social navigation models learned from human demonstrations may repli-

cate societal biases; thus, frameworks for diminishing bias through “relearning” mechanisms are essential to promote equitable social relationships and safe interactions [522]. Collectively, these advances in protocols, metrics, and interactive frameworks form the backbone of robust safety evaluations. However, even with robust social protocols and calibrated dynamics, embodied agents remain susceptible to internal cognitive failures. The subsequent section addresses these intrinsic limitations by exploring advanced detection frameworks and mitigation techniques designed to prevent hallucinations and ensure the semantic fidelity of the agent’s perception-cognition-action loop.

8.6 Advanced Detection Frameworks and Mitigation Techniques

8.6 Advanced Detection Frameworks and Mitigation Techniques

While robust social protocols and calibrated dynamics form the foundation of safe navigation, they primarily address external interactions and physical constraints. As highlighted at the conclusion of the previous section, embodied agents remain susceptible to intrinsic cognitive failures, particularly hallucinations, where the perception-cognition-action loop generates responses or plans inconsistent with physical reality or visual inputs. In open-world environments, such internal discrepancies can be as detrimental as physical collisions, necessitating a shift from reactive safety measures to sophisticated, real-time detection frameworks and proactive mitigation strategies. These approaches leverage internal model states, semantic reconstruction, and targeted alignment techniques to ensure that agents operate reliably without constant human supervision.

A prominent direction in advanced detection involves mining the dense semantic information retained within the internal states of large models, moving beyond superficial logit-level uncertainty which often suffers from information loss during token decoding. Analyzing internal representations offers a more robust signal for early intervention. For instance, the INSIDE framework proposes utilizing the eigenvalues of response covariance matrices within internal states to measure semantic consistency, effectively detecting hallucinations before they manifest in the output [482]. Similarly, unsupervised frameworks like MIND leverage these internal activations for real-time detection without requiring manual annotations, demonstrating superior performance over state-of-the-art post-processing techniques [523]. Probing specific layers and hidden states has also revealed that extrinsic hallucinations—those contradicting external facts—are often more salient in transformer representations than intrinsic errors, allowing for efficient, ecologically valid detection probes [524]. Furthermore, weakly supervised scanning methods can identify anomalous patterns in activations indicative of hallucination without prior knowledge of the error type, offering a scalable auditing tool for deployed agents [525].

Complementing internal monitoring, semantic reconstruction techniques provide a powerful mechanism for grounding generated text back to visual inputs, thereby closing the loop between perception and generation. The ESREAL framework introduces an unsupervised approach where the model reconstructs an image based on its generated caption and aligns it with the original input. By penalizing tokens that lead to semantic mismatches in the reconstructed regions, this method significantly reduces hallucination rates in vision-language models without needing paired image-text training data [526]. Alongside reconstruction, decoding strategies that manipulate the generation process itself have shown promise in preventing errors at the source. The OPERA method addresses “over-trust” penalties in self-attention matrices, where models focus too heavily on summary tokens while neglecting visual details. By introducing a penalty term and a retrospection-allocation strategy during beam search, OPERA mitigates hallucinations without additional training or external

knowledge [527]. Similarly, Multi-Modal Mutual-Information Decoding (M3ID) amplifies the influence of the visual prompt over the language prior during sampling, ensuring generated tokens maintain high mutual information with the input image [528].

To further fortify agents against cognitive drift, mitigation strategies have evolved towards targeted instruction tuning and preference optimization designed to align model behaviors strictly with physical reality. Recognizing that generic instruction datasets often fail to address model-specific hallucination tendencies, the DFTG framework diagnoses specific hallucinatory concepts unique to a model and generates targeted instruction data to correct them, proving more effective than broad-spectrum datasets [529]. Expanding on this, the LRV-Instruction dataset incorporates negative instructions—such as nonexistent object manipulation—to train models to resist hallucinating under adversarial prompts, leading to more robust visual instruction tuning [530].

Preference optimization techniques, particularly Direct Preference Optimization (DPO), have emerged as a potent tool for refining agent reliability by directly encoding trustworthiness into the model’s policy. The Silkie model demonstrates that distilling preference feedback from AI annotators regarding visual faithfulness can significantly reduce hallucinations while enhancing fine-grained perception capabilities [531]. Building on this, RLHF-V utilizes fine-grained, segment-level human corrections to perform dense preference optimization, achieving substantial reductions in hallucination rates with relatively small annotated datasets [532]. Advanced frameworks like FGAIF further refine this by employing fine-grained AI feedback to train specialized reward models for different hallucination types (object, attribute, relationship), which are then integrated into reinforcement learning algorithms to align modalities more precisely [533]. Additionally, the POVID approach automates the creation of preference data by injecting plausible hallucinations and distorting images, enabling scalable alignment without human labeling [534].

Finally, hybrid approaches that combine detection with active retrieval or logical consistency checks offer comprehensive safety nets for complex reasoning tasks. The Rowen framework activates external retrieval only when semantic-aware detection identifies inconsistencies, balancing parametric knowledge with external evidence to mitigate both internal and external hallucinations [535]. LogicCheckGPT employs a logical closed-loop strategy, probing the model with logically correlated questions to verify consistency; if the responses fail to form a closed loop, the system identifies potential object hallucinations and mitigates them without external models [481]. Together, these advanced detection and mitigation techniques form a critical layer of defense, ensuring that the semantic fidelity of the agent’s cognition matches the robustness of its social and physical safety protocols. With these intrinsic cognitive safeguards in place, the focus can now shift to the broader system-level challenges of deploying these capable agents in real-world infrastructure and long-horizon tasks.

9 Open Challenges and Future Directions for Generalist Embodied Agents

9.1 Causal Reasoning and Intuitive Physics Integration

9.1 Causal Reasoning and Intuitive Physics Integration

The transition from static perception to dynamic embodied reasoning represents a pivotal frontier in artificial intelligence, serving as the foundational prerequisite for the adaptive capabilities discussed in subsequent sections. While current Vision-Language Models (VLMs) have achieved remarkable proficiency in recognizing objects and describing scenes, they fundamentally operate on correlational patterns derived from vast datasets rather than a genuine understanding of cause and

effect. This reliance on statistical association creates a critical fragility when agents are deployed in open-world physical environments where conditions frequently deviate from training distributions. As highlighted in recent surveys, deep learning models often struggle to generalize beyond their initial data distribution because they fail to learn mechanisms that are independent of specific data contexts [536]. To bridge this gap and enable the robust, long-term adaptability required for lifelong learning, future generalist embodied agents must integrate causal reasoning frameworks with intuitive physics engines, allowing them to infer underlying physical mechanisms and perform robust counterfactual planning.

The core limitation of contemporary embodied AI lies in its inability to distinguish between spurious correlations and true causal relationships. In complex robotic tasks, an agent might learn that a specific visual texture correlates with a successful grasp, rather than understanding the physical properties of friction and mass that actually govern the interaction. This issue is exacerbated by the fact that standard deep learning approaches primarily analyze associations, effectively functioning as sophisticated curve-fitting exercises that lack the capacity for intervention and imagination [537]. Consequently, when an embodied agent encounters a novel object or an unexpected physical perturbation, it often fails catastrophically because its policy is not grounded in the invariant laws of physics. To overcome this, researchers argue that off-the-shelf deep learning is insufficient; instead, there is a pressing need for Bayesian methods and causal models that incorporate prior physical knowledge to handle confounding factors and ensure robust performance in experimental domains [538]. Without such grounding, the stability-plasticity balance essential for continual adaptation becomes impossible to maintain, as the agent lacks a stable mechanistic model upon which to build new skills.

A promising direction for addressing these limitations is the development of causal deep learning frameworks that explicitly model the structural, parametric, and temporal dimensions of causality. Such frameworks allow agents to leverage partial causal knowledge and quantitatively characterize relationships over time, which is essential for tasks requiring long-horizon planning and dynamic adaptation [539]. In the context of embodied reasoning, this implies moving beyond passive observation to active intervention: agents must be capable of performing experiments within their environment to discover causal factors, a concept akin to “causal curiosity” where intrinsic rewards drive agents to execute actions that reveal the underlying dynamics of the world [540]. By treating actions as interventions, agents can construct internal causal graphs that describe how their manipulations affect state transitions, thereby facilitating zero-shot transfer to new tasks and environments [156]. This active discovery process lays the groundwork for the continuous skill acquisition central to lifelong learning paradigms.

Integrating intuitive physics into these causal frameworks is equally critical. Human infants rapidly develop an understanding of physical laws through interaction, forming prior knowledge that guides future learning. Inspired by this, embodied agents can utilize graphical physics networks integrated with deep reinforcement learning, employing intrinsic motivation to continuously refine their physical models through active engagement with objects [541]. Furthermore, the ability to simulate counterfactual scenarios—asking “what would have happened if I had pushed the object harder?”—is a hallmark of true physical understanding. Recent advances in counterfactual learning of physical dynamics, such as the CoPhy benchmark, demonstrate that models can predict alternative futures based on altered initial conditions, capturing latent physical properties like mass and elasticity without explicit supervision [542]. This capability allows agents to engage in “hindsight for foresight” reasoning, simulating potential outcomes before committing to physical actions, thus reducing the risk of catastrophic failures during the exploration phases of lifelong learning.

However, realizing this vision requires overcoming significant challenges in representation learning and model architecture. High-dimensional sensory data, such as video streams from egocentric cameras, must be distilled into concise causal representations. Techniques for causal disentanglement in multimodal data are essential here, allowing agents to identify independent causal factors from complex visual inputs using physics-based constraints [543]. Moreover, the integration of neural networks with physics-informed architectures can drastically improve data efficiency and generalization. By embedding known physical principles, such as conservation laws or kinematic constraints, directly into the network structure or loss functions, agents can learn dynamics models that are both accurate and interpretable [544]. This approach ensures that the learned representations respect the fundamental laws of the physical world, reducing the sample complexity required for mastering new manipulation skills and providing a stable substrate for the neuromodulatory and memory mechanisms discussed in the following section.

Looking forward, the path toward generalist embodied intelligence necessitates a shift from purely data-driven correlation mining to hybrid systems that combine the pattern recognition strengths of deep learning with the mechanistic rigor of causal inference. Future research must focus on developing scalable methods for causal discovery from indefinite data types, such as unstructured video and natural language dialogue, which are prevalent in human-robot interaction [545]. Additionally, the creation of comprehensive benchmarks that specifically evaluate causal reasoning and intuitive physics, such as X-VoE and CRAFT, will be instrumental in driving progress [484; 546]. Only by equipping agents with the ability to reason about causes, simulate interventions, and understand the intuitive physics of their surroundings can we achieve truly autonomous systems capable of adapting seamlessly to the complexities of the real world. The ultimate goal is to create agents that do not merely react to sensory inputs but actively construct and test causal hypotheses about their environment, mirroring the scientific inquiry process inherent in human cognition [547], thereby enabling the resilient, open-ended growth characteristic of biological intelligence.

9.2 Lifelong Learning and the Stability-Plasticity Dilemma

9.2 Lifelong Learning and the Stability-Plasticity Dilemma

While the integration of causal reasoning and intuitive physics discussed in the previous section provides the mechanistic foundation for understanding *how* the world works, the realization of truly generalist embodied agents further hinges on their capacity for lifelong learning: the ability to continuously acquire new skills and adapt to dynamic environments without erasing previously mastered knowledge. This challenge is fundamentally rooted in the stability-plasticity dilemma: the need to maintain stable representations of past experiences (stability) while remaining flexible enough to integrate novel information (plasticity). In the context of embodied reasoning, where agents interact with non-stationary physical worlds governed by the causal laws identified earlier, failing to balance these competing demands leads to catastrophic forgetting, rendering long-term deployment impossible. Recent research increasingly turns to biological inspiration to resolve this tension, proposing architectures that mimic the brain’s sophisticated mechanisms for memory consolidation, synaptic regulation, and adaptive control, thereby creating the resilient substrate necessary for the data-efficient exploration strategies detailed in the following section.

A primary avenue for addressing this dilemma involves the development of neuromodulatory mechanisms that dynamically regulate learning rates and synaptic plasticity based on task context and uncertainty—factors often derived from the agent’s causal models of the environment. Biological nervous systems utilize neuromodulators to fine-tune synaptic dynamics in real-time, allowing for

efficient adaptation without destabilizing existing circuits. Inspired by this, researchers have introduced neural networks that learn a compact set of parameters to neuromodulate the activity of unchanging, randomized weights, effectively eliminating catastrophic forgetting while maintaining a minimal spatial footprint [548]. Furthermore, uncertainty has emerged as a critical signal for modulating lifelong learning; algorithms that employ uncertainty-based neuromodulation can distinguish between situations requiring the retention of old knowledge and those demanding rapid adaptation to new conditions. Such approaches have demonstrated robust performance in closed-loop embodied settings, such as simulated drone navigation, where agents must learn continuously under changing environmental constraints without forgetting prior capabilities [549]. By integrating uncertainty-guided modulation, agents can achieve a form of “soft” parameter isolation, adapting learning rates according to the probability distribution of weights to identify precisely what to remember and what to update [550].

Parallel to neuromodulation, dual-memory architectures offer a structural solution to the stability-plasticity trade-off by emulating the complementary learning systems (CLS) theory observed in mammalian brains, which posits an interplay between a fast-learning hippocampal system and a slow-learning neocortical system. This architectural duality allows agents to separate episodic memory, which captures fine-grained spatiotemporal details of specific interactions, from semantic memory, which extracts generalized categories and rules—potentially including the causal graphs and physical priors discussed previously. For instance, dual-memory self-organizing architectures have been shown to successfully learn object instances and categories concurrently, using episodic replay to consolidate knowledge into semantic structures without catastrophic interference [401]. Extending this concept, Triple Memory Networks (TMNs) model the complex interplay between the hippocampus, prefrontal cortex, and sensory cortex using a triple-network architecture of generative adversarial networks, achieving state-of-the-art performance by balancing specific data representations with generalized task-solving knowledge [551]. Similarly, frameworks like LLEDA draw directly from CLS theory to coordinate a fast-adapting domain adaptation network with a slowly learning self-supervised network, utilizing latent replay to stabilize long-term generalization across shifting domains [552]. These biologically inspired designs suggest that embodied agents require distinct memory stores with different plasticity rates to manage the influx of continuous sensory data effectively.

Beyond structural and modulatory changes, the mechanism of forgetting itself is being re-evaluated. Rather than viewing forgetting solely as a failure, emerging perspectives argue that “active forgetting” is essential for efficient lifelong learning, allowing agents to shed irrelevant or conflicting information to make room for new concepts. The SMART framework, for example, introduces novel short-term and long-term forgetting mechanisms inspired by neuroscience to enhance memory efficiency and robustness against noisy data, positing that reasonable forgetting is crucial for closing the performance gap between AI and human learners [553]. This aligns with the Active Forgetting with synaptic Expansion-Convergence (AFEC) approach, which dynamically expands parameters for new tasks and selectively combines them, mimicking biological processes where old knowledge that interferes with new learning is actively suppressed [554]. Moreover, the incorporation of sleep-like mechanisms, including veridical and generative replay alongside synaptic downscaling, has been shown to improve continual learning by tuning signal-to-noise ratios and protecting early task accuracy, highlighting the potential of offline consolidation phases for embodied agents [555].

Despite these advances, significant challenges remain in deploying these strategies within the high-dimensional, continuous control loops required for embodied generalists. Current methods often

rely on simplified benchmarks that do not fully capture the complexity of physical interaction or the sheer duration of real-world operation. Future research must focus on scaling these bio-inspired mechanisms to large-scale foundation models, ensuring that neuromodulatory signals and dual-memory interfaces can operate efficiently within the computational constraints of mobile robots. Crucially, the successful resolution of the stability-plasticity dilemma serves as the enabling condition for the next frontier: data efficiency. By synthesizing neuromodulation, dual-memory structures, and adaptive forgetting protocols, the field moves closer to creating embodied agents capable of open-ended growth. These mechanisms ensure that the diverse experiences generated through the intrinsic motivation and active exploration discussed in the subsequent section can be accumulated and refined over a lifetime, mirroring the resilience and adaptability of biological intelligence in an ever-changing world.

9.3 Data Efficiency through Intrinsic Motivation and Active Exploration

9.3 Data Efficiency through Intrinsic Motivation and Active Exploration

While the mechanisms discussed in the previous section address how agents retain knowledge over time, a prerequisite for viable lifelong learning is the ability to acquire that knowledge efficiently. A paramount challenge in realizing generalist embodied agents remains the profound sample inefficiency inherent in current deep learning paradigms. Unlike biological systems, which acquire complex sensorimotor repertoires through relatively sparse interactions, artificial agents often require millions of trials to master tasks that a human child could learn in minutes. To bridge this gap and enable the continuous adaptation required for lifelong deployment, research must pivot towards frameworks that emulate human-like developmental curricula, driven by intrinsic motivation and active exploration. These mechanisms enable agents to autonomously select goals, generate their own training data, and construct robust world models without reliance on exhaustive external supervision or hand-crafted reward functions, thereby laying the data-rich foundation necessary for the stability-plasticity balance.

The core of this paradigm shift lies in the transition from extrinsic, task-specific rewards to intrinsic drives that foster open-ended skill acquisition. Intrinsically Motivated Goal Exploration Processes (IMGEP) have emerged as a foundational architecture for this purpose, allowing agents to self-generate, select, and order learning goals based on internal metrics of progress rather than external validation [556]. By treating goals as parameterized fitness functions, agents can systematically reuse information acquired during the pursuit of one objective to improve performance on others, effectively building a curriculum of increasing complexity. This approach mirrors the autotelic nature of human development, where play serves as a mechanism for discovering diverse skills that act as stepping stones for more sophisticated behaviors, such as nested tool use [556].

A critical component of efficient exploration is the structure of the goal space itself. Early implementations relied on engineered feature spaces, which limited autonomy and scalability. Recent advancements demonstrate that leveraging deep representation learning to discover disentangled goal spaces significantly enhances exploration efficiency in complex, multi-object environments [557]. When the latent structure of the goal space reflects the underlying semantics of the environment, agents can modularly sample goals that maximize learning progress, thereby avoiding the pitfalls of exploring entangled or irrelevant state dimensions [557]. Furthermore, the integration of language as a cognitive tool allows agents to imagine and target out-of-distribution goals, expanding the horizon of discoverable skills beyond the distribution of previously experienced states [558]. This “goal imagination,” supported by the compositionality of natural language, enables agents to for-

mulate abstract objectives and decompose them into achievable sub-goals, a capability essential for tackling long-horizon tasks in open worlds [559].

The mechanism driving this exploration is often rooted in the concept of “learning progress” or curiosity. Agents are motivated to engage with regions of the state space where their predictive models are neither too certain nor too uncertain, optimizing the rate of knowledge acquisition [560]. This principle of targeting the “frontier” of competence ensures that exploration remains directed and productive. For instance, algorithms that prioritize goals maximizing the epistemic uncertainty of the value function have been shown to sample tasks that are optimally challenging, leading to continual improvement and preventing stagnation [561]. Moreover, the combination of curiosity with empowerment—the measure of an agent’s potential control over its future—provides a unified strategy for balancing the drive to learn new dynamics with the desire to maintain agency over the environment [562].

Active exploration is further enhanced by the construction of structured world models. Instead of using models solely to compute intrinsic rewards, next-generation agents utilize these models for planning within imagined rollouts, enabling zero-shot generalization to downstream tasks after a phase of task-agnostic play [563]. The Latent Explorer Achiever (LEXA) framework exemplifies this by training an explorer policy to discover surprising states through foresight, which are then used as targets for an achiever policy, effectively decoupling the discovery of goals from their execution [564]. This separation allows for the accumulation of a rich repertoire of skills that can be recombined to solve novel problems without additional environment interaction, directly addressing the data hunger that plagues traditional reinforcement learning.

The role of social interaction and environmental feedback cannot be overstated in refining these curricula. While early development is often Piagetian, relying on individual interaction with physics, the Vygotskian perspective emphasizes the value of social scaffolding. Protocols like “Help Me Explore” demonstrate how minimal social interventions from a partner can guide autotelic agents toward complex configurations they would otherwise fail to discover, effectively bootstrapping the learning process [565]. Additionally, active probing strategies allow agents to clarify uncertainties about human intentions or environmental dynamics by choosing actions that maximally challenge their current beliefs, thereby accelerating the convergence of accurate world models [566].

Looking forward, the integration of these principles into unified embodied agents creates a synergistic loop with the challenges of lifelong learning and generalization. The data efficiency gained through intrinsic motivation provides the steady stream of diverse experiences necessary to test and refine the neuromodulatory and dual-memory systems discussed previously, ensuring that agents can adapt without catastrophic forgetting. Simultaneously, the structured world models and disentangled goal spaces cultivated through active exploration serve as the precursors to the grounded, compositional representations required for robust generalization. Approaches that utilize graph-based representations to organize sub-goals and memories offer a promising direction for maintaining a coherent, lifelong curriculum [565], bridging the gap between efficient data collection and the structured reasoning needed for the next stage of embodied intelligence. Ultimately, achieving human-level data efficiency will depend on synthesizing curiosity-driven exploration, language-guided goal imagination, and structured world modeling into a cohesive architecture that allows agents to learn autonomously in the unstructured, dynamic reality of the physical world.

9.4 Grounded Representation and Compositional Generalization

9.4 Grounded Representation and Compositional Generalization

While Section 9.3 detailed how intrinsic motivation and active exploration generate the diverse, data-rich experiences necessary for lifelong learning, the utility of this data hinges on how it is represented. A paramount challenge in the pursuit of truly generalist embodied agents lies in their persistent inability to generalize beyond the specific distributions encountered during training. Even with abundant exploration data, contemporary Vision-Language Models (VLMs) often falter when required to apply learned concepts to novel physical configurations or unseen environmental dynamics. This limitation stems fundamentally from a reliance on statistical correlations within training data rather than an understanding of the underlying causal and relational structures that govern the physical world. To bridge this gap, and to provide the structured foundation required for the introspective capabilities discussed in the following section, research must pivot towards architectures that prioritize grounded semantic representations and compositional generalization. These mechanisms enable agents to decompose complex scenarios into known entities and recombine them to solve unprecedented tasks, transforming raw sensory streams into robust, manipulable knowledge.

The core of this challenge is the “binding problem,” where neural networks struggle to dynamically and flexibly bind distributed information to form symbol-like entities, which is crucial for systematic generalization [567]. Current deep learning approaches often treat inputs as holistic patterns, failing to disentangle the explanatory factors of variation such as object identity, spatial relations, and physical properties. Consequently, when an agent encounters a scenario with a different number of objects or a novel arrangement, performance degrades significantly because the model has not learned the abstract rules governing these interactions. As noted in broader analyses of artificial intelligence, achieving human-like abilities requires a shift towards structured representations and computations that facilitate combinatorial generalization [568]. Without such structural priors, agents remain confined to interpolating within their training manifolds, unable to extrapolate to out-of-distribution settings, regardless of the volume of data acquired through the exploratory curricula described previously.

To address these shortcomings, the integration of relational inductive biases through graph-based architectures emerges as a critical direction. Graph Neural Networks (GNNs) and related frameworks offer a natural mechanism for encoding object-centric and relation-centric representations, allowing agents to reason about the structure of the world explicitly. Research has demonstrated that incorporating relational graphs into reinforcement learning agents can significantly improve both in-distribution and out-of-distribution generalization compared to standard convolutional baselines [569]. By mapping grid structures to relational graphs, agents can leverage the topology of interactions to infer dynamics that were not explicitly programmed but learned from data. Furthermore, graph networks have shown surprising efficacy as learnable physics engines, capable of predicting future states and generalizing across distinct physical systems by capturing the universal laws of interaction [570]. This ability to learn the dynamics and structure of complex systems suggests that graph-based models can serve as the backbone for embodied agents needing to understand intuitive physics and causal mechanisms, effectively structuring the data collected via active exploration.

However, simply adopting graph architectures is insufficient; the nature of the relations learned must be robust to distribution shifts. A significant issue in current graph learning is the tendency to exploit spurious correlations present in the training environment, leading to failure when these correlations change in testing scenarios. To mitigate this, causal inference techniques must be em-

ployed to distinguish stable, invariant relations from environment-specific noise. Frameworks like StableGNN propose extracting high-level subgraph representations and utilizing causal variable distinguishing regularizers to focus learning on stable correlations, thereby enhancing robustness against distribution shifts [571]. Similarly, methods that enforce physically motivated inductive biases, such as conserving energy or adhering to Newtonian laws within the message-passing mechanisms of graph networks, have been shown to recover explicit symbolic forms of physical laws and improve zero-shot generalization to systems with varying numbers of entities [572]. These approaches underscore the necessity of grounding representations in physical reality rather than purely statistical patterns, ensuring that the agent’s world model reflects true causality.

Compositional generalization also demands that agents can learn abstract relational semantics that are independent of specific entity instances. In the context of knowledge graphs and relational reasoning, models that reason over local subgraph structures have demonstrated the ability to generalize to entirely unseen entities and graphs, a capability known as inductive learning [573]. This contrasts with embedding-based methods that are often limited to transductive settings. For embodied agents, this implies the need for architectures that can abstract “concepts” such as “pushing,” “stacking,” or “supporting” as relational predicates that can be applied to any object possessing the relevant affordances. Recent work in neuro-symbolic approaches highlights the importance of balancing symbolic generalization with neural grounding, where inductive concepts are expressed as symbolic compositions over grounded neural action concepts [574]. This hybrid strategy allows agents to sketch a programmatic plan, execute it via grounded policies, and then generalize the underlying logic to novel heights or object configurations.

Moreover, the multimodal nature of embodied interaction offers unique opportunities to enhance compositional generalization. Studies indicate that while pure vision models may struggle to generalize compositionally, the introduction of multimodal inputs can significantly improve an agent’s ability to infer high-level relationships and generalize across object counts and attributes [575]. This suggests that language can act as a scaffold for structuring visual perceptions into compositional units, reinforcing the role of language as a cognitive tool introduced in the context of goal imagination. However, current large language models still exhibit limitations in abstract linguistic generalization, often failing to generalize argument structures to contexts not observed during pre-training [576]. Therefore, future embodied agents must integrate linguistic priors with visual and physical grounding more tightly, ensuring that the semantic representations are not just linguistically coherent but physically executable.

In conclusion, overcoming the generalization barrier in embodied AI requires a fundamental rethinking of representation learning, building upon the data efficiency gained through exploration to construct robust internal models. The path forward involves moving away from monolithic, end-to-end black boxes towards modular, structured architectures that explicitly model entities, relations, and rules. By leveraging relational inductive biases [577], employing causal reasoning to filter spurious correlations [578], and integrating symbolic planning with neural perception, we can develop agents capable of true compositional generalization. Such agents will not merely recall past experiences but will possess the cognitive flexibility to construct novel solutions by composing known concepts. Crucially, these grounded, compositional representations serve as the prerequisite for the introspective architectures discussed next; only when an agent possesses a structured understanding of its environment can it meaningfully monitor its own confidence, quantify uncertainty, and engage in the self-mediated adaptation required for safe deployment in open worlds.

9.5 Introspective Architectures and Self-Mediated Adaptation

9.5 Introspective Architectures and Self-Mediated Adaptation

While Section 9.4 established the necessity of grounded, compositional representations for extrapolating to novel physical configurations, the deployment of such structured agents in open worlds demands an additional layer of cognitive capability: introspection. The transition from reactive embodied agents to truly adaptive, generalist intelligence hinges on the development of introspective architectures. Unlike traditional systems that operate as black boxes, introspective agents possess the capability to monitor their internal states, assess their own confidence levels, and dynamically adjust their strategies based on a deep awareness of their limitations. This self-mediated adaptation is critical for deploying robots in open-world environments where uncertainty is pervasive and pre-defined reward signals are often sparse or misleading. The core of this architectural shift involves moving beyond simple perception-action loops to include a meta-cognitive layer that evaluates the quality of the agent’s own reasoning and predictions, thereby bridging the gap between static structural understanding and dynamic operational reliability.

A fundamental component of introspective architecture is the ability to quantify uncertainty and translate it into actionable confidence measures. For an agent to safely interact with humans or manipulate delicate objects, it must distinguish between aleatoric uncertainty, which arises from inherent noise in the environment, and epistemic uncertainty, which stems from a lack of knowledge. Research has demonstrated that ensembles of deep generative models can effectively quantify these uncertainties when forecasting task outcomes, allowing agents to communicate their competencies accurately [579]. Furthermore, agents must be able to forecast these uncertainties over long-horizon trajectories, as errors compound over time in dynamic settings. By developing world models that predict full agent trajectories while accounting for both types of uncertainty, systems can produce calibrated outcome estimates that are essential for reliable autonomy [580]. This capacity for self-evaluation extends to control mechanisms, where mathematical frameworks allow robots to derive closed-form expressions for control confidence. Such metrics enable agents to make better-informed decisions, such as selecting the most appropriate tool for a task by balancing performance, effort, and self-confidence [581].

Beyond mere uncertainty quantification, advanced introspective agents employ self-mediated learning strategies to refine their policies without external supervision. This involves the agent acting as its own critic, analyzing past trajectories to generate succinct “tips” or insights that guide future decision-making. For instance, the concept of “Introspective Tips” allows Large Language Models (LLMs) to self-optimize by examining their own execution histories and integrating expert demonstrations or past failures directly into their context, thereby improving performance in few-shot and zero-shot scenarios without parameter fine-tuning [582]. Similarly, introspective planning frameworks guide language-enabled agents to identify ambiguity in natural language instructions and proactively seek clarification before executing potentially unsafe or misaligned plans. This systematic approach to refining uncertainty significantly enhances both success rates and safety compared to standard planning methods [583].

The biological inspiration for these architectures often draws from the interdependence of strategy, physiology, and sensing. True adaptation requires an agent to consider its own physical and computational limitations within the context of its environment, rather than relying solely on increasing model complexity. An introspective agent recognizes that in certain environments, enhancing physical capabilities or sensing modalities may yield greater returns than extending planning horizons, mirroring the evolutionary adaptations seen in biological organisms [4]. This self-awareness is

further deepened by models that learn internal states directly from neuron activations. By reconstructing feature extraction network activations via Variational Autoencoders, agents can build internal representations that accelerate convergence in actor-critic models, effectively reducing the sample complexity required to master robotic tasks [584].

Moreover, the development of self-models allows agents to adversarially challenge their own world models, driving curiosity-driven exploration. In this paradigm, a separate explicit self-model tracks the error map of the agent’s predictive world model, encouraging the system to explore novel interactions that maximize information gain and reduce prediction errors [366]. This aligns with broader theories suggesting that agents should predict internally generated signals regarding their own learning processes, such as confidence in their predictions, to construct knowledge in a bottom-up, constructivist fashion [585]. The integration of such introspective capabilities is also vital for social interaction and human-robot teaming. Agents that can assess their own capabilities through task-based dialogues before, during, and after missions foster greater trust and facilitate more effective collaboration with human partners [586].

Looking forward, the realization of generalist embodied agents will depend on synthesizing these introspective mechanisms into unified cognitive architectures. Future research must address the stability-plasticity dilemma, ensuring that agents can continuously adapt to new environments via self-mediated updates without catastrophic forgetting. Techniques such as confidence-aware moving averages for continual learning offer promising pathways for updating agent knowledge based on internal confidence scores rather than rigid task boundaries [587]. Additionally, the emergence of architectures that combine LLMs with global neuronal workspace theories suggests a route toward agents capable of genuine agency and meta-cognition, moving beyond static pattern matching to dynamic self-motivation [588]. Ultimately, by endowing agents with the ability to introspect on their own cognitive processes, we lay the essential groundwork for the next evolutionary step: a shift from isolated autonomy to symbiotic collaboration. As explored in the following section, this self-awareness is the prerequisite for the hybrid intelligence ecosystems where humans and machines co-evolve within a shared cognitive loop to navigate the complexities of the real world.

9.6 Hybrid Intelligence and Human-AI Symbiosis in Open Worlds

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Building directly upon the introspective capabilities and self-assessment dialogues established in Section 9.5, the trajectory of embodied artificial intelligence is increasingly shifting away from the pursuit of isolated, fully autonomous generalists toward the development of hybrid intelligence ecosystems. While introspection provides the necessary internal mechanism for an agent to recognize its own limitations, the logical culmination of this self-awareness is the ability to seamlessly integrate human oversight and collaboration. In these emerging ecosystems, embodied agents do not merely operate alongside humans but collaborate through shared narratives, calibrated symbolic interfaces, and the strategic integration of complementary strengths to solve complex, open-world problems. This paradigm recognizes that while modern foundation models possess immense reasoning capabilities, they often lack the grounded physical intuition, ethical alignment, and contextual adaptability inherent to biological agents. Consequently, the future of generalist embodied agents lies in symbiotic relationships where humans and machines co-evolve within a shared cognitive loop, transforming the agent’s internal confidence metrics into actionable signals for human-AI teamwork.

The conceptual foundation for this transition rests on the recognition that true general intelligence

may not be achievable by scaling data alone but requires a framework of “shared intelligence” where humans are integral participants in a cyber-physical ecosystem [11]. This approach leverages active inference to create a common ground for belief sharing, allowing agents and humans to align their generative models of the world. Rather than viewing AI as a replacement for human cognition, hybrid intelligence systems aim to amplify human capabilities by treating AI as a collaborative partner that can learn from and adapt to human strategies [589]. This symbiosis is particularly critical in open worlds where environments are dynamic, unstructured, and fraught with uncertainties that purely data-driven models struggle to navigate without catastrophic failures or misalignment [590]. Here, the introspective agent’s ability to quantify uncertainty becomes the bridge that invites human intervention precisely when epistemic confidence drops, ensuring safety without sacrificing autonomy.

A central mechanism for enabling this symbiosis is the development of calibrated symbolic interfaces that facilitate robust communication and mutual understanding, extending the self-mediated adaptation discussed previously into the social domain. Traditional natural language interactions often suffer from ambiguity and a lack of grounding in physical reality. To address this, recent frameworks propose integrating large language models with symbolic reasoning layers to create interpretable and reliable interaction channels [13]. Furthermore, the concept of “symmetrical reality” suggests a unified representation framework that accommodates the dual perceptual centers of both humans and AI agents, allowing for seamless collaboration across physical and virtual domains [591]. By establishing such shared semantic spaces, agents can better interpret human intent, while humans can more effectively monitor and guide agent behavior based on the agent’s exposed internal states, thereby reducing the risk of hallucinations and execution errors [592].

The dynamics of human-AI collaboration are further enriched by the adoption of shared narratives and social learning protocols, mirroring the way introspective agents utilize their own history for self-improvement. Just as human cognitive development is deeply rooted in social interaction and cultural transmission, embodied agents can accelerate their learning trajectories by engaging in socially guided exploration [565]. This Vygotskian perspective emphasizes that higher cognitive functions emerge from the internalization of socio-cultural processes, suggesting that agents should be designed to actively seek human feedback and clarification when facing novel situations, much like the proactive clarification strategies seen in introspective planning [593]. In practice, this manifests as agents capable of “theory of mind” modeling, where they infer the beliefs and goals of their human partners to maintain common ground during collaborative tasks [594]. Such capabilities enable agents to anticipate human needs, resolve ambiguities proactively, and adapt their communication styles to match the expertise level of their human collaborators [595].

Moreover, the architecture of hybrid intelligence systems must support variable autonomy, allowing the degree of human involvement to fluctuate based on task complexity and the agent’s self-assessed confidence levels. Frameworks like the Autonomous Cognitive Entity (ACE) model propose layered architectures where high-level strategic goals set by humans are decomposed into executable tasks by AI, with mechanisms for failure handling and adaptive replanning that leverage the agent’s introspective error maps [596]. This flexibility ensures that humans remain “in the loop” for critical decision-making while delegating routine or high-speed operations to the agent, optimizing the overall efficiency and safety of the system [597]. The integration of wireless collective intelligence networks further extends this symbiosis, enabling distributed agents to share knowledge and reasoning outcomes with human operators in real-time, thus creating a scalable network of collective intelligence [598].

Looking forward, the realization of human-AI symbiosis in open worlds requires addressing funda-

mental challenges in trust, ethics, and long-term alignment, building upon the stability-plasticity solutions proposed for individual introspective agents. Theories of parenting and radical nurturing principles offer a novel lens for designing AI systems that are not only competent but also aligned with human values through continuous, interactive development [599]. Additionally, the concept of “open-ended intelligence” posits that intelligence is a formative process of self-organization, suggesting that hybrid systems should be designed to evolve their own structures and goals in coordination with human society [600]. By fostering environments where humans and agents co-create knowledge and solve problems together, we move closer to a future where embodied AI serves as a true extension of human agency, capable of navigating the complexities of the real world with wisdom, adaptability, and shared purpose [17]. This transition from isolated autonomy to deep symbiosis represents not just a technical evolution, but a fundamental reimagining of the role of intelligent machines in human society, completing the arc from structured representation to self-aware adaptation, and finally to collaborative co-evolution.

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